

Chapter Drafting Worksheet

Topic
Understanding Social Scientific Research

Parenthetical disambiguation
(Mis)Information

Title movement
From certainty to curiosity

One big shift
This chapter moves students from the common-sense belief that research gives us clear, universal answers about how people communicate toward the (un)common-sense understanding that research is a method of sensemaking that shapes what we notice, believe, and do. Research claims do not stay in articles. They become advice, habits, diagnoses, workplace trainings, relationship expectations, social media scripts, AI answers, and everyday explanations for why people act the way they do. A single study does not transfer final truth about communicating. It offers partial, situated evidence that has to be interpreted in context, connected to other evidence and lived experience, evaluated for credibility and limits, and used to build flexible heuristics rather than rigid rules. The goal is to help students use research more carefully so they can improve their communicating in personal relationships, professional settings, and the job skills they will need in an AI-shaped information world.

Master annotated bibliography

Master Annotated Bibliography: Research Literacies, AI, and the (Un)Common-Sense Approach

1. The "Ivory Tower," Public Opinion, and the Peer Review Process

Connecting to the brainstorming notes on how students view research as an intimidating "ivory tower" product, leading to a disconnect between their daily lives and the scholarly quality-control process.

National Science Board. (2024). *Science and technology: Public perceptions, awareness, and information sources (NSB-2024-4)*. National Science Foundation. <https://ncses.nsf.gov/pubs/nsb20244>

- **Annotation:** This is one of the best starting points for your chapter because it gives an updated, official overview of U.S. public perceptions of science, scientists, and research processes. It synthesizes long-running survey indicators rather than a single dataset, which makes it especially useful for textbook framing. The strongest contribution is its combination of trust, familiarity with research logic, and information-source use. For your purposes, it supports a nuanced claim: the public is not broadly anti-science, but confidence is neither universal nor evenly distributed, and understanding of research design remains incomplete. A caution: this source is about science and technology perceptions broadly, not about peer review narrowly, and it should not be used to claim that the public understands peer-reviewed scholarship in detail .

Miller, J. D. (2004). *Public understanding of, and attitudes toward, scientific research: What we know and what we need to know*. Public Understanding of Science, 13(3), 273–294. <https://doi.org/10.1177/0963662504044908>

- **Annotation:** Miller is a foundational review for explaining the difference between favorable attitudes toward research and deep understanding of research. The article uses long-term national survey evidence and argues that public support for science can coexist with limited scientific literacy. That is especially helpful if you want to resist both simplistic cynicism (“the public distrusts science”) and simplistic optimism (“people trust science, so they understand it”). A limitation is its age: it does not speak to platformed misinformation, contemporary social media, or generative AI. It is also focused on “scientific research” generally, not on social-science peer review in particular. What you should not overstate is causality; Miller shows coexistence of support and limited literacy, not that one straightforwardly produces the other .

Sturgis, P., & Allum, N. (2004). *Science in society: Re-evaluating the deficit model of public attitudes*. Public Understanding of Science, 13(1), 55–74. <https://doi.org/10.1177/0963662504042690>

- **Annotation:** This article is especially useful for textbook writing because it lets you complicate a common teaching shortcut. Students often hear that “people reject science because they don’t know enough,” but this article argues that although knowledge matters, attitudes toward science are also shaped by contextual understanding and broader social settings. Methodologically, it uses survey data and quantitative modeling, making it stronger than purely anecdotal discussions of public skepticism. Its main caution is scope: it is about public attitudes toward science, not specifically about peer review or U.S. social-science publishing. It also cannot be stretched into the claim that more knowledge automatically solves polarization or mistrust .

Gauchat, G. (2011). *The cultural authority of science: Public trust and acceptance of organized science*. Public Understanding of Science, 20(6), 751–770. <https://doi.org/10.1177/0963662510365246>

- **Annotation:** Gauchat is one of the most conceptually helpful public-opinion sources in this set because it moves beyond a simple trust/distrust frame. It shows that public acceptance of science depends partly on how people imagine science itself—its methods, institutional setting, and cultural role. That makes it especially relevant to textbook writing about peer-reviewed scholarship, because lay judgments are often mediated by images of science rather than by detailed reading of studies. The main limitation is that the article addresses “organized science” broadly and predates the current platform environment. It should not be used to imply that attitudes toward all fields, including communication research, are identical .

Burnham, J. C. (1990). *The evolution of editorial peer review*. JAMA, 263(10), 1323–1329. <https://doi.org/10.1001/jama.1990.03440100023003>

- **Annotation:** Burnham remains one of the classic concise histories of editorial peer review. Its core value for your chapter is that it helps dispel the myth that peer review has always existed in its current form. Burnham shows that pre-publication editorial review developed unevenly and historically, which is exactly the kind of contextual knowledge students need when they hear “peer-reviewed” treated as a timeless guarantee. The limitation is scope: the article is older, historically broad, and weighted toward medicine/science rather than the social sciences specifically. It should not be overstated into a claim that one unified model of peer review emerged at once across all disciplines .

Baldwin, M. (2018). *Scientific autonomy, public accountability, and the rise of “peer review” in the Cold War United States*. Isis, 109(3), 538–558. <https://doi.org/10.1086/700070>

- **Annotation:** Baldwin is arguably the strongest history source for your purposes because it directly answers the question of when modern peer review became central and why. The article’s major contribution is to connect peer review’s rise to public accountability, funding institutions, and Cold War U.S. science policy rather than portraying it as a simple extension of old scholarly customs. That historical framing is especially good for a textbook: it lets students see peer review as part of professionalization and legitimacy-building. The limitation is that it is focused on the United States and on science broadly, not on communication journals specifically .

Smith, R. (2006). *Peer review: A flawed process at the heart of science and journals*. Journal of the Royal Society of Medicine, 99(4), 178–182.

- **Annotation:** This is one of the most widely cited concise critiques of peer review. Its continuing usefulness is rhetorical and pedagogical: it names core problems—bias, inconsistency, slowness, weak evidence of effectiveness, vulnerability to abuse—without claiming that peer review should simply disappear. Because Smith writes from within journal culture, the essay is often a good classroom counterweight to naïve views of peer review as a guarantee. The limitation is genre. This is not a new systematic review; it is a critique essay, and parts of it are intentionally sharp. It should not be overstated into “peer review is useless” .

Tennant, J. P., & Ross-Hellauer, T. (2020). *The limitations to our understanding of peer review*. Research Integrity and Peer Review, 5, Article 6. <https://doi.org/10.1186/s41073-020-00092-1>

- **Annotation:** This is a particularly useful source because it does not merely list peer review’s problems; it argues that the evidence base about peer review is itself limited. That is a powerful meta-level lesson for your chapter: even academic quality-control systems need critical scrutiny. The article is strongest when you want to teach humility and methodological caution. A limitation is that it is about the literature on peer review overall, not specifically about social-science journals or U.S. communication publishing. It should not be overstated into a claim that we know nothing about peer review; the point is that evidence is uneven and incomplete .

Larivière, V., Haustein, S., & Mongeon, P. (2015). *The oligopoly of academic publishers in the digital era*. PLoS ONE, 10(6), e0127502. <https://doi.org/10.1371/journal.pone.0127502>

- **Annotation:** This is the strongest scholarly source I verified for discussing concentration and corporate power in scholarly publishing. Its central claim is not about peer review quality per se, but about market structure: a small number of firms increasingly control a large share of scholarly output. That matters for your chapter because access, pricing, prestige, and dissemination are shaped by the publishing ecosystem, not only by authors and reviewers. The limitation is that the article focuses on market concentration rather than directly on profit margins, reviewer labor, or communication journals specifically. It should not be overstated into a claim that all publishing by large firms is substantively poor; its strongest point is structural concentration .

RELX PLC. (2026). *2025 annual report*. <https://www.relx.com/~media/Files/R/RELX-Group/documents/reports/annual-reports/relx-2025-annual-report.pdf>

- **Annotation:** This is a useful primary source for the economics of major academic publishing, because RELX is Elsevier’s parent company and publishes official financial statements. Its strongest use is evidentiary: if you want to discuss the profitability of large commercial STM publishing infrastructure, this is a direct, citable corporate document rather than a secondary news story. The main caution is scope. RELX is a diversified information/analytics company, so its company-wide margin is not identical to the profitability of every journal or every publishing subunit. It should not be used to claim that all profits come specifically from peer review or from social-science journals .

Informa PLC. (2026). *Financial summary*. <https://www.informa.com/investors/financial-summary/>

- **Annotation:** This source is especially useful because it gives a more publishing-specific financial snapshot than a broad parent-company total. Taylor & Francis is a major social-science and communication-journal publisher, so this source is directly relevant to the sector you are writing about. Its value is not argumentative but documentary: it lets you cite a current official margin figure for a major journal publisher. The limitation is that corporate reporting categories are managerial constructs, not neutral research categories. A margin figure does not, by itself, prove exploitation or invalidate the scholarly value of the journals .

2. Social Media, "Research Says" Myths, and Misinterpreted Science

Connecting to the brainstorming notes on how oversimplified claims, wellness culture, and the "common-sense" flattening of research cause real-world harm.

Fahnestock, J. (1998). *Accommodating science: The rhetorical life of scientific facts*. Written Communication, 15(3), 330–350.

- **Annotation:** This classic rhetorical analysis demonstrates what happens when scientific reports travel from expert to lay publications. Fahnestock explains how the genre shifts from forensic (establishing the validity of observations) to epideictic (celebrating discoveries for a lay audience). In this accommodation process, hedges, qualifications, and limitations vanish, creating an illusion of exaggerated certainty—such as translating cautious data on testing differences into the absolute societal myth of a "math gene." This perfectly supports the "common sense vs. understanding" framework by showing how original research nuance is frequently stripped away before it ever reaches the public .

Yavchitz, A., et al. (2012). *Misrepresentation of randomized controlled trials in press releases and news coverage: A cohort study*. PLoS Medicine, 9(9), e1001308. <https://doi.org/10.1371/journal.pmed.1001308>

- **Annotation:** This is one of the most useful sources for your “research says” chapter argument, even though it focuses on press releases and news rather than TikTok or Instagram. Its value is causal structure: it shows that distortion frequently enters the communication chain between the original article, the press release, and the public-facing story. The paper’s framing of spin is especially helpful for teaching because it names a recognizable mechanism by which nuance gets converted into apparent certainty. A limitation is topical scope: it addresses randomized trials and health reporting, not interpersonal communication research. You should not overstate it into “media always distort research”; rather, it shows that misrepresentation is a documented and consequential risk in the translation pipeline .

Suarez-Lledo, V., & Alvarez-Galvez, J. (2021). *Prevalence of health misinformation on social media: Systematic review*. Journal of Medical Internet Research, 23(1), e17187. <https://doi.org/10.2196/17187>

- **Annotation:** This is a strong umbrella source for arguing that misinformation about research-linked topics is not marginal on social media. Because it is a systematic review, it is better for textbook synthesis than for making a single-topic claim. Its contribution is breadth: it documents that misinformation has been studied across many platforms and content areas, and that the field itself is methodologically diverse. A limitation is that it centers health misinformation, not interpersonal communication specifically, and it does not isolate the rhetorical formula “research says.” It also should not be overstated into a claim that most social-media health information is false; the review maps prevalence across studies, not a universal platform-wide percentage .

Mehrabian, A., & Wiener, M. (1967). *Decoding of inconsistent communications*. Journal of Personality and Social Psychology, 6(1), 109–114.

- **Annotation:** The original, highly specific lab study that was overgeneralized into the widespread communication myth that “93% of communication is nonverbal,” leading people to falsely use eye movements as absolute lie detectors. This serves as a primary example of how failing to read original research results in a widespread, brittle myth .

Fei, Y. Y., et al. (2022). *Maladaptive or misunderstood? Dopamine fasting as a potential intervention for behavioral addiction*. Lifestyle Medicine, 3, e54.

- **Annotation:** Wellness culture often uses neurotransmitter language to create certainty where the research requires complexity. This source addresses the "dopamine detox" trend, showing how wellness culture misreads neuroscience by turning behavioral interventions into false claims about literally lowering or resetting brain chemistry. It provides a real-world example of students applying rigid “rules” based on misinterpretations .

Gilmore, D., et al. (2024). *Building community and identity online: A content analysis of highly viewed #Autism TikTok videos*. Autism in Adulthood, 6(1), 95–105. <https://doi.org/10.1089/aut.2023.0019>

- **Annotation:** This source is especially valuable because it gets close to the kind of interpersonal and identity-adjacent content your students are likely to encounter. Instead of generic misinformation talk, it shows how lived-experience content, identity formation, and educational claims can blur together on a platform. That blur matters pedagogically: students may mistake visibility, authenticity, or relatability for evidence quality. The main caution is interpretive fairness. The study does not show that first-person neurodivergent content is useless or deceptive; rather, it shows that highly viewed videos are often doing community/identity work rather than providing comprehensive or clinically aligned education .

Lupton, D., & Southerton, C. (2024). *#ActuallyAutistic: Competing cultures of expertise and knowledge in relation to autism and ADHD self-diagnosis on TikTok*. Historical Social Research, 49(3), 188–212. <https://doi.org/10.12759/hsr.49.2024.29>

- **Annotation:** This article is especially useful if you want your chapter to avoid a simplistic “experts good / TikTok bad” framing. Its contribution is to show that social-media claims are often struggles over who gets to define knowledge: clinicians, researchers, creators, affected communities, or algorithms. That perspective is highly relevant to interpersonal communication because many viral claims about attachment, neurodivergence, conflict, and “toxic” behavior operate as identity-making discourse as much as as research summaries. The limitation is that the article is interpretive, not a prevalence study or accuracy audit. It should not be overstated into a claim that most self-diagnosis content is false; its stronger contribution is conceptual, showing competing cultures of expertise .

3. AI, Automation Bias, and the Future of Work

Connecting to the brainstorming notes on how AI gives fast answers but forces us to evaluate them, and why durable skills matter in an automated workforce.

Logg, J. M., Minson, J. A., & Moore, D. A. (2019). *Algorithm appreciation: People prefer algorithmic to human judgment*. Organizational Behavior and Human Decision Processes, 151, 90–103. <https://doi.org/10.1016/j.obhdp.2018.12.005>

- **Annotation:** This is a core source for explaining why “ChatGPT says…” can feel rhetorically powerful even when it should not settle an argument. The article’s central contribution is the concept of algorithm appreciation, which complicates the older assumption that people instinctively distrust algorithms. For your chapter, that means students may overvalue AI-generated answers simply because they appear computational, efficient, or data-driven. A limitation is contextual scope: this source studies judgment and advice in experimental settings, not classroom research habits or scholarly reading directly. It should not be overstated into “people always trust algorithms more than humans”; the evidence is conditional and task-dependent .

Skitka, L. J., Mosier, K. L., & Burdick, M. (1999). *Does automation bias decision-making?* International Journal of Human-Computer Studies, 51(5), 991–1006. <https://doi.org/10.1006/ijhc.1999.0252>

- **Annotation:** This is an older source, but it remains foundational because it defines the mechanism of automation bias that later AI work continues to document. The relevance to your chapter is straightforward: students may defer to AI-generated answers not because they have checked the evidence, but because automation itself cues trust and reduces scrutiny. The study predates large language models, so it should not be used as if it tested ChatGPT-like systems directly. Its strongest use is theoretical: it explains why overreliance on automated outputs is a predictable human tendency rather than just an individual student failure .

Sundar, S. S., & Kim, J. (2019). *Machine heuristic: When we trust computers more than humans with our personal information*. Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (pp. 1–9). <https://doi.org/10.1145/3290605.3300768>

- **Annotation:** This source sharpens your AI argument by giving you a communication concept that maps well onto everyday rhetoric. People may not just “like” AI; they may use a heuristic in which machine-produced output feels objective, neutral, or less biased. That is highly relevant in classroom and interpersonal settings where someone invokes AI output as if it were a neutral third-party judge. The limitation is that the study’s immediate context concerns personal-information disclosure and trust, not scholarly interpretation. It should therefore be used as a transferable mechanism, not as direct evidence about students reading journal articles .

Long, D., & Magerko, B. (2020). *What is AI literacy? Competencies and design considerations*. Proceedings of the CHI Conference on Human Factors in Computing Systems. doi:10.1145/3313831.3376727

- **Annotation:** Long and Magerko define AI literacy as the competencies people need to understand, use, and critically evaluate artificial intelligence. This source is foundational because it treats AI literacy as a civic and educational issue, not only a technical one. Students do not need to become computer scientists to need AI literacy; they need to understand what AI systems can do, what they cannot do, how they shape information, and how to evaluate their outputs. For your bibliography, this article helps connect information literacy and scientific literacy to AI: students need to question AI-generated claims, recognize limitations, and understand that fluent language is not the same thing as evidence .

Ng, D. T. K., et al. (2021). *Conceptualizing AI literacy: An exploratory review*. Computers and Education: Artificial Intelligence, 2, 100041. doi:10.1016/j.caeai.2021.100041

- **Annotation:** Ng and colleagues review peer-reviewed AI literacy literature and identify key dimensions of AI literacy, including knowing and understanding AI, using and applying AI, evaluating and creating with AI, and understanding ethical issues. This source is useful because it gives a structured vocabulary for talking about AI literacy in education. It also helps show that information literacy and

scientific literacy are not replaced by AI literacy; they become part of it. Students need to understand how AI systems generate outputs, how to evaluate those outputs, and how to use them ethically. For research reading, this means students should use AI as a tool for support, not as a substitute for interpreting evidence

Lo, L. S. (2023). *The CLEAR path: A framework for enhancing information literacy through prompt engineering*. The Journal of Academic Librarianship, 49(4), 102720. doi:10.1016/j.jcalib.2023.102720

- **Annotation:** Lo connects information literacy directly to prompt engineering and generative AI. The article proposes the CLEAR framework—Concise, Logical, Explicit, Adaptive, and Reflective—as a way to help users create and evaluate AI-generated content. This source is helpful because it does not treat prompting as a purely technical trick. Instead, it frames prompt engineering as an information literacy practice that requires clarity, reflection, evaluation, and revision. For your project, Lo supports the argument that AI-era students need to know how to ask better questions, check outputs against credible sources, and reflect on whether an answer is useful, accurate, biased, incomplete, or unsupported

Chu-Ke, C., & Dong, Y. (2024). *Misinformation and literacies in the era of generative artificial intelligence: A brief overview and a call for future research*. Emerging Media, 2(2). doi:10.1177/27523543241240285

- **Annotation:** Chu-Ke and Dong examine misinformation in the age of generative AI and argue for evolving literacy education. Their article is important because it connects AI-generated misinformation to broader literacy needs: users must understand, evaluate, interpret, and respond to AI-generated content. This source is especially useful for tying your bibliography to public communication. Generative AI can produce persuasive text, images, explanations, and claims at scale, which makes “just checking a source” more complicated. Students need information literacy to assess credibility, scientific literacy to evaluate evidence-based claims, and AI literacy to understand how generated content can imitate authority without actually being grounded in reliable research

López-Borrull, A., & Lopezosa, C. (2025). *Mapping the impact of generative AI on disinformation*. Publications, 13(3), 33.

- **Annotation:** López-Borrull and Lopezosa provide a scoping review of peer-reviewed research on generative AI and disinformation. Their review identifies themes including political disinformation, scientific disinformation, fact-checking, journalism, media literacy, education, and deepfakes. This source is valuable because it shows that AI-related disinformation is not limited to fake news headlines. It can affect scientific communication, public trust, research interpretation, and the circulation of fabricated or distorted evidence. For your bibliography, this source supports the argument that students need to read research carefully because AI-generated content can make false or weak claims look polished, cited, and authoritative. Scientific and information literacy become safeguards against being persuaded by style instead of evidence

Fulsher, A., Pagkratidou, M., & Kendeou, P. (2026). *GenAI and misinformation in education: A systematic scoping review of opportunities and challenges*. AI & Society, 41, 1373–1385. doi:10.1007/s00146-025-02536-y

- **Annotation:** Fulsher, Pagkratidou, and Kendeou review how generative AI creates both risks and opportunities for misinformation in education. They explain that misinformation can operate at individual, community, and system levels, and they identify approaches such as misconception detection, prebunking, and misinformation correction. This source is especially strong for your AI-age argument because it positions education as a major defense against misinformation. Students need critical thinking and information literacy because generative AI can amplify misinformation while also being used to address it. The key is not whether students use AI, but whether they have the literacy skills to evaluate AI-mediated claims against evidence, expertise, and research

World Economic Forum. (2025). *The future of jobs report 2025*. https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf

- **Annotation:** This is not a research-literacy source in the narrow sense, but it is useful for your chapter’s forward-looking stakes. It gives you a credible institutional basis for saying that AI does not eliminate the need for human judgment; instead, it raises the value of analytical and technological-literacy skills. The limitation is obvious: it is an employer/skills forecast, not a peer-reviewed education study on reading research. It should not be overstated into a claim that the labor market directly proves your pedagogy. Its proper use is as contextual support showing that the skills your chapter emphasizes are widely treated as durable and future-relevant

4. The (Un)Common-Sense Literacies

Connecting to the shift from simply "knowing" facts to practicing active sensemaking through distinct, interconnected literacies.

Association of College and Research Libraries. (n.d.). *Framework for information literacy for higher education*. <https://www.ala.org/acrl/standards/ilframework>

- **Annotation:** This is the single best source in this set for your broadest pedagogical vocabulary. It defines information literacy not as a checklist of library skills but as a conceptual orientation toward authority, inquiry, creation, and scholarly conversation. That is very close to your distinction between “knowing” and “understanding.” Its strongest contribution for a textbook chapter is the idea that research is a process and that authority is contextual rather than absolute. A limitation is that it is not a peer-reviewed empirical study and it is not specific to science or statistics. It should not be overstated into a claim that information literacy alone covers all issues of scientific method or quantitative interpretation

Sample, A. (2020). *Historical development of definitions of information literacy: A literature review of selected resources*. The Journal of Academic Librarianship, 46(2), 102116. doi:10.1016/j.jcalib.2020.102116

- **Annotation:** Sample’s literature review is useful because it shows that information literacy has never had one simple definition. Instead, definitions have shifted over time from library-search skills toward broader ways of thinking, learning, evaluating, and participating in information environments. This source would work well early in a bibliography because it gives historical context for why information literacy is more than “knowing how to Google.” For your project, Sample helps establish that students need to understand information as something created, circulated, interpreted, and evaluated—not merely retrieved. That matters in research reading because students must judge not only whether a source exists, but how it was produced and what kind of authority it has

Lloyd, A. (2006). *Information literacy landscapes: An emerging picture*. Journal of Documentation, 62(5), 570–583. doi:10.1108/00220410610688723

- **Annotation:** Lloyd defines information literacy as a situated practice rather than a neutral checklist of skills. Her article argues that information-literate people are “engaged, enabled, enriched, and embodied” in relation to information practices, and she describes information literacy as a way of knowing and a meta-competency. This is valuable because it moves information literacy away from a simple “find, evaluate, cite” model and toward a more realistic understanding of how people use information in communities, workplaces, classrooms, and everyday life. For research reading, Lloyd helps explain why students need to understand how knowledge works within a field: research articles are not just containers of facts; they are part of a disciplinary conversation

Mackey, T. P., & Jacobson, T. E. (2011). *Reframing information literacy as a metaliteracy*. College & Research Libraries, 72(1), 62–78. doi:10.5860/crl-76r1

- **Annotation:** Mackey and Jacobson argue that information literacy needs to be reframed as “metaliteracy” because digital environments require people not only to consume information, but also to produce, remix, share, and evaluate it. This is an especially important bridge to the AI age. Generative AI tools make students into information producers even when they are only prompting, summarizing, or revising. This source helps support the argument that students must be reflective participants in information systems, not passive receivers of answers. For your bibliography, this article connects information literacy to digital participation, authorship, ethical use, and critical evaluation—the same skills students need when working with AI-generated content

Laugksch, R. C. (2000). *Scientific literacy: A conceptual overview*. Science Education, 84(1), 71–94. doi:10.1002/(SICI)1098-237X(200001)84:1<71::AID-SCE6>3.0.CO;2-C

- **Annotation:** Laugksch’s article is a foundational overview of scientific literacy as a complex and contested concept. Rather than treating scientific literacy as memorizing scientific facts, Laugksch reviews how the term has been used across education, public understanding of science, policy, and citizenship. This source is useful because it explains why scientific literacy can mean different things depending on the audience: scientists, educators, policymakers, and the general public may emphasize different outcomes. For your project, this helps frame scientific literacy as a civic and interpretive skill. Students need to read research not because every student will become a scientist, but because scientific claims shape public health, technology, climate, relationships, communication, and everyday decision-making

Norris, S. P., & Phillips, L. M. (2003). *How literacy in its fundamental sense is central to scientific literacy*. Science Education, 87(2), 224–240. doi:10.1002/sci.10066

- **Annotation:** Norris and Phillips are essential for tying scientific literacy directly to reading. Their central argument is that scientific literacy cannot be separated from literacy in the ordinary sense: reading, writing, interpreting, and evaluating texts. Many discussions of scientific literacy emphasize understanding scientific concepts, but Norris and Phillips show that science is communicated through texts, graphs, arguments, methods, and evidence. This source is especially strong for your claim that students need to read research rather than rely only on summaries. If students cannot interpret scientific writing, methods, and evidence, then they are dependent on someone else’s explanation of science. In the AI age, that “someone else” may be a model that sounds confident but cannot guarantee accuracy

Holbrook, J., & Rannikmäe, M. (2009). *The meaning of scientific literacy*. International Journal of Environmental & Science Education, 4(3), 275–288.

- **Annotation:** Holbrook and Rannikmäe emphasize that scientific literacy should be understood broadly, not as simple science-content knowledge. Their work is useful because it connects scientific literacy to social contexts, decision-making, and the ability to use science meaningfully in life. This article helps support a student-facing argument that scientific literacy is not reserved for STEM majors. Students need scientific literacy when they interpret health advice, environmental claims, technology claims, risk communication, public policy, and media reports about research. For your project, this article helps explain why reading research matters: scientific literacy includes knowing how scientific knowledge is built, communicated, debated, and applied outside the lab .

Klucsevsek, K. M. (2017). *The intersection of information and science literacy*. Communications in Information Literacy, 11(2), 354–365. doi:10.15760/comminfolit.2017.11.2.7

- **Annotation:** Klucsevsek is one of the most useful sources for your specific topic because it directly connects information literacy and scientific literacy. She argues that stronger scientific literacy requires discipline-specific information literacy, especially the ability to find, evaluate, access, and understand scientific research. The article also emphasizes that reading primary scientific research is not just an advanced academic exercise; it is part of understanding how science works. This source is especially strong for explaining why students and the public need access to research and the skills to interpret it. For your AI-age argument, Klucsevsek helps show that evaluating scientific claims requires more than trusting a summary—it requires understanding the research process behind the claim .

Keser Aschenberger, F., & Pfeffer, T. (2021). *Concept of research literacy in academic continuing education: A systematic review*. European Journal of University Lifelong Learning, 5(1), 43–58.

- **Annotation:** This is probably the most directly on-point source for the term research literacy itself. Its most useful finding is not a neat, closed definition, but the opposite: the literature does not yet offer a single comprehensive research-literacy concept. That helps you write honestly and defensibly. You can argue that research literacy is emerging and that it commonly bundles together skills such as understanding research purposes, methods, reading practices, and academic discourse. The limitation is that the review is centered on continuing and higher education rather than everyday public information environments. It should not be overstated into a consensus definition of research literacy .

Carlson, J., et al. (2011). *Determining data information literacy needs: A study of students and research faculty*. portal: Libraries and the Academy, 11(2), 629–657.

- **Annotation:** This is a useful bridge source because it locates data literacy inside research practice rather than treating it only as workplace dashboard fluency. Its value for your chapter is that it makes visible the competencies surrounding data itself: collection, management, curation, interpretation, and responsible use. Methodologically, it is a needs-assessment study rather than a philosophical definition piece, so it grounds data literacy in actual higher-education and research workflows. A limitation is that the article comes from library/data-management contexts and is not focused on students as everyday consumers of social-media claims. It should not be overstated into a complete account of statistical reasoning or scientific argumentation .

Ng, W. (2012). *Can we teach digital natives digital literacy?* Computers & Education, 59(3), 1065–1078.

- **Annotation:** Ng is useful because it prevents “digital literacy” from collapsing into “can use apps.” The article explicitly frames digital literacy as broader than mere technical use, encompassing technical, cognitive, and social-emotional dimensions of working with digital technologies. That matters for your chapter because students often assume that fluency with platforms or chatbots equals evaluative competence. A limitation is that the article is older than TikTok and generative AI, and it is situated in higher education technology adoption rather than research credibility specifically. It should not be used as if digital literacy automatically includes scientific reasoning or research-method evaluation .

National Center for Education Statistics. (2024). *PIAAC highlights of U.S. national results*. U.S. Department of Education, Institute of Education Sciences.

- **Annotation:** National assessments show concerns in reading, science performance, adult literacy, numeracy, and adaptive problem solving. This supports the framing that students need explicit support with the literacies that make careful thinking possible: reading complex texts, evaluating sources, interpreting data, understanding scientific claims, and revising interpretations when evidence changes .

5. Reading Research, Annotating, and Sense-Making

Connecting to the practical tools students use to process and connect information rather than passively copying or memorizing facts.

Carey, M. A., Steiner, K. L., & Petri, W. A., Jr. (2020). *Ten simple rules for reading a scientific paper*. PLOS Computational Biology, 16(7), e1008032.

- **Annotation:** A practical guide to help undergraduates tackle the complexity of primary research articles .

Porter, J. A., et al. (2010). *Integration of information and scientific literacy: Promoting literacy in undergraduates*. CBE—Life Sciences Education, 9(4), 536–542. doi:10.1187/cbe.10-01-0006

- **Annotation:** Porter and colleagues describe an undergraduate biology project that integrated information literacy and scientific literacy by having students access, retrieve, analyze, and evaluate primary scientific literature. This article is valuable because it shows that the connection between information literacy and scientific literacy can be taught explicitly. Students improved in their understanding of the relevance of both skill sets. For your bibliography, this source supports the practical claim that reading research helps students develop literacy skills that are transferable beyond one assignment. It also helps argue that research reading should not be saved only for advanced students; students benefit from guided exposure to research early in their academic development .

Krontiris-Litowitz, J. (2013). *Using primary literature to teach science literacy to introductory biology students*. Journal of Microbiology & Biology Education, 14(1), 66–77. doi:10.1128/jmbe.v14i1.538

- **Annotation:** Krontiris-Litowitz focuses on a common teaching problem: undergraduate students often struggle to read scientific literature. The article describes a strategy for using peer-reviewed articles and structured assignments to help introductory biology students build scientific literacy. The study found gains, while also noting that higher-order interpretive skills remained challenging. This source is useful because it supports a balanced argument: reading research is essential, but students need scaffolding to do it well. For your AI-age focus, that matters because students may be tempted to outsource difficult reading to AI summaries. This article helps argue that the difficulty of reading research is not a reason to avoid it; it is the reason students need practice .

Perry, H. B. (2017). *Information literacy in the sciences: Faculty perception of undergraduate student skill*. College & Research Libraries, 78(7), 964–977.

- **Annotation:** Perry’s study examines science faculty perceptions of undergraduate students’ information literacy skills. Faculty emphasized that students need to find relevant, high-quality, peer-reviewed primary research articles and understand gaps in the literature. The article is useful because it provides evidence that faculty view research reading and source evaluation as central undergraduate skills, not optional library add-ons. For your project, Perry helps connect information literacy to disciplinary expectations: students are not just being asked to “use sources”; they are being asked to enter a research conversation. In the AI age, this matters because tools may generate plausible research language without helping students understand whether the evidence is current, peer-reviewed, methodologically sound, or relevant .

Goudsouzian, L. K., & Hsu, J. L. (2023). *Reading primary scientific literature: Approaches for teaching students in the undergraduate STEM classroom*. CBE—Life Sciences Education, 22(3), es3. doi:10.1187/cbe.22-10-0211

- **Annotation:** Goudsouzian and Hsu review approaches for teaching undergraduate students to read primary scientific literature. Their work is especially useful because it identifies reading primary scientific literature as a goal across STEM classrooms and connects it to cognitive and affective benefits. The article helps support the argument that research reading builds more than content knowledge: it can support critical thinking, data interpretation, and scientific/information literacy. For your AI-age argument, this source helps show why students should not only ask AI for simplified summaries. They need practice with the original research genres—abstracts, methods, results, figures, and limitations—because those are where evidence is made visible .

Kiewra, K. A. (1989). *A review of note-taking: The encoding-storage paradigm and beyond*. Educational Psychology Review, 1, 147–172. doi: 10.1007/BF01326640.

- **Annotation:** Kiewra’s review is a useful foundation because it explains two major reasons notes may help learning: the encoding function and the storage function. Encoding refers to the learning benefit that may occur while students are taking notes, especially when they process, organize, or transform information. Storage refers to the later benefit of having notes available for review. This source is helpful because it avoids the simplistic idea that “writing things down” automatically creates understanding. Instead, it frames note-taking as useful when it supports generative processing and later study. The limitation is that this is an older review and not specifically about handwritten versus typed notes, but it gives the conceptual language needed to explain why notes can help .

Mueller, P. A., & Oppenheimer, D. M. (2014). *The pen is mightier than the keyboard: Advantages of longhand over laptop note taking*. Psychological Science, 25(6), 1159–1168. doi: 10.1177/0956797614524581.

- **Annotation:** This is the widely cited study behind many claims that handwritten notes can support deeper understanding. Across three studies, Mueller and Oppenheimer found that laptop note-takers tended to transcribe more verbatim, while longhand note-takers were pushed to process and reframe information in their own words. The key finding was stronger performance for longhand note-takers on

conceptual questions, not simply that pens are magically better than keyboards. This source is useful for arguing that handwritten notes may help because they slow students down enough to select, condense, and paraphrase. Use it carefully: this study should not be presented as final proof that handwriting always beats typing, especially because later replication work has produced more mixed results .

Morehead, K., Dunlosky, J., & Rawson, K. A. (2019). *How much mightier is the pen than the keyboard for note-taking? A replication and extension of Mueller and Oppenheimer (2014).* Educational Psychology Review, 31(3), 753–780. doi: 10.1007/s10648-019-09468-2.

- **Annotation:** This replication and extension is essential because it prevents overclaiming. Morehead and colleagues directly replicated and extended Mueller and Oppenheimer’s work, including eWriter and no-note groups. They found some trends favoring longhand notes, but performance did not consistently differ between groups, and a combined analysis of direct replications found small, nonsignificant effects favoring longhand. Their conclusion was that it was premature to claim one method is clearly superior for all note-taking functions. This is a strong source to include because it shows intellectual honesty: handwritten notes may help, but the evidence is not uniform, and the benefit likely depends on task, review, note quality, and learner behavior .

Flanigan, A. E., et al. (2024). *Typed versus handwritten lecture notes and college student achievement: A meta-analysis.* Educational Psychology Review, 36(3), Article 78. doi: 10.1007/s10648-024-09914-w.

- **Annotation:** This is one of the strongest recent sources for a cautious pro-handwriting claim. The meta-analysis synthesized 24 studies across 21 articles and found that taking and reviewing handwritten notes was associated with higher achievement among college students, with a small effect size, while typed notes produced a much larger note-volume advantage. In practical terms, this suggests handwritten notes may help achievement somewhat, but typed notes let students capture more material. The most careful interpretation is not “typing is bad,” but “handwriting may better support studying and memory for many college students when the goal is achievement, while typing may support quantity and accessibility.” This is currently one of the best sources for a balanced literature review because it includes many studies rather than relying on a single experiment .

Van der Weel, F. R. R., & Van der Meer, A. L. H. (2024). *Handwriting but not typewriting leads to widespread brain connectivity: A high-density EEG study with implications for the classroom.* Frontiers in Psychology, 14, Article 1219945. doi: 10.3389/fpsyg.2023.1219945.

- **Annotation:** This neuroscience study compared university students handwriting visually presented words with a digital pen versus typewriting them on a keyboard while researchers recorded high-density EEG activity. The authors reported more elaborate theta/alpha connectivity patterns during handwriting, especially in brain regions associated in prior literature with memory formation and encoding. This is useful for explaining a plausible brain-based mechanism: handwriting involves visual, motor, and proprioceptive information, which may create richer neural engagement than simple key pressing. Use this source carefully. The study measured brain connectivity during a controlled task; it did not directly test classroom comprehension, course grades, or long-term learning from reading or lecture content .

Pinet, S., & Longcamp, M. (2025). *Commentary: Handwriting but not typewriting leads to widespread brain connectivity: A high-density EEG study with implications for the classroom.* Frontiers in Psychology, 15, Article 1517235. doi: 10.3389/fpsyg.2024.1517235.

- **Annotation:** This reviewed commentary is important because it directly cautions against overstating the Van der Weel and Van der Meer EEG study. Pinet and Longcamp argue that the original EEG protocol did not include a learning task, used adult participants in a lab, and involved an artificial typing condition, including one-finger typing and no visual feedback. They also note that the study did not report behavioral learning measures, which makes it risky to infer classroom learning directly from connectivity differences. This source does not dismiss handwriting; instead, it helps you make a more defensible claim: neuroscience findings are suggestive, but brain activation/connectivity should not be treated as automatic proof that students understand content better .

Umejima, K., et al. (2021). *Paper notebooks vs. mobile devices: Brain activation differences during memory retrieval.* Frontiers in Behavioral Neuroscience, 15, Article 634158. doi: 10.3389/fnbeh.2021.634158.

- **Annotation:** Umejima and colleagues compared participants who recorded schedule information using a paper notebook, tablet, or smartphone, then tested recognition memory after a one-hour delay while using fMRI. The paper-notebook group completed the writing task faster and showed higher accuracy on easier questions; the authors also reported stronger activation in memory-related regions for the paper condition and interpreted this as evidence that physical paper may provide richer spatial and retrieval cues. This source is useful for explaining why physical layout, spatial memory, and tactile cues may matter. Use it narrowly: the task involved schedule memory, not complex academic reading, and it does not prove that paper is always better than tablets for all students or disciplines .

James, K. H., & Engelhardt, L. (2012). *The effects of handwriting experience on functional brain development in pre-literate children.* Trends in Neuroscience and Education, 1(1), 32–42. doi: 10.1016/j.tine.2012.08.001.

- **Annotation:** James and Engelhardt studied pre-literate children and found that handwriting experience recruited brain regions involved in letter processing in ways that typing or tracing did not. This source is valuable for the brain-science portion of the bibliography because it links handwriting to visual-motor integration and early reading-related brain networks. The careful use of this study is developmental: it supports the idea that physically producing letters can matter for early literacy and symbol recognition. It should not be used to claim that handwriting lecture notes automatically improves college-level conceptual understanding .

Lloyd, Z. T., et al. (2022). *Using the annotating strategy to improve students’ academic achievement in social studies.* Journal of Research in Innovative Teaching & Learning, 15(2), 218–231. doi: 10.1108/JRIT-09-2021-0065.

- **Annotation:** This mixed-method experimental study examined annotation as a reading comprehension strategy in an eighth-grade social studies class. The authors reported that annotation increased engagement, reading comprehension, and academic achievement, and they explain that annotation helped students identify key points, break down complex historical texts, and slow down while reading. This source is especially useful for arguing that annotation supports understanding when it makes students actively process complex disciplinary texts. The limitation is clear: the study was conducted with eighth-grade students in one middle school in South Georgia, so it should not be generalized to all ages, subjects, or annotation practices .

Yeh, H.-C., Hung, H.-T., & Chiang, Y.-H. (2017). *The use of online annotations in reading instruction and its impact on students’ reading progress and processes.* ReCALL, 29(1), 22–38. doi: 10.1017/S0958344016000021.

- **Annotation:** Yeh and colleagues studied 54 university English language learners using online annotations within a reciprocal teaching framework. Students annotated through predicting, clarifying, questioning, and summarizing, and the study found improved reading comprehension after the intervention. The most useful part of this study is its explanation of how annotation helps: students who made more progress reviewed earlier annotations and gave feedback to peers. This supports the claim that annotations are most powerful when they are not just marks in the margins, but prompts for metacognition, peer interaction, and repeated engagement with the text .

Johnson, T. E., Archibald, T. N., & Tenenbaum, G. (2010). *Individual and team annotation effects on students’ reading comprehension, critical thinking, and meta-cognitive skills.* Computers in Human Behavior, 26(6), 1496–1507. doi: 10.1016/j.chb.2010.05.014.

- **Annotation:** This study examined online social annotation in second-semester first-year English classes. It is useful because it shows that annotation by itself is not automatically effective. In Study 1, the authors found no significant difference between the social annotation approach and the control condition. In Study 2, students showed better reading comprehension and metacognitive skill, but not critical thinking, when annotation activities included small-team collaboration. This source supports a careful claim: annotation can help when it is embedded in structured, social, reflective learning activities, but it should not be treated as a guaranteed comprehension intervention .

Chen, C.-M., & Chen, F.-Y. (2014). *Enhancing digital reading performance with a collaborative reading annotation system.* Computers & Education, 77, 67–81. doi: 10.1016/j.compedu.2014.04.010.

- **Annotation:** Chen and Chen tested a collaborative reading annotation system with Grade 5 students in Taiwan. The experimental group used a digital system that combined reading annotation with interactive discussion scaffolds, while the comparison group used traditional paper-based reading annotation and face-to-face discussion. The experimental group outperformed the control group in direct comprehension, inferential comprehension, and use of reading strategies. This source is useful because it shows that digital annotation can also support understanding when it is structured and collaborative. Use carefully: the study involved young students, a particular digital tool, and a specific instructional design, so it is not evidence that all digital annotation tools work equally well .

Lee, S., et al. (2023). *Annotations serve as an on ramp for introductory biology students learning to read primary scientific literature.* Journal of Microbiology & Biology Education, 24(1). doi: 10.1128/jmbe.00214-22.

- **Annotation:** Lee and colleagues studied annotations as a support for first-year students learning to read primary scientific literature. The qualitative findings suggested that annotations helped students break scientific articles into manageable pieces, summarize text, identify key information, and distinguish article sections. However, the quantitative results did not show a significant influence of annotations on the

development of reading strategies. This is a valuable source precisely because the findings are mixed. It supports a modest claim: annotations may help novice readers make difficult scientific texts more approachable, especially with scaffolding, but annotations alone may not be enough to produce measurable strategy growth .

What students think the topic is

- The "Ivory Tower" Disconnect: Students often view academic research as an intimidating, overly technical product of the "ivory tower." Because of this, they assume it is only read by a few people and has absolutely no relevance to their daily lives.
- The Common-Sense Approach: When they do engage with research, they treat a single study as a machine that transfers absolute, universal truths rather than an opportunity for complex sense-making.
- Either/Or Thinking: They believe a study is either completely right or completely wrong, and they read simply to "know" or memorize facts rather than to understand nuances.
- The Generalizability Trap: They assume all research must be generalizable; if a study contradicts their beliefs, they will quickly dismiss it by claiming the sample size is too small.

How Research Shapes Their Behavior

- Unknowing Application: Even though they feel disconnected from academic studies, their behaviors are deeply shaped by research as it filters down to them through wellness influencers, social media algorithms, and AI tools.
- Rigid Behavioral Rules: They take oversimplified scientific claims and turn them into rigid rules for their lives, losing the flexibility needed for real-world interactions.
- "Hacking" and Self-Diagnosis: They attempt to "hack" their moods based on distorted science (e.g., eating beans to boost serotonin or doing tasks for dopamine), and they may even self-diagnose conditions based on an influencer vaguely claiming "research says".
- Misinterpreting Relationships: They apply overgeneralized myths directly to their relationships, such as closely watching a partner's eyes during an argument because they falsely believe nonverbal research gives them a foolproof way to catch a liar.

What this chapter offers as an alternative

The (Un)Common-Sense Perspective: Shifting from Knowing to Understanding

- Demystifying Peer Review: The chapter defines peer review as a historically constructed quality-control process rather than a timeless, magical stamp of absolute truth. By examining its known flaws—such as bias and inconsistency—students learn to respect the publication process while still evaluating studies with evidence-sensitive skepticism.
- Identifying Types of Research: Students learn to distinguish between original scientific reports (which carefully validate observations and acknowledge limitations) and accommodated or popularized science (which often celebrates discoveries while stripping away nuance and hedges). Understanding what different texts achieve helps students spot when a complex study has been converted into an oversimplified myth.
- Research is Sense-Making, Not Universal Truth: Instead of treating a single study as a machine that produces absolute facts, students learn that research findings are always partial, incomplete, and highly dependent on context.
- Flexibility Over Rigid Rules: By accepting that scientific claims are not simply 100% right or 100% wrong, students learn to replace rigid behavioral rules with curiosity, asking when and where a claim is actually true.
- Literacies as the Engines of Understanding: Having access to information does not automatically create understanding. Instead of treating literacies as textbook trivia, the chapter frames them as the active "street smarts" required to evaluate where claims come from, navigate uncertainty, and connect evidence to lived experiences.
- Reorienting Reading and Note-Taking Strategies: The (un)common-sense approach completely changes how students study. Instead of passively highlighting or transcribing verbatim facts just to "know" them, students use note-taking and annotation for active, generative processing—paraphrasing, questioning, and connecting ideas to build durable understanding.

Short version for the chapter planning document

What students should notice differently

Students should notice that research is not a magic fact machine. Research claims are produced in contexts, shaped by methods, limited by samples, interpreted through theories, circulated through media, and applied inside real relationships. They should notice when research is being treated as proof, a weapon, a rule, or a social media spell. They should also notice that misinformation often begins when partial, local findings are stripped of context and turned into universal advice. In the AI era, students should notice that easy access to information does not equal understanding.

What students should try differently

What students should try differently

Students should practice reading research as sense-making. They should move around articles strategically, ask what a study helps them understand, check source-to-claim fit, look for context and limits, synthesize across sources, and apply findings heuristically rather than algorithmically. They should use AI as a support tool without outsourcing judgment. Most importantly, they should leave with the habit of asking, “When is this true, for whom, under what conditions, and what does it help me understand about communicating?”

Below is a **chapter-level answer** you can use as drafting guidance. I’m treating the whole chapter as moving students from **research as knowing** to **research as understanding**, with misinformation, social media, AI, and literacies as the stakes.

1. What students should notice differently

The biggest noticing shift

Students should begin to notice that research is not just something they **find, cite, summarize, or use to prove a point**. Research is something they have to **interpret**. A study does not hand them a complete truth. It gives them partial, contextual evidence that has to be connected to methods, participants, limits, other studies, lived experience, and the communication situation they are trying to understand. The chapter notes frame this as the major shift from research as “truth transfer” to research as “sense-making with partial, contextual, and local evidence.”

Students should notice that research claims are communication events

A research claim does not just sit quietly in an article. It travels. It moves from research article to press release, headline, classroom, TikTok, AI summary, workplace training, argument, or relationship conflict. By the time a student hears “research says,” the claim may have been shortened, exaggerated, stripped of limits, or turned into a rule.

Students should notice the **pathway** of a claim:

- Original study
- Researcher interpretation
- Journal article
- Press release or summary
- News story or social media post
- Influencer explanation
- AI-generated summary
- Everyday advice
- Personal or relational judgment

That pathway matters because misinformation does not always begin with fake research. Sometimes it begins with real research that gets flattened.

Students should notice that “there is a source” is not the same as “the source supports this claim”

This is one of the most important chapter-level outcomes. Students should stop treating a citation like a magic truth stamp.

They should notice the difference between:

- a source existing
- a source being credible
- a source being relevant
- a source actually supporting the specific claim being made
- a source being strong enough to justify the way the claim is being applied

This matters especially online, where “sources?” and “research says” can become authority moves rather than invitations to understand the evidence. The notes identify Reddit “Sources?” culture and “research says” claims as strong examples of how research can become a proof object instead of something interpreted carefully.

Students should notice the appeal of certainty

The common-sense approach to research feels good because it offers certainty. It lets students think:

- I found the answer.
- I know what the study says.
- I can cite this.
- I can prove my point.
- I can win the argument.
- I can apply this rule.

The opening story captures this perfectly. Knowing asks whether the study is true, useful, memorable, or able to prove a point, while understanding asks what the study helps us see, where it applies, where it does not apply, and what other research complicates it.

Students should notice that certainty is not always a sign of understanding. Sometimes certainty is the moment understanding stops.

Students should notice the common-sense pattern of reading, evaluating, and applying research

The common-sense approach shapes three habits:

1. **Reading research linearly**
Students assume they must start at the beginning and move sentence by sentence to the end, as if truth must be received in order.
2. **Evaluating research monologically**
Students ask whether the study is right or wrong, true or false, useful or useless.
3. **Applying research algorithmically**
Students turn findings into rules. If the study says X, then they should do X, and Y should happen.

The chapter notes name this as the three-part bridge into the alternative approach: common-sense reading becomes linear, evaluation becomes true/false, and application becomes algorithmic. The (un)common-sense alternative makes reading interactive, evaluation dialectical, and application heuristic.

Students should notice when research is being turned into a rule

Students should become sensitive to moments when careful research becomes rigid advice.

Examples:

- Eye movement means someone is lying.
- Crossed arms mean resistance.
- Dopamine is the motivation chemical.
- Five positive interactions repair one negative interaction.
- Mirroring body language creates trust.
- This ADHD hack will work for you.
- AI summarized it, so now I understand it.

The pattern underneath these examples is the same: a partial finding becomes a universal rule.

Students should notice context instead of only conclusions

Students should learn to notice the parts of research that often disappear in everyday summaries:

- Who was studied?
- Who was not studied?
- What was the sample?
- What was the method?
- What was measured?
- What was not measured?
- What kind of claim did the authors actually make?
- What limits did the authors name?
- What other studies complicate this finding?
- What situation is the finding being applied to now?

This makes **context** visible. Research is not just a finding. It is a finding produced under conditions.

Students should notice local truth

Students should notice that a study can be true without being universally true.

This is a key tension for the chapter. The common-sense approach often treats research as complete, stable, and universal. The (un)common-sense approach treats findings as partial and local. A local truth is still a truth, but it is true under specific conditions, for a specific question, with specific participants, using specific methods.

This helps students avoid two weak responses:

- “This study proves it for everyone.”
- “This study has limits, so it is useless.”

The stronger response is:

- “What does this study help us understand, and under what conditions?”

Students should notice contradiction as useful

Students often experience contradiction as a problem. If one study says one thing and another study complicates it, the common-sense approach asks, “Which one is right?”

The (un)common-sense approach asks better questions:

- Why might both findings make sense?
- Were the studies asking different questions?
- Were the contexts different?
- Were the participants different?
- Were the methods different?
- What does the contradiction help us see?

Contradiction should not automatically shut down trust in research. It can deepen understanding.

Students should notice how misinformation becomes interpersonal

This chapter should make students see misinformation as more than a media problem. Misinformation about communication shapes how individuals interpret relationships, bodies, behavior, credibility, deception, mental health, intelligence, and everyday decision-making.

That means misinformation can affect:

- what we accuse someone of
- what we think counts as honesty
- how we interpret silence
- how we judge emotion
- how we diagnose ourselves or others
- how we decide whether someone is lying
- how we evaluate care, respect, confidence, or connection

The point is not only “bad information is bad.” The point is that bad information becomes part of how we communicate.

Students should notice AI as a literacy problem, not just a technology problem

AI intensifies the common-sense approach because it makes information feel easy to receive. It can summarize, organize, and produce fluent explanations. But fluent output is not the same thing as understanding.

The chapter notes frame AI carefully: AI can support access, especially for dyslexia and ADHD, but it cannot do understanding for the reader. Understanding still requires sense-making, uncertainty tolerance, credibility evaluation, and connection across contexts.

Students should notice this tension:

- AI can help us work with information.
- AI can also make us feel like we understand before we have actually evaluated anything.

Students should notice literacies as practical judgment

Research literacy, information literacy, scientific literacy, data literacy, digital literacy, and AI literacy should not feel like separate academic hoops. Students should notice them as overlapping forms of judgment.

They are ways of asking:

- Where did this claim come from?
- How was it produced?
- What evidence supports it?
- What are its limits?
- What context does it assume?
- Who benefits if I believe it?
- What happens if I apply it too broadly?

This makes literacies feel less like school skills and more like intellectual street smarts.

2. What students should try differently

The biggest practice shift

Students should leave the chapter practicing this move:

Do not ask research to give you a rule before you ask what the research helps you understand.

That should be the chapter’s practical center.

1. Read research recursively, not obediently

Students should stop assuming they must start at the first word and march straight to the end.

They can try this instead:

1. Read the title and abstract.
2. Skim the introduction for the research problem.

3. Jump to the discussion or conclusion to see what the authors think they found.
4. Go back to the methods to see how they studied it.
5. Look at the findings or results.
6. Return to the literature review to understand what conversation the study entered.
7. Re-read the abstract after they understand the study better.

This practice fits the (un)common-sense approach because reading becomes interaction with the article rather than passive reception of expert truth.

2. Use the “what does this help me understand?” question

Instead of beginning with “What did the study prove?” students should ask:

- What does this study help me understand?
- What does it help me notice?
- What problem was it trying to explain?
- What does it clarify?
- What does it complicate?
- What does it leave unresolved?

This keeps research from becoming a fact machine.

3. Use the “where is this true?” heuristic

Students need a simple way to practice local truth.

They can ask:

- When is this true?
- Where is this true?
- For whom is this true?
- Under what conditions is this true?
- What would make this less true?
- What would I need to know before applying this?

This is one of the most important heuristics for the chapter because it prevents both over-trust and cynical dismissal.

4. Treat findings as guides, not scripts

Students should leave with the difference between algorithms and heuristics.

Algorithmic use of research says:

- Do X and Y will happen.
- Follow the rule and get the result.
- Apply the finding the same way everywhere.

Heuristic use of research says:

- Let this finding guide what you notice.
- Let it sensitize you to a possibility.
- Let it help you ask better questions.
- Adapt it to the situation.

The notes describe heuristics as flexible guides that help us notice, anticipate, contextualize, and adapt, while algorithms generalize, predict, and standardize.

5. Convert rules into functions

This is a very practical student skill.

When students encounter advice, they should ask:

What is the function underneath the rule?

Example:

- Rule: Buy a craft cart for ADHD organization.
- Function: Create a visible container for materials connected to a hobby, task, or hyperfixation.
- Heuristic: Make important materials visible, contained, and easy to return to.
- Adaptation: A basket, shelf, bin, or phone reminder might serve the same function.

This lets students preserve what is useful without becoming trapped by the literal rule.

6. Practice source-to-claim fit

Students should not only ask whether a source exists. They should ask whether the source fits the claim.

They can use this check:

- What is the exact claim?

- What source is being used to support it?
- Does the source actually study that claim?
- Does the source support the claim strongly, weakly, indirectly, or not at all?
- Is the claim broader than the evidence?
- Has the claim been stretched from “may” to “does,” or from “some” to “all”?

This practice directly addresses “research says” culture.

7. Pause when research agrees with them

Students should learn that agreement can feel like evidence even when it is just familiarity.

When a study agrees with what they already believe, they should ask:

- Why do I like this finding?
- Am I accepting it more easily because it supports me?
- Would I trust the same method if the finding challenged me?
- What are the limitations?
- What other research might complicate it?

This is the practical response to confirmation bias.

8. Pause when research challenges them

Students should also notice when they become suddenly very “methodologically rigorous” only because a study challenges them.

They can ask:

- Am I critiquing the study fairly?
- Am I applying the same standards I would apply to a study I liked?
- Is the limitation fatal, or does it simply narrow the claim?
- What could this study still help me understand?

This keeps critique from becoming disguised defensiveness.

9. Summarize, then synthesize

Students should not stop at summary.

Summary asks:

- What did this article say?

Synthesis asks:

- How does this article connect to other research?
- How does it complicate what I thought?
- How does it connect to lived experience?
- What question does it help answer?
- What question does it raise?

The chapter notes name this as a major shift: common sense collects and summarizes; (un)common sense connects and synthesizes.

10. Use AI as support, not substitution

Students should be allowed to use AI in ways that support access and comprehension, but they should not outsource understanding.

A useful AI heuristic:

1. Use AI to clarify vocabulary, generate questions, or summarize a passage.
2. Return to the original source.
3. Check whether the AI summary matches the article.
4. Look for what the AI left out.
5. Make the final interpretation yourself.

The chapter’s stance should be balanced: AI can help students work with information, but it cannot evaluate, connect, contextualize, and understand for them.

11. Ask what got flattened

When students encounter misinformation or oversimplified advice, they should not only ask, “Is this fake?”

They should ask:

- What got simplified?
- What context disappeared?
- What limitation was removed?
- What population was generalized?
- What uncertainty became certainty?

- What careful claim became a rule?
- What is this post trying to make me feel or do?

This is especially useful for social media, wellness content, mental health content, and communication advice.

12. Check the interpersonal consequence before applying a claim

Because this is an interpersonal communication textbook, students should ask what happens when a research claim enters a relationship.

Before applying a claim to another person, they should ask:

- What might this claim make me notice?
- What might it make me over-notice?
- What might it make me accuse someone of?
- What might it make me ignore?
- Could this claim make me more curious, or only more certain?
- Could this claim harm the relationship if I am wrong?

This is where the chapter comes back to communicating intentionally.

13. Use research to ask better questions, not just make stronger claims

Students should leave with the habit of turning research into better questions.

Instead of:

Research says this is true.

Try:

This study helps me ask whether this pattern might be happening here.

Instead of:

This proves my point.

Try:

This gives me one way to understand the situation, but I need to know more.

Instead of:

The research says to do this.

Try:

The research suggests one possible guide, but I need to adapt it to this context.

14. Practice curiosity as a discipline

Curiosity should not be framed as a personality trait. It should be framed as a research literacy practice.

Students should practice asking:

- What else could be true?
- What am I missing?
- What would change my mind?
- What does this study not tell me?
- What would I need to understand before using this claim?
- How could this be both useful and incomplete?

The notes repeatedly frame curiosity as the alternative to certainty. Certainty closes interpretation too early; curiosity keeps sense-making open.

Here is a more concrete version of those steps, using the specific examples from your chapter materials so you can easily copy and paste them:

What Students Should Try to Do Differently (Concrete Examples)

- **Stop Using Research as a "Vending Machine":** Instead of grabbing a study just to win an argument (like claiming "science proves liars avoid eye contact"), they should read the research to discover that most deception studies actually show humans guess no better than random chance.
- **Ask Better Questions (Use Flexible Heuristics):** When a wellness influencer claims "eating beans boosts serotonin for happiness," students should stop and ask, "What did the original study actually measure?" instead of blindly accepting the "research says" rule.
- **Choose Curiosity Over Certainty:** If a study contradicts their worldview, instead of instantly dismissing it by claiming "the sample size is too small," they should pause and ask *under what specific conditions* the finding might actually be true.
- **Practice "Street Smarts" (Connect Evidence to Experience):** Just like using the memory of David Bowie's eye injury to question a vet's assumption about a dog's blindness, students should combine formal research claims with their own lived experiences to make better contextual judgments.
- **Critically Evaluate AI Summaries:** When an AI summarizes an article, students must fight "automation bias"—the urge to trust a claim just because a machine wrote it—and actively check if the AI stripped away the original researchers' doubts or limitations.

Definition of the approach

How does common sense as knowing define research in this topic?

The **common-sense-as-knowing** approach defines research as a way to **get expert facts**. In this view, a research article is treated like a container of findings: researchers discover something, write it down, students read it, and now students “know” it. The report transfers expert knowledge from researchers to readers. The student’s job is to receive the finding accurately, summarize it, remember it, decide whether it is true, and use it to answer or prove something.

This approach extends the common-sense model of communication, where communication is imagined as the transfer of information from one person to another. In Chapter 1, the common-sense approach treats communication as moving information between individuals, and good communication is often judged by clarity, efficiency, intention, and success. Applied to research, the study becomes the “message,” the researchers become the “senders,” and the student becomes the “receiver.”

Student-facing version:

Common sense as knowing says research gives us facts to collect, cite, and apply.

It asks:
What did the study find?
Is it true?
Can I remember it?
Can I use it?
Does it prove my point?

That way of reading is not useless. Students do need accurate information. But it is incomplete because it treats research as if the goal is simply to possess the right answer. The opening story draft names this directly: knowing asks whether the study is true, memorable, useful, or able to prove a point, while understanding asks what the study helps us see and what its limits are.

Overview narrative

2) Overview narrative

How does this research operate in everyday life? What does it ask students to locate, know, fix, control, predict, or prove?

In everyday life, this approach shows up whenever people say things like “**research says**,” “**studies show**,” “**science proves**,” or “**sources?**” The problem is not asking for sources. The problem is expecting a source to settle the matter completely. In the opening story draft, a study becomes a “truth machine,” a citation becomes a weapon, and a finding becomes a rule.

For students, common sense as knowing asks them to approach research like a hunt for the answer. They look for the main finding, the statistic, the conclusion, or the quote that can be carried into an argument. It asks them to locate what the study found, know the correct claim, fix uncertainty, control interpretation, predict what will happen, or prove that one side is right.

The D_03 analysis gives a strong structure for this: the common-sense approach shapes how students **read, evaluate, and apply** research. Students read linearly, as if truth must be received in order; evaluate the report as true or false; and apply the finding as a universal rule or algorithm.

A student might read one article on conflict, body language, relationship repair, ADHD, motivation, or deception and then try to turn it into a rule:

If the study says X, then I should do X, and Y should happen.

That is the algorithmic fantasy. It promises control. It says communication problems can be solved if we can just find the right finding, the right script, the right ratio, the right trick, or the right evidence-backed move.

Communication consequences

3) Communication consequences

What does this approach help students do, and what does it cause them to miss, flatten, over-trust, or misread?

This approach helps students in real ways. It encourages them to value evidence, ask for sources, learn research vocabulary, summarize findings, and avoid treating every opinion as equally supported. Those are useful habits. The chapter materials are clear that the problem is not knowledge itself. Knowing matters; accurate information matters. The issue is treating knowing as the whole goal.

But the common-sense-as-knowing approach also creates several communication problems.

First, it can make students **flatten research into true or false**. Good research usually does not simply “prove” something once and for all. It supports, complicates, qualifies, and raises more questions. The analysis notes emphasize that a single study offers partial and local truth, not universal truth.

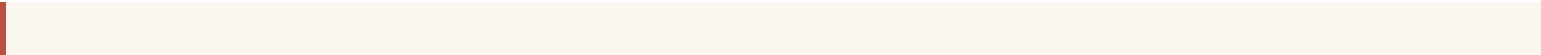
Second, it can make students **over-trust one study, one statistic, one AI summary, or one “research says” claim**. A partial finding can become a universal life rule. The opening story draft gives examples: someone might treat eye movements as proof of lying, crossed arms as resistance, dopamine as a simple explanation for unfinished work, or “research says” as a way to make a complicated study sound like a rule for everyone’s life.

Third, it can turn research into misinformation even when the original study was not misinformation. This is important: the danger is not only fake research. Real research can become misleading when it is detached from context, oversimplified, generalized too widely, or applied too rigidly. The analysis notes identify examples such as social media self-diagnosis, self-treatment, dopamine advice, the 5-to-1 relationship ratio, mirroring in sales, and ADHD craft-cart advice.

Fourth, it can make students miss the interpersonal stakes. Misinformation about communication does not just sit in someone’s head. It shapes what people notice, accuse, trust, dismiss, and interpret as honesty, care, deception, disrespect, confidence, or connection.

In short:
Common sense as knowing helps students collect evidence, but it can cause them to mistake a research finding for a universal rule.

Definition of the approach



5) What problem does (Un)Common Sense help students understand?

What complexity, relation, context, history, power dynamic, or tension becomes visible in research?

The **(un)common-sense approach to research** helps students understand that the problem is not simply that they need more research facts. The problem is that students are often taught to relate to research as if a study gives them a finished truth. That creates a fragile relationship to evidence: students may accept a study as totally true, reject it as totally false, or turn one finding into a rule for life.

The (un)common-sense approach makes research visible as **partial, contextual, situated, and connected evidence**. A research article is not a magic answer machine. It is one piece of a larger sense-making process. The D_03 analysis names this directly: the common-sense approach treats research as a "transfer of truth," while the (un)common-sense approach treats reading research as "sense-making with partial, contextual, and local evidence."

This approach helps students see several things that common sense often hides.

First, it makes **context** visible. A finding may be useful, but students still need to ask where it came from, who was studied, what methods were used, what question the researchers asked, and where the finding may or may not apply. The notes describe this as **local truth**: a finding can be true under specific parameters, in a specific context, for a specific question or population, without becoming a universal rule.

Second, it makes **relation** visible. A single study does not stand alone. It connects to other studies, lived experience, theory, methods, limitations, contradictions, and the situation where someone wants to apply it. This is why the notes frame synthesis as different from summary: summary reports what one article says; synthesis connects that article to other research, contradictory findings, lived experience, and the specific question being asked.

Third, it makes **power and authority** visible. "Research says" can sound neutral, but research claims often carry authority. That authority can be used responsibly, or it can be used to shut people down, sell communication hacks, overstate evidence, or make one interpretation seem unquestionable. The chapter notes describe communication charlatans as people who use research as simplified facts, hacks, rules, or authority moves rather than as a complex body of inquiry.

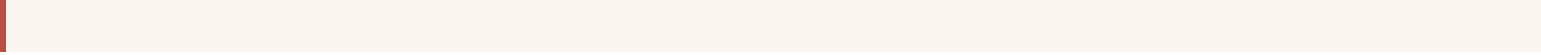
Fourth, it makes **tension** visible. Two findings may seem to contradict each other, but that does not automatically mean one is right and the other is wrong. From an (un)common-sense approach, contradiction becomes a reason to ask better questions: When is this finding useful? For whom? Under what conditions? What does this contradiction help us understand? The notes use mirroring in sales as an example: one finding may suggest mirroring helps, while another may suggest it feels manipulative. The point is not to force one universal truth, but to understand the contexts where each claim may make sense.

So the problem (un)common sense helps students understand is this:

Research becomes misleading when students treat partial evidence like complete truth.

The goal is not to make students distrust research. The goal is to help them use research more carefully.

What problem does (Un)Common Sense help students understand?



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Overview narrative

6) Overview narrative

How does this approach operate in everyday life? What does it ask students to notice, interpret, hold open, revise, test, or try?

In everyday life, the **(un)common-sense approach to research** asks students to interact with research rather than simply receive it.

Instead of reading an article as if truth is being delivered from expert to reader, students read it as something they are making sense with. The notes describe this shift as moving from **passive reception to active interaction** and from **reporting to sense-making**. Research reports do not transfer complete truth; they help students connect concepts, ideas, evidence, contexts, limitations, other research, and lived experience.

This changes how students move through a research article. They do not have to begin at the first sentence and march straight through to the end like the article is a sacred hallway. They can read strategically. They might begin with the abstract, move to the discussion, check the conclusion, return to the rationale, look at the literature review, and then examine the methods and findings. The notes describe this as **interactive reading** rather than linear reading.

This also changes what students ask research to do. They stop asking only:

What did the study find?
Is it true?
Can I use it to prove my point?

They start asking:

What does this study help me understand?
What question was it trying to answer?
Who or what did it study?
What does it explain well?
What does it not explain?
What other research complicates it?
Where might this finding apply?
Where might it fail?
How could I use this as a flexible guide instead of a rigid rule?

The approach also asks students to revise how they apply research. Instead of applying research **algorithmically**, they apply it **heuristically**. An algorithm promises: if you do X, you will get Y. A heuristic says: this finding can guide your attention, sensitize you to a possibility, and help you adapt to a situation. The D_03 notes specifically recommend moving from "algorithms to heuristics" as one of the central (un)common-sense shifts.

The 5-to-1 ratio example shows the difference. Read algorithmically, students might think: "For every negative interaction, I need five positive ones, and then the relationship should be repaired." Read heuristically, the research becomes more flexible: "Relationships often need more positive, affirming, connecting interaction than negative or damaging interaction, but what counts as positive depends on the people, the conflict, the history, and the size of the rupture." The notes emphasize that the problem is not that the 5-to-1 ratio is simply wrong; the problem is applying it like a one-to-one rule without flexible, adaptive thinking.

So, in everyday life, this approach asks students to hold research open long enough to understand it. It asks them to test claims, revise interpretations, compare findings, notice limits, and try research-informed action without pretending one study can control the outcome.

Communication consequences

7) Communication consequences

How does this approach change what students can notice, interpret, respond to, repair, or practice when using research?

The **(un)common-sense approach to research** changes students from people who collect findings into people who can use research with judgment.

It changes what they **notice**. Students begin to notice the parts of a study that common-sense reading often skips: the research question, the sample, the method, the limitations, the context, the assumptions, the terms, and the difference between what the study actually found and what someone online claims it found.

It changes what they **interpret**. Students stop treating a study as simply true or false and begin interpreting findings as local, partial, and connected. The notes are clear that local truth is not relativism. A local truth is still a truth; it is just true under particular conditions. This helps students avoid both naïve acceptance and cynical rejection.

It changes how they **respond** to research claims in everyday communication. When someone says, "Research says..." students do not have to either bow down before the citation or reject it because it feels inconvenient. They can ask better questions: What research? Conducted with whom? In what context? Compared to what? What were the limits? What does this actually help us understand?

It changes how they **repair** misinformation. A major point in the notes is that misinformation does not always come from fake research. Sometimes it comes from real research being oversimplified, detached from context, generalized too far, or turned into a rigid rule. The chapter notes connect this to social media self-diagnosis, self-treatment, dopamine advice, ADHD craft-cart advice, mirroring in sales, and communication hacks.

It changes what students can **practice**. They can practice synthesis instead of summary. They can practice local truth instead of universal truth. They can practice dialectical thinking instead of either/or evaluation. They can practice using research heuristically instead of algorithmically. The notes summarize the major leap this way: from reporting to sense-making, from summarizing to synthesizing, from universal truths to local truths, from algorithms to heuristics, from passive reception to active interaction, and from certainty to curiosity.

The ADHD craft-cart example makes the consequence concrete. Algorithmically, the advice becomes: **buy this craft cart**. Heuristically, the advice becomes: **create a visible container that helps organize scattered materials related to crafts or hyperfixations**. Once students understand the function of the advice, they can adapt it. A cart, basket, shelf, bin, or other system might work. The research-informed idea becomes more usable because it is no longer trapped inside one rigid instruction.

In one sentence:

The (un)common-sense approach to research helps students use studies as tools for careful sense-making, not weapons for certainty or recipes for control.

Opening Story: The Myth in Motion

Section goal

Pull students in with a vivid, relatable story that makes the common sense myth feel plausible, shows its everyday cost, and previews the chapter's movement from knowing toward understanding.

Purpose of the story

This story should show that even the author who has over 15 years of reading research and conducting didn't escape the common sense approach to research. the primary goal is to show what the common sense approach to research looks like and the secondary prupose is to show that expertise doesn't necessarily mean understanding and that the common-sense approach isn't because people are stupid and uneducated. Drafting note: the story should lead cleaning into the introduction of the common sense approach and and unccomonsense approach and then lead into the topic overview. Below is a drafting guide for the section **immediately after the opening story**. The section's job is to turn the story from "I was overwhelmed by a research article" into the chapter's central argument: **the problem is not just that research is hard to read; the problem is how we have been taught to relate to research**.

Core purpose of the section

After the opening story, the chapter should make this move:

The story shows the common-sense approach before the chapter names it.

In the story, the article seems to "know things," and the narrator does not. The article feels like a container of expert knowledge. The reader's job seems to be absorbing the truth, extracting the finding, and carrying it into the world. That is exactly the **common-sense approach to research**: research as a method of knowing, where findings are transferred from researchers to readers. Your analyzed notes frame this as the difference between treating research as "truth transfer" and treating research as contextual sense-making.

The transition should make students feel this recognition:

Oh. I do this too. I treat research like it is supposed to hand me the answer.

That recognition gives you a natural entry into the two approaches.

1. How the opening story leads into the common-sense approach to research

The opening story dramatizes three features of the common-sense approach.

A. Research feels like expert truth

The narrator assumes the article knows something complete and the reader's job is to receive it. That connects directly to the common-sense model of communication as transfer: someone has information, sends it, and someone else receives it.

Applied to research, the model becomes:

Researchers know something → the article contains it → the student reads it → now the student knows it.

This is the common-sense approach to research.

A student-facing definition could be:

The common-sense approach to research treats a study like a container of expert truth. The researchers found something, wrote it down, and the reader's job is to extract the finding, remember it, and use it correctly.

That definition should be followed immediately by a soft qualification:

This is not useless. Students do need to know what a study found. They need facts, evidence, sources, and accurate summaries. But the problem begins when this becomes the whole relationship to research.

B. Research becomes a tool for certainty

The opening story already has the perfect line of development: research promises "answers, facts, truths, certainty." That is the emotional appeal of the common-sense approach. Research feels powerful because it seems to settle things.

This is where you can introduce the "citation as folding chair" idea. The common-sense approach makes research feel like something students can use to win.

That is funny, but it is also the danger.

If research becomes a way to win, students may stop asking what a study helps them understand. They may only ask whether the study proves them right.

C. Research gets flattened into "true or false"

The common-sense approach encourages students to ask:

- Is this study true?
- Is it false?
- Can I use it?
- Does it prove my point?
- Can I cite it?

Those questions are not wrong, but they are too small. Your voice notes argue that good research usually supports, complicates, qualifies, and raises more questions. The danger is treating one study as a complete unit of truth instead of partial, contextual evidence.

This is the key sentence the section needs to build toward:

The common-sense approach does not make research useless. It makes research too simple.

2. The problem the common-sense approach creates

The next move should explain why this approach matters outside the classroom. This is where the chapter connects the opening story to social media, misinformation, and AI.

The common-sense approach creates a danger chain:

Research as transfer → research as certainty → research as proof → research as rule → research as misinformation.

A. "Research says" becomes a shortcut for authority

Students should recognize the online version immediately. Someone says:

- "Research says..."

- “Studies show...”
- “Science proves...”
- “Sources?”
- “Do you have a study?”

The problem is not asking for sources. Asking for sources is good. The problem is treating a source as if it automatically settles the conversation.

The section should explain that “research says” is not a complete explanation. It leaves out:

- What research?
- Conducted with whom?
- Using what method?
- In what context?
- Compared to what?
- With what limitations?
- Connected to what larger body of evidence?

This connects beautifully to the ACRL Framework, especially the ideas that authority is constructed and contextual, information creation is a process, research is inquiry, and scholarship is conversation. Those concepts support your (un)common-sense approach because they frame research as something students interpret within context, not something they simply receive as final truth. ([American Library Association](#))

B. Real research can become misinformation

This is one of the strongest points for the chapter.
Misinformation does not always begin with fake research. Sometimes it begins with real research that gets simplified too far.

- A finding loses its context.
- A limitation disappears.
- A correlation becomes a cause.
- A study with a narrow sample becomes advice for everyone.
- A cautious claim becomes a TikTok rule.

Sumner et al. (2014) are useful here because their BMJ study found that exaggeration in health-related news was strongly associated with exaggeration already present in academic press releases. That helps you make a more nuanced point: misinformation can enter the communication chain long before a claim reaches social media. ([BMJ](#))

This supports a key chapter claim:

The problem is not only that people misunderstand research. The problem is that research often travels through systems that reward simplification, confidence, and shareability.

C. Social media turns partial findings into identity, advice, and rules

This section should then move into social media examples. Mental health, ADHD, dopamine, attachment styles, body language, trauma language, and “communication hacks” all work well because students already see these claims.
Starvaggi et al. (2024) support the claim that mental health misinformation is common on social media, though prevalence varies across disorders and treatment types. Yeung et al. (2022) provide a specific ADHD/TikTok example by examining popular ADHD videos and classifying content quality. ([PubMed](#))
The point is not “students are gullible.” The better point is:
Students are living in information environments where research language gets turned into highly confident, highly shareable claims.
That connects directly to the common-sense approach. If students already think research should give clean answers, they are more likely to trust content that presents research as a clean answer.

D. The common-sense approach encourages algorithmic application

This is where the chapter can introduce one of your strongest distinctions:

Common sense applies research algorithmically.
(Un)common sense applies research heuristically.

Algorithmic use says:

If the research says X, then I should do X, and Y should happen.

Heuristic use says:

This finding can guide my attention, but I still need to interpret the situation.

This matters for interpersonal communication because human interaction is not a vending machine. Research can inform our judgment, but it does not remove the need for judgment.
Examples:

- “Five positive interactions fix one negative interaction.”
- “Eye movements reveal lying.”
- “Crossed arms mean resistance.”
- “Dopamine is the motivation chemical.”
- “This ADHD hack will work for you.”
- “AI summarized it, so now I understand it.”

The common-sense danger is not applying research. The danger is applying research as if it were a universal rule.

3. How AI intensifies the common-sense problem

The AI section should not feel like a separate topic. It should feel like the newest and most powerful version of the same problem.
The common-sense approach says:
Information can be transferred to me.
AI makes that fantasy feel even easier:
The answer can be generated for me.

That is why AI belongs in this transition section.

Define automation bias carefully

Automation bias is the tendency to over-rely on automated systems or recommendations. Goddard et al. define it as over-reliance on automation and review how this shows up across fields, especially when people fail to catch errors introduced by automated systems. ([OUP Academic](#))

For this chapter, do **not** say:

AI decreases critical thinking.

That is too broad and would need stronger causal evidence.

Say this instead:

AI can invite automation bias when students accept machine-generated answers without enough checking, interpretation, or contextual judgment.

That phrasing is safer and stronger.

Add algorithm appreciation

Logg et al. (2019) are useful because their experiments found that people sometimes adhere more to advice when they think it comes from an algorithm than when they think it comes from a person. ([IDEAS/RePEc](#))

That gives you a direct bridge to students:

A polished AI summary may feel objective, neutral, or more reliable because it came from a machine. But fluency is not evidence. Confidence is not credibility. A clean answer is not the same thing as understanding.

AI makes literacies more important, not less important

Long and Magerko (2020) define AI literacy as the competencies needed to understand, use, and critically evaluate AI. Chu-Ke and Dong (2024) connect generative AI to misinformation and argue that AI-era information environments require stronger literacy education. ([ACM Digital Library](#))

This supports the chapter's balanced stance:

This is not an anti-AI chapter. It is an anti-outsourcing-understanding chapter.

AI can summarize, organize, translate, brainstorm, and reduce access barriers. But it cannot do the student's sense-making for them. Students still need to evaluate evidence, notice limitations, compare claims, identify context, and decide whether an answer is useful for the situation.

4. Introduce literacies as the reason students should care

After social media and AI, students need a constructive reason to keep reading. This is where literacies come in.

The section should define literacies as **practiced forms of judgment**, not as academic vocabulary students are supposed to memorize.

Information literacy

Information literacy is the ability to find, evaluate, interpret, and use information responsibly. In this chapter, it should be framed as more than "finding sources." It includes understanding how authority works, how information is created, how claims circulate, and how evidence fits into a larger conversation.

ACRL (2016), Lloyd (2006), and Mackey and Jacobson (2011) are useful here. The ACRL Framework is especially aligned with your approach because it treats scholarship as conversation and research as inquiry, not answer extraction. ([American Library Association](#))

Scientific literacy

Scientific literacy is not just knowing science facts. It includes reading scientific texts, evaluating evidence, understanding methods, interpreting claims, and recognizing limits.

Norris and Phillips (2003) are especially useful because they argue that reading and writing are central to scientific literacy. Kluevsek (2017) connects information literacy and scientific literacy directly, which helps you argue that students need to understand how research claims are made before they can responsibly use them.

AI literacy

AI literacy means understanding what AI can do, what it cannot do, how to evaluate its outputs, and how to use it ethically and reflectively. Long and Magerko (2020) and Ng et al. (2021) can support this. ([ACM Digital Library](#))

Digital literacy

Digital literacy is the ability to navigate, evaluate, create, and participate responsibly in digital information environments. In this chapter, digital literacy matters because so many research claims reach students through social media, search engines, influencers, screenshots, and AI-generated summaries.

Data literacy

Data literacy is the ability to understand and evaluate data-based claims. Students do not need to become statisticians to ask good questions:

- What is being measured?
- Who was included?
- What comparison is being made?
- What does the number actually show?
- What does it not show?

Because your current annotated bibliography is stronger on information, scientific, and AI literacy than data literacy specifically, I would define data literacy briefly here and add a dedicated data-literacy source later if this becomes a major key term.

Research literacy

Research literacy can function as the umbrella concept for this chapter.

A student-facing definition could be:

Research literacy is the ability to read, evaluate, interpret, and use research responsibly without treating one study as the final word.

This keeps the section grounded in the chapter's purpose.

5. Introduce the (un)common-sense approach to research

Once students understand the dangers of the common-sense approach, introduce the alternative.

The **(un)common-sense approach to research** treats research as a method of understanding.

A student-facing definition could be:

The (un)common-sense approach to research treats reading research as sense-making. A study does not hand us the final answer. It helps us understand a question, a pattern, a context, a relationship, or a problem more carefully.

This approach asks students to read research by asking:

- What does this study help me understand?
- What question were the researchers asking?
- Who did they study?
- What did they not study?
- What methods did they use?
- What does this finding explain well?
- Where might it not apply?
- What other research complicates it?
- What assumptions am I bringing to it?
- How could this inform my judgment without becoming a rigid rule?

This is where you can introduce **partial truth** and **local truth**.

Partial truth

A finding may be useful and supported, but still incomplete.

Local truth

A finding may be true in a particular context, with a particular population, using particular methods, for a particular question.

This helps students avoid two bad options:

- Naively accepting research as universal truth
- Cynically rejecting research because it has limits

The (un)common-sense approach gives them a third option:

Use the study carefully.

6. Benefits of the (un)common-sense approach

The section should make the payoff concrete.

A. It improves students' academic lives

Students learn to read research with less panic and more strategy. They stop treating a hard article as proof that they are not smart enough. They learn that even experienced researchers need time to understand research.

That matters because the opening story begins with intimidation. The alternative should offer relief:

You do not have to know everything immediately. You have to learn how to make sense.

B. It improves students' online lives

Students become better at evaluating "research says" claims. They can recognize when a claim is missing context, overgeneralizing, or using scientific language as decoration.

They can ask:

- Is this a study, a summary, a headline, or an influencer interpretation?
- What evidence is actually being cited?
- What did the study really examine?
- Is the claim broader than the evidence?
- Is the source selling something?

C. It improves students' AI use

Students can use AI as a support tool without confusing output with understanding.

They can ask:

- Where did this answer come from?
- Does it cite real sources?
- Does it overstate the evidence?
- What does it leave out?
- What would I need to check?
- How does this connect to the actual research?

This keeps the chapter balanced. AI is not the villain. Passive certainty is the problem.

D. It improves personal relationships

This is the interpersonal communication payoff.
If students learn to read research carefully, they are less likely to turn oversimplified claims into relational judgments.
They may be less likely to think:

- “Their eyes moved, so they lied.”
- “They crossed their arms, so they are resisting.”
- “They have this attachment style, so I know what is wrong with them.”
- “My brain just lacks dopamine, so this is a personal failure.”
- “Research says this script should work, so if it did not work, the other person is the problem.”

Instead, they can hold research as a guide, not a weapon.

E. It improves professional judgment

Professionally, students will encounter research-based claims in training, leadership, workplace wellness, AI tools, assessment reports, customer communication, management advice, and organizational decision-making.
The (un)common-sense approach prepares them to ask better questions before accepting a claim as “evidence-based.”
That is a real workplace skill.

7. Suggested structure for the textbook section after the story

Paragraph 1: Name what the story revealed

The first paragraph should say that the opening story was not only about one hard article. It was about a relationship to research.
Main idea:
I thought research was something experts knew and I was supposed to absorb.

Paragraph 2: Define the common-sense approach to research

Main idea:
The common-sense approach treats research as a transfer of expert truth from researchers to readers.
Use language from the opening:

- answers
- facts
- truths
- certainty
- proving a point
- citing research to win

Paragraph 3: Clarify that this approach is useful but incomplete

Main idea:
Knowing what a study found matters, but knowing is not the same as understanding.
This prevents the chapter from sounding anti-research.

Paragraph 4: Explain how this approach shows up online

Main idea:
The same approach shows up in “research says,” “studies show,” and “sources?” culture.
This is where social media examples enter.

Paragraph 5: Explain how misinformation happens

Main idea:
Research becomes misinformation when partial findings are stripped of context and turned into universal rules.
Use examples:

- ADHD TikTok
- mental health misinformation
- wellness claims
- dopamine hacks
- body language myths

Paragraph 6: Explain why AI raises the stakes

Main idea:
AI makes information easier to access, but easier access does not equal understanding.
Define automation bias and algorithm appreciation carefully.

Paragraph 7: Introduce literacies

Main idea:
In this information environment, students need research literacy, information literacy, scientific literacy, digital literacy, data literacy, and AI literacy.
Keep the list short and practical. Do not let it become a glossary dump.

Paragraph 8: Define the (un)common-sense approach to research

Main idea:
The (un)common-sense approach treats research as sense-making with partial, contextual, connected evidence.

Paragraph 9: Explain the payoff

Main idea:
This approach helps students use research more responsibly in school, online, with AI, in relationships, and at work.
End with the chapter promise:
This chapter will not teach students to worship research or dismiss it. It will teach them to read research carefully enough to use it well.

8. Source-to-claim map for drafting

Claim for the section	Best source support
Information literacy is more than finding sources; it includes evaluating authority, inquiry, and scholarship as conversation.	Association of College and Research Libraries (2016); Lloyd (2006); Mackey & Jacobson (2011)
Scientific literacy requires reading, interpreting, and evaluating scientific texts.	Norris & Phillips (2003); Klucevsek (2017)
Research can be exaggerated as it moves from article to press release to public communication.	Sumner et al. (2014)
Mental health misinformation circulates on social media and varies by topic/disorder.	Starvaggi et al. (2024)
ADHD content on TikTok provides a concrete example of variable/misleading social media health information.	Yeung et al. (2022)
Social media health/mental health claims can involve competing forms of expertise, not just “truth vs. ignorance.”	Lupton & Southerton (2024)
Automation bias involves over-reliance on automated systems.	Goddard et al. (2012)
People may sometimes prefer algorithmic advice over human advice.	Logg et al. (2019)
AI literacy involves understanding, using, and critically evaluating AI.	Long & Magerko (2020); Ng et al. (2021)
Generative AI makes misinformation and literacy education more urgent.	Chu-Ke & Dong (2024)
Reading research should be framed as sense-making, not answer extraction.	Your analyzed voice notes + ACRL Framework + Klucevsek (2017)

9. Drafting cautions

Do not say research is unreliable

Say:
Research is powerful because it is careful, but careful does not mean universal, simple, or final.

Do not say social media is always wrong

Say:
Social media often rewards claims that are simple, confident, and shareable, even when the research is complex.

Do not say AI makes students less intelligent

Say:
AI can encourage over-reliance when students treat generated answers as substitutes for evaluation, connection, and judgment.

Do not make students feel accused

Say:
This is not a personal failure. It is a predictable response to information environments that reward certainty.

Do not overload the introduction with every literacy

Introduce literacies as a cluster, then define only what students need right away.

A good sentence:

These literacies are not separate hoops to jump through. They are overlapping forms of judgment that help us ask better questions about information.

10. Strongest transition sentence for the section

I would build the whole section around this sentence:

The problem was not that I did not know enough research yet. The problem was that I thought research was supposed to work by giving me certainty.

That sentence cleanly ties the opening story to the chapter framework.

Then the section can unfold like this:

Common sense asks research to give us certainty.

That certainty makes research feel useful, powerful, and citeable.

But it also makes research easier to flatten, weaponize, automate, and misuse.

The (un)common-sense approach asks research to do something better: help us understand.

Place for the story

Opening Story Draft

The first time someone placed a 30-page academic research article in front of me, I did what any brave, intellectually serious, emerging scholar would do.

I counted the pages and hoped some of them were pictures.

I flipped through the article slowly, trying to look curious instead of mildly afraid. Maybe there would be a few illustrations. Maybe a generous callout box. Maybe a table that took up three pages and asked very little of me emotionally.

No.

Nearly every page was packed with impossibly small text. The only page that offered any visual relief had a diagram full of arrows, labels, circles, and numbers pointing in enough directions that it could have been a road map, a poor rendition of a constellation, or a child’s drawing of a very stressful airport.

I remember staring at it and thinking, surely this means something.

I just had no idea what.

Still, I read the article. Or, more honestly, I performed the ritual of reading. I started at the top of the first page because that is what a reasonable person does. I moved sentence by sentence, underlining things I did not understand with the confidence of someone who believed underlining was a form of comprehension. I wrote little question marks in the margins. After a while, the margins were mostly question marks.

At some point, I realized I had been reading the same paragraph for several minutes. My eyes were moving. My hand was holding a pen. My body was technically present. But the article and I were not exactly in a relationship of mutual understanding.

The article seemed to know things.

I did not.

And I assumed that was the point.

In those days, I thought research was supposed to give me answers. Facts. Truths. Certainty. Research was where knowledge lived before I collected it and carried it into the world like a very responsible academic squirrel.

If I could read enough research, I thought, I would know things. And if I knew things, I could win arguments. I could cite a study. I could say “Actually, research shows…” and then watch the room fall respectfully silent.

I am embarrassed to admit how satisfying that fantasy was.

There are few pleasures in life quite like citing a study to win an argument. It feels clean. It feels efficient. It feels like dropping a tiny academic folding chair into the middle of a conversation.

Someone says something vague. You raise one finger. You clear your throat. You say, “Well, according to a study…”

Powerful.

Deeply annoying, but powerful.

This is not just a graduate school problem. You can see the same pattern all over social media. Someone makes a claim, and someone else replies, “Sources?” Someone says, “Research says…” or “Studies show…” or “Science proves…” as if Research were a person who had walked into the room wearing a lab coat and carrying the final answer.

I do not know who named their child Research, but he seems very busy online.

The problem is not that people are asking for sources. Asking for sources can be a good thing. The problem is what we often expect a source to do. We sometimes treat research like it is supposed to settle the matter. A study becomes a truth machine. A citation becomes a weapon. A finding becomes a rule.

From the common-sense approach, this makes sense. The common-sense approach treats communication like the transfer of information. Someone has information. They send it. Someone else receives it. Communication complete.

When we apply that same logic to research, we start to treat a research article as if it transfers truth from researchers to readers. The researchers found something. They wrote it down. We read it. Now we know it.

Beautiful.

Not useless.

Also wildly incomplete.

That way of reading research pulls us toward knowing. Knowing asks: What did the study find? Is it true? Can I remember it? Can I use it? Does it prove my point? Can I cite it at Thanksgiving when Uncle Rob starts saying something confident and incorrect?

Understanding asks different questions.

Understanding asks: What does this study help me see? What kind of question were the researchers asking? Who did they study? What did they not study? What does this finding explain well? Where might it not apply? What other research complicates it? What assumptions am I bringing to it? What does this help me understand about communicating?

That shift matters because research is rarely as simple as “true” or “false.” Good research does not usually walk into the room, slam its hand on the table, and announce, “I have solved human behavior.”

Good research is usually more irritating than that.

It supports. It complicates. It qualifies. It says, “Here is what we found in this context, with these participants, using these methods, while also acknowledging these limitations, and please do not turn this into a universal life rule on TikTok.”

Research often gives us evidence, but it also gives us more questions. That is not a weakness. That is part of what makes research useful.

But if we read research only through knowing, we can miss that. We can take one study and turn it into a rule. We can take a partial finding and treat it like a universal truth. We can accept research too quickly when it agrees with us and reject it too quickly when it does not. We can use research to become more certain instead of more thoughtful.

And certainty is where a lot of misinformation starts to feel comfortable.

This matters especially in interpersonal communication because misinformation about communicating does not just sit harmlessly in someone's head. It shows up in relationships. It shapes what we notice. It shapes what we accuse people of. It shapes what we think counts as honesty, care, deception, confidence, connection, or disrespect.

A person can watch a partner's eye movements and think they have detected a lie. A manager can read one article about body language and start treating every crossed arm as resistance. A student can hear that dopamine is the motivation chemical and start interpreting every unfinished assignment as a personal neurotransmitter failure. Someone online can say "research says" and make a complicated study sound like a rule for everyone's life.

That is the danger of reading research as knowing.

It can make us feel informed while making the world smaller.

The goal of this chapter is not to make research seem impossible. Research can be difficult, yes. Sometimes it is dense. Sometimes it is full of technical language. Sometimes one sentence contains so many clauses that by the end you are no longer sure where you live.

But research is not beyond you.

I have met many of the people who write research in interpersonal communication. They are human beings. They own mugs. They forget passwords. Some of them probably have unread emails from 2018. You can understand research.

But understanding research does not mean treating every article like a container of facts you have to absorb in order. It means learning how to interact with research. It means learning how to move around inside an article, ask better questions, notice limits, connect findings, and stay curious long enough for the study to become useful.

That is the (un)common-sense approach to research.

Instead of treating research as a method of knowing, this chapter treats research as a method of understanding. Instead of asking research to hand us certainty, we will ask it to help us practice sense-making. Instead of reading studies as universal truth, we will read them as partial, contextual, and connected pieces of evidence that can help us understand communicating more carefully.

That shift is especially important now because we live in an age where information is everywhere. Social media gives us claims. Influencers give us summaries. AI gives us answers. Search engines give us snippets. Everyone seems to have a source, a statistic, a study, or a screenshot.

But access to information is not the same thing as understanding.

AI can produce information for us. It can summarize, organize, translate, and generate. It can be tremendously useful. I use it. Without tools that help me manage dyslexia and ADHD, writing this book would be much harder than it already is, and it is already a little rude how hard it is.

So this is not an anti-AI chapter.

It is an anti-outsourcing-understanding chapter.

AI can help us work with information, but it cannot understand for us. Understanding is something we have to practice. We have to evaluate. We have to connect. We have to notice context. We have to ask when something is true, when it is not true, and what its limits are.

That is why reading research belongs in a book about interpersonal communicating.

Because communication problems are rarely solved by collecting one more fact. They usually require better judgment, better questions, and more flexible understanding.

So yes, this chapter is about research.

But it is also about how not to get trapped by certainty.

It is about how to read without flattening complexity.

It is about how to use research without turning it into a weapon, a rule, or a social media spell.

And maybe, if all goes well, it is about how to look at a 30-page article full of tiny text and arrows pointing every which way and think:

Okay.

I do not have to know everything yet.

I just have to start making sense.

Annotated bibliography box

References to use in the drafting packet

Association of College and Research Libraries. (2016). *Framework for information literacy for higher education*. Association of College and Research Libraries.

Chu-Ke, C., & Dong, Y. (2024). Misinformation and literacies in the era of generative artificial intelligence: A brief overview and a call for future research. *Emerging Media*, 2(2). doi:10.1177/27523543241240285

Goddard, K., Roudsari, A., & Wyatt, J. C. (2012). Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association*, 19(1), 121–127. doi:10.1136/amiajnl-2011-000089

Kluccevsek, K. M. (2017). The intersection of information and science literacy. *Communications in Information Literacy*, 11(2), 354–365. doi:10.15760/comminfolit.2017.11.2.7

Lloyd, A. (2006). Information literacy landscapes: An emerging picture. *Journal of Documentation*, 62(5), 570–583. doi:10.1108/00220410610688723

Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151, 90–103. doi:10.1016/j.obhdp.2018.12.005

Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. *Proceedings of the CHI Conference on Human Factors in Computing Systems*. doi:10.1145/3313831.3376727

Lupton, D., & Southerton, C. (2024). #ActuallyAutistic: Competing cultures of expertise and knowledge in relation to autism and ADHD self-diagnosis on TikTok. *Historical Social Research*, 49(3), 188–212. doi:10.12759/hsr.49.2024.29

Mackey, T. P., & Jacobson, T. E. (2011). Reframing information literacy as a metaliteracy. *College & Research Libraries*, 72(1), 62–78. doi:10.5860/crl-76r1

Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, 100041. doi:10.1016/j.caeai.2021.100041

Norris, S. P., & Phillips, L. M. (2003). How literacy in its fundamental sense is central to scientific literacy. *Science Education*, 87(2), 224–240. doi:10.1002/sce.10066

Starvaggi, I., Dierckman, C., & Lorenzo-Luaces, L. (2024). Mental health misinformation on social media: Review and future directions. *Current Opinion in Psychology*, 56, 101738. doi:10.1016/j.copsyc.2023.101738

Sumner, P., et al. (2014). The association between exaggeration in health related science news and academic press releases: Retrospective observational study. *BMJ*, 349, g7015. doi:10.1136/bmj.g7015

Yeung, A., Ng, E., & Abi-Jaoude, E. (2022). TikTok and attention-deficit/hyperactivity disorder: A cross-sectional study of social media content quality. *Canadian Journal of Psychiatry*, 67(12), 899–906. doi:10.1177/07067437221082854

Callout box ideas

Communication Myth: "Research Says"

Placement: After the "Research seems very busy online" paragraph

Content:
"Research says" sounds powerful, but it leaves out almost everything you need to evaluate the claim.

- Questions to ask:
- What research?
 - Studying whom?
 - Using what method?
 - In what context?
 - Compared to what?
 - With what limitations?

Takeaway:
A source can help a conversation. A vague appeal to research can shut one down.

Illustration ideas

1. The One Diagram That Was Supposed to Help

Placement: After the "stressful airport" paragraph

- Visual:**
- Tiny academic article text everywhere
 - One giant diagram with arrows, circles, boxes, labels, numbers
 - Looks equally like:
 - a road map
 - a constellation
 - an airport terminal
 - an overconfident flowchart

Caption:

Figure X.1: The One Diagram That Was Supposed to Help
Sometimes research tries to clarify complexity. Sometimes the clarifying diagram appears to be managing air traffic.

H1: Overview of [Topic]

Section goal

Define the topic in everyday terms, explain why it matters to students now, introduce Common Sense and (Un)Common Sense as two approaches to the same reality, and establish the main ideas students need before the chapter begins shifting their assumptions.

Framing box: section claim

The purpose of the this overview section is to help students overcome common barriers to reading research by understanding where it comes from (history), what it means to be peer-reviewed and why it is held up as credibility and its criticisms, and what forms research takes. Drafting note: In the introductory paragraph of this section peer-reviewed research needs to be defined, then the chapter progresses by asking "where does it come from" (history); "how does it work" (an overview of the peer review process what it does well and its criticism), and finally "what does it look like" this is where the chapter will define different types of research and what they do.

Specific help request box

This section needs to be an overview but descriptive with examples.

H1: Overview of [Topic] / Overview Main Idea 1

Main idea box

The history main idea is just that students need to know where the peer-review process came from and how it took the form that it did because when we know where something comes from we can understand it better. Drafting note: this should be relatively brief to the other section but include enough to show where peer-reviewed research came from and why

Idea development + evidence box

The rise of modern peer review was driven by practical and political necessities rather than ancient scholarly ideals.

- **Public Accountability and Autonomy:** During the Cold War, unprecedented government funding required scientists to prove they were accountable to the public. Peer review allowed researchers to self-govern, building institutional trust while protecting their work from political interference.
- **Organizational Survival:** In the social sciences, higher education expanded rapidly and manuscript submissions skyrocketed. Editors simply needed a logistical system to manage the overwhelming workload, distribute scarce editorial attention, and prevent reviewer burnout.

Here is a more descriptive, in-depth breakdown of how it actually evolved:

1. The Early Days: The Myth of Ancient Origins While academic journals have been around for centuries, routine external peer review did not actually become widespread until after World War II. Before this, editorial review developed very unevenly. Often, the decision to publish a study simply came down to the journal editor's personal judgment or a quick conversation with a colleague down the hall. There was no standardized, global system of anonymous experts vetting every claim.

2. The Cold War Boom: Funding and Prestige Modern peer review gained its massive prestige and symbolic authority during the Cold War in the United States. During this era, the government began pouring unprecedented amounts of funding into research, which brought demands for public accountability. Scientists needed a way to demonstrate to the public and politicians that their work was legitimate, while simultaneously defending their own scientific autonomy from government interference.

- **Example:** Imagine the government handing millions of dollars to universities for research. Politicians wanted proof the money was well-spent. By establishing formal peer review, scholars could say, "Trust us, we are policing ourselves using independent experts," which secured institutional trust and kept funding flowing without political meddling.

3. The Social Science Squeeze: A Practical Necessity In the social sciences, such as sociology and communication, the modern peer-review system did not emerge from a single, high-minded philosophical design. Instead, it evolved piecemeal as a practical response to severe organizational constraints. During the mid-to-late 20th century, higher education expanded rapidly, and the volume of submitted manuscripts skyrocketed.

- **Example:** Imagine an editor of a major communication journal who used to read 50 submissions a year suddenly receiving 500. It became physically impossible for one editor to screen everything. Journals had to manage scarce editorial attention and distribute the overwhelming workload to prevent reviewer burnout. Standardized, anonymous external review diffused gradually simply because editors needed a logistical system to survive the flood of papers.

4. The Takeaway: An Evolving Filter, Not a Magic Stamp Understanding this history shows students that peer review is a family of evolving evaluative practices—like single-blind, double-blind, or open review—rather than a single, timeless institution. When a student reads a peer-reviewed communication study, they should see it as a paper that survived a practical, historically developed quality-control filter, rather than a document certified as an absolute, unquestionable truth.

Does this level of depth and description give you what you need for the drafting packet, or would you like to explore the specific modern critiques of this system (like bias and unpaid labor) next?

Annotated bibliography box

Here is the in-depth annotated bibliography focusing specifically on public opinion about research, the history of peer review, and research literacy. It is formatted cleanly so you can easily copy and paste it into your drafting packet.

Annotated Bibliography: Public Opinion, Peer Review History, and Research Literacy

1. Public Opinion About Research

Focusing on how the public views research, trust in scientists, and the gap between favorable attitudes and deep methodological understanding.

National Science Board. (2024). *Science and technology: Public perceptions, awareness, and information sources (NSB-2024-4)*. National Science Foundation.

- Annotation:** This report synthesizes long-running survey indicators, showing that Americans have generally expressed confidence in science and scientists over time, but comprehension of scientific processes remains uneven. Better understanding of the scientific method is associated with greater confidence in scientists. This supports the nuanced claim that the public is not broadly anti-science, but confidence is neither universal nor evenly distributed, and understanding of research design is incomplete.

Cologna, V., et al. (2025). *Trust in scientists and their role in society across 68 countries*. *Nature Human Behaviour*, 9(4), 713–730.

- Annotation:** A massive 68-country survey finding that in most countries, the majority of people trust scientists and support them engaging more in society and policymaking. However, trust in scientists is not the same thing as understanding peer-reviewed scholarship, its methods, or its limits. This source supports a nuanced opening claim: many people retain meaningful trust in scientists, but trust alone does not equal research literacy.

Miller, J. D. (2004). *Public understanding of, and attitudes toward, scientific research: What we know and what we need to know*. *Public Understanding of Science*, 13(3), 273–294.

- Annotation:** A foundational review using decades of U.S. survey data to illustrate that public support for scientific research can strongly coexist with relatively low levels of civic scientific literacy. This helps explain the paradox where people endorse research as socially valuable while still misunderstanding what it can legitimately claim, helping justify the need to teach students to understand inquiry rather than just "know" findings.

Gauchat, G. (2011). *The cultural authority of science: Public trust and acceptance of organized science*. *Public Understanding of Science*, 20(6), 751–770.

- Annotation:** Demonstrates that public acceptance of science depends not only on knowledge but also on public beliefs about science's methods, institutional location, and its relation to common sense. This helps explain why audiences and students may treat "research" as a cultural badge of authority or a university product, which directly affects their credibility judgments.

Sturgis, P., & Allum, N. (2004). *Science in society: Re-evaluating the deficit model of public attitudes*. *Public Understanding of Science*, 13(1), 55–74.

- Annotation:** Testing the "deficit model," this study finds that while knowledge matters, the relationship between knowledge and attitudes is complex and interactive,. It complicates the teaching shortcut that people reject science simply because they do not know enough, showing that attitudes are also shaped by contextual understanding and broader social settings.

2. The History and Critique of Peer Review

Focusing on the historical development of research validation and the structural limitations of the modern publishing ecosystem.

Burnham, J. C. (1990). *The evolution of editorial peer review*. *JAMA*, 263(10), 1323–1329.

- Annotation:** This classic historical account dispels the myth that peer review has always existed in its current form,. It shows that pre-publication editorial review developed unevenly and historically, helping students understand that peer review is a constructed quality-control practice rather than a timeless, magical stamp of truth,.

Baldwin, M. (2018). *Scientific autonomy, public accountability, and the rise of "peer review" in the Cold War United States*. *Isis*, 109(3), 538–558.

- Annotation:** Baldwin connects the rise of modern peer review to public accountability, funding institutions, and Cold War U.S. science policy rather than portraying it as a simple extension of old scholarly customs,. It shows students that peer review matters socially as well as intellectually, and became deeply tied to professional legitimacy and institutional trust during the late twentieth century,.

Merriman, B. (2021). *Peer review as an evolving response to organizational constraint: Evidence from sociology journals, 1952–2018*. *The American Sociologist*, 52(2), 341–366.

- Annotation:** Argues that many features of current peer review emerged as responses to organizational constraints in social science journals—balancing scarce attention, submission growth, and reviewer burden—rather than from a single principled design,. This demystifies the review system for students without dismissing it.

Horbach, S. P. J. M., & Halffman, W. (2018). *The changing forms and expectations of peer review*. *Research Integrity and Peer Review*, 3, 8.

- Annotation:** Reviews how peer review has changed across time and argues that current debates cannot be understood as if there were ever one stable model. It supports the claim that today's system is better understood as a family of evolving arrangements than as a single timeless institution.

Smith, R. (2006). *Peer review: A flawed process at the heart of science and journals*. *Journal of the Royal Society of Medicine*, 99(4), 178–182.

- Annotation:** This widely cited critique names core problems of peer review—such as bias, inconsistency, slowness, and weak evidence of effectiveness—without claiming it should disappear. It provides a balanced view so students learn to respect the screening process without romanticizing it as a flawless certification of absolute truth.

3. Research Literacies and Understanding

Focusing on what it means to be literate in evaluating, reading, and making sense of academic research.

Keser Aschenberger, F., & Pfeffer, T. (2021). *Concept of research literacy in academic continuing education: A systematic review*. *European Journal of University Lifelong Learning*, 5(1), 43–58.

- Annotation:** This systematic review found that the literature does not yet offer a single comprehensive definition of "research literacy" ,. Instead, it shows that "research literacy" functions as an emerging concept that bundles together skills such as understanding research purposes, methods, reading practices, and academic discourse.

Brody, J. L., et al. (2012). *Conceptualizing the role of research literacy in advancing societal health*. *Journal of Health Psychology*.

- Annotation:** Proposes "research literacy" as a key concept, treating it as a distinct competence involving understanding what research is, how it works, and how to engage it responsibly. This supports naming your chapter's specific target: students need to become better readers of studies, methods, and evidence claims.

Klucsevsek, K. M. (2017). *The intersection of information and science literacy*. *Communications in Information Literacy*, 11(2), 354–365.

- Annotation:** Directly connects information and science literacy, arguing that evaluating scientific claims requires finding, accessing, and understanding original research. It proves that trusting a summary is insufficient; evaluating a claim requires understanding the research process behind it.

Norris, S. P., & Phillips, L. M. (2003). *How literacy in its fundamental sense is central to scientific literacy*. *Science Education*, 87(2), 224–240.

- Annotation:** Argues that reading, interpreting, and evaluating texts are central to scientific literacy. This emphasizes that students must read research themselves to interpret scientific writing, methods, and evidence, rather than relying strictly on someone else's simplified summaries.

Goudsouzian, L. K., & Hsu, J. L. (2023). *Reading primary scientific literature: Approaches for teaching students in the undergraduate STEM classroom*. *CBE—Life Sciences Education*, 22(3), es3.

- Annotation:** Reviews approaches for teaching undergraduate students to read primary scientific literature. It identifies reading original research genres (abstracts, methods, results, limitations) as a goal that builds critical thinking, data interpretation, and research literacy.

Would you like me to use this new bibliography to generate a fresh Tailored Report for your Studio tab, or should we move on to drafting the final piece of your chapter?

Main idea box

The purpose of this main point is

Idea development + evidence box

Below is copy-ready textbook prose with parenthetical APA citations and suggested callout boxes.

Peer Review: What Happens After Research Is Written?

After researchers finish a project, they usually write a formal report called a manuscript. A manuscript explains the research question, what earlier studies have already found, how the researcher collected or analyzed evidence, what the researcher found, and what those findings might mean. At this point, the research is not automatically "official." It still needs to be evaluated by other scholars before it becomes part of the peer-reviewed research record (Tumin & Tobias, 2019; Horbach & Halfman, 2018).

The researcher then submits the manuscript to an academic journal. A journal is a publication that specializes in a particular field or topic. For example, one journal might publish research on interpersonal communication, another might publish research on psychology, and another might publish research on education. The journal editor first checks whether the manuscript fits the journal's topic, follows the journal's rules, and seems strong enough to send out for review. If it clearly does not fit, the editor may reject it before review. This is sometimes called a desk rejection (Tumin & Tobias, 2019).

Academic journals are different from many magazines, news sites, or social media platforms because their basic purpose is not to attract advertisers or generate clicks. In most fields, journals are supposed to publish work because it meets scholarly standards, not because it pleases advertisers. That does not mean journals are free from financial problems or bias. It means the main business model is usually built around subscriptions, library packages, publisher agreements, or publication fees rather than advertising revenue (Bergstrom, 2001; Larivière et al., 2015; Shu et al., 2018).

If the editor thinks the manuscript is a possible fit, the editor sends it to expert reviewers. For this class, imagine the editor sends it to three experts in the field. These reviewers are called peers because they have expertise in the same area as the author. They read the manuscript carefully and ask questions like these: Is the research question important? Does the study connect to existing scholarship? Are the methods appropriate? Do the findings support the claims? Is the writing clear enough for readers to follow? What needs to change before this work should be published? (Bormmann, 2011; Tumin & Tobias, 2019).

The reviewers do not usually make the final decision. Instead, they send comments and recommendations back to the editor. The editor reads the reviewers' reports, compares their advice, and makes the decision. The three basic recommendations students should know are reject, accept with minor revisions, and revise and resubmit (Tumin & Tobias, 2019).

Reject means the journal will not publish the manuscript. A rejection does not always mean the project is worthless. It might mean the manuscript does not fit that journal, the evidence is not strong enough, the argument needs major rethinking, or the study has a problem that cannot be fixed in revision. Researchers often revise rejected manuscripts and send them to a different journal (Tumin & Tobias, 2019; Smith, 2006).

Accept with minor revisions means the journal is very close to publishing the article, but the author still needs to make small changes. These changes might include clarifying a paragraph, fixing a table, adding a citation, tightening the conclusion, or correcting formatting. This is usually a good sign, but the author still has to respond carefully to the editor and reviewers (Tumin & Tobias, 2019).

Revise and resubmit means the journal sees potential in the manuscript, but the author must make bigger changes before the editor will decide whether to publish it. This is not a final acceptance. The author revises the manuscript, writes a response explaining how each reviewer comment was addressed, and sends the new version back to the journal. The revised manuscript may go back to the same reviewers for another round. Among manuscripts that are not rejected outright and remain under consideration, some form of revision is much more common than immediate acceptance; acceptance with no revision is rare (Huisman & Smits, 2017; Tumin & Tobias, 2019).

Peer review is considered highly credible because it puts research through several layers of expert judgment before publication. A single person does not simply declare, "My study is true." Instead, other experts inspect the logic, evidence, methods, citations, and conclusions. The author often has to revise the work in response to criticism. This process does not make research perfect, but it does make peer-reviewed research much more credible than unsupported claims, opinion posts, or "research says" statements that do not show where the research came from (Bormmann, 2011; Horbach & Halfman, 2018; Smith, 2006).

Peer review is also credible because it is part of a larger scholarly conversation. Researchers are expected to explain how their work connects to earlier studies, what evidence supports their claims, and what limits their study has. Other researchers can later challenge, extend, replicate, or correct the work. In that sense, peer review is not a magic stamp that says "true forever." It is a quality-control process that helps knowledge become more careful, public, and revisable (Bormmann, 2011; Horbach & Halfman, 2018).

[CALLOUT BOX: THE BIG IDEA]

Peer review does not mean "perfect." It means "checked by experts before publication." That makes peer-reviewed research one of the strongest kinds of evidence students can use, but it should still be read carefully.

Criticism 1: Access and Publisher Profit

One major criticism of academic publishing is accessibility. Even though scholarly journals are usually not built around advertising, many of them do make money by selling expensive subscriptions to university libraries, institutions, and databases. This creates a strange situation: researchers often do the study, write the article, submit it, and review other people's articles without being paid by the publisher, while commercial publishers can sell access back to universities and readers (Bergstrom, 2001; Larivière et al., 2015).

This is especially important in the social sciences, where a small number of large publishers control a large share of journal publishing. Larivière, Haustein, and Mongeon found that the top five publishers accounted for a particularly high share of social science papers, showing how concentrated the publishing system has become (Larivière et al., 2015). That concentration matters because people who are not enrolled in a college or employed by a university may lose access to research when they leave school. They may click on an article and find that reading one study costs more than they can reasonably pay (Shu et al., 2018).

This means peer-reviewed research can be credible and difficult to access at the same time. That is the contradiction. The process is designed to protect quality, but the publishing system can still keep high-quality research behind paywalls (Bergstrom, 2001; Shu et al., 2018).

[CALLOUT BOX: WHERE CAN I FIND PEER-REVIEWED RESEARCH AFTER I LEAVE SCHOOL?]

After you leave college, try these options before paying for an article:

- 1. Search the article title in Google Scholar.
- 2. Try Unpaywall, which helps locate legal open-access copies of scholarly articles.
- 3. Search the Directory of Open Access Journals, or DOAJ, for open-access journals and articles.
- 4. Search PubMed or PubMed Central for health, medicine, psychology, and life-science topics.
- 5. Check a public library. Some public libraries provide access to research databases.
- 6. Email the author politely. Many researchers are allowed to share a copy of their article.

DOAJ describes itself as an index of open access journals committed to making quality content openly available, and Unpaywall describes its database as a way to locate free scholarly articles. ([Directory of Open Access Journals](#)) ([Unpaywall](#)) PubMed Central is also a free full-text archive for biomedical and life-sciences journal literature. ([PMC](#))

Criticism 2: Peer Review Can Still Be Biased

Another criticism is that peer review is not always as blind as it claims to be. In single-blind review, the reviewers know who the author is, but the author does not know who the reviewers are. In double-blind review, the author and reviewers are supposed to be hidden from each other. Double-blind review can help reduce some forms of bias, but it is not a perfect shield. Reviewers may still guess the author's identity from the topic, writing style, citations, methods, institution, conference presentations, or earlier versions of the paper (Blank, 1991; Lee et al., 2013; Tomkins et al., 2017).

Bias can show up in several ways. Reviewers may be influenced by an author's university, status, country, gender, theoretical approach, method, or reputation. Research on peer review has found that reviewer bias is a real concern across scholarly fields, even when reviewers are trying to be fair (Lee et al., 2013; Tomkins et al., 2017).

Peer review can also involve gatekeeping. Gatekeeping means deciding what gets admitted into the scholarly conversation and what gets kept out. Some gatekeeping is necessary because journals cannot publish everything. But gatekeeping becomes a problem when unusual methods, new theories, interdisciplinary work, or ideas that challenge dominant assumptions are treated as "too risky" simply because they do not look familiar. Studies of peer review suggest that work challenging established perspectives can face especially intense criticism and revision demands (Siler et al., 2015; Siler & Strang, 2017).

This does not mean peer review is useless. It means peer review is a human process. Humans can be careful, generous, skeptical, defensive, rushed, biased, insightful, or wrong. Peer review is valuable because it creates expert scrutiny, but it still needs transparency, accountability, and ongoing improvement (Smith, 2006; Tennant et al., 2017).

Criticism 3: Predatory Journals

A predatory journal is a publication that pretends to be a legitimate scholarly journal but does not provide real scholarly quality control. Predatory journals often claim to be peer reviewed, but their real goal is to collect money from authors. They may charge publication fees, promise unrealistically fast review, send flattering spam emails, list fake editors, invent fake impact factors, or publish almost anything as long as the author pays (Eriksson & Helgesson, 2017; Shamseer et al., 2017).

Predatory journals are dangerous because they imitate the appearance of real scholarship. A website may look academic. The journal title may sound official. The article may have an abstract, keywords, citations, and a PDF. But if the journal is not actually reviewing the work, then the article has not gone through the process that makes peer-reviewed research credible (Eriksson & Helgesson, 2017; Shamseer et al., 2017).

It is important not to confuse predatory journals with open-access journals. Open access simply means readers can access the article without a paywall. Many open-access journals are legitimate, rigorous, and peer reviewed. The problem is not openness. The problem is deception: pretending to provide peer review while really selling publication space (Piwowar et al., 2018; Shamseer et al., 2017).

[CALLOUT BOX: HOW DO I KNOW WHETHER A JOURNAL IS CREDIBLE?]

Ask these questions before trusting a journal:

1. Is the journal indexed in a respected database for its field?
2. Does the journal clearly explain its peer-review process?
3. Are the editor and editorial board real scholars with university or research affiliations?
4. Are publication fees explained clearly before submission?
5. Does the journal publish articles that actually match its stated topic?
6. Does the timeline sound realistic, or does it promise acceptance in a few days?
7. Does the email invitation sound spammy, overly flattering, or urgent?
8. Does the journal claim a suspicious "impact factor" that is not from a recognized source?
9. Can you find the journal in DOAJ if it is open access?
10. Would a librarian, professor, or experienced researcher recognize it as credible?

A single warning sign does not automatically prove a journal is predatory, but several warning signs together should make you slow down and investigate (Eriksson & Helgesson, 2017; Shamseer et al., 2017; Strinzel et al., 2019).

How Researchers Are Responding

Researchers, librarians, editors, and publishers are trying to fix these problems in several ways. Open-access publishing and open repositories make more research available to people outside universities, although open access can create new problems when publication fees shift costs onto authors (Piwowar et al., 2018; Tennant et al., 2016). Some journals are experimenting with more transparent forms of peer review, including open peer review, published reviewer reports, and clearer conflict-of-interest policies (Horbach & Halfman, 2018; Tennant et al., 2017). Retraction tracking also helps correct the scholarly record when published research turns out to be unreliable; Crossref now makes Retraction Watch data publicly available and says the database is updated every working day. (www.crossref.org) Researchers are also building and comparing lists of credible and predatory journals so authors and readers can better evaluate where research is published (Strinzel et al., 2019). None of these solutions is perfect. But together, they show that scholarship is not just a pile of published articles. It is an ongoing system of checking, correcting, debating, and improving what we think we know.

How the Process Works

1. **Submission & Screening:** A researcher submits an article to a journal. The journal's editor first reviews it to see if it fits the publication's focus.
2. **Expert Review:** If it passes the initial screening, the editor sends it to independent experts (referees) in the same field. These reviewers evaluate the study's methods, evidence, and logic to ensure the researchers are not making absolute, conclusive claims beyond what their data supports. Reviews are typically single-blind (reviewers know the author) or double-blind (neither knows the other).
3. **Recommendation:** Reviewers recommend whether to publish, revise, or reject the study. They often force authors to add limitations and soften overly broad claims so the research is not treated as a universal rule.

Criticisms of the Process While peer review improves clarity and checks for obvious methodological errors, it has significant structural flaws (Smith, 2006):

- **Missed Errors:** Reviewers are busy, unpaid volunteers who rarely have the time to re-analyze the raw data. Because of this, they can easily miss hidden coding errors or fabricated data.
- **Prestige Bias:** The process is vulnerable to bias; reviewers may subconsciously favor submissions from famous researchers or elite universities over unknown scholars (Lee et al., 2013).
- **System Overload:** In the social sciences, the modern peer-review system did not emerge from a flawless philosophical design; it evolved as a logistical necessity to manage overwhelming submission volumes and prevent reviewer burnout (Merriman, 2021).

Ultimately, when a student sees an article is "peer-reviewed," it simply means the paper survived an initial quality-control filter and entered the ongoing scholarly conversation—not that its findings are a permanent, unquestionable truth.

References

Bergstrom, T. C. (2001). Free labor for costly journals? *Journal of Economic Perspectives*, 15(4), 183–198. doi: 10.1257/jep.15.4.183

Blank, R. M. (1991). The effects of double-blind versus single-blind reviewing: Experimental evidence from *The American Economic Review*. *The American Economic Review*, 81(5), 1041–1067.

Bormmann, L. (2011). Scientific peer review. *Annual Review of Information Science and Technology*, 45(1), 197–245. doi: 10.1002/aris.2011.1440450112

Eriksson, S., & Helgesson, G. (2017). The false academy: Predatory publishing in science and bioethics. *Medicine, Health Care and Philosophy*, 20, 163–170. doi: 10.1007/s11019-016-9740-3

Fang, F. C., & Casadevall, A. (2011). Retracted science and the retraction index. *Infection and Immunity*, 79(10), 3855–3859. doi: 10.1128/IAI.05661-11

Horbach, S. P. J. M., & Halfman, W. (2018). The changing forms and expectations of peer review. *Research Integrity and Peer Review*, 3, Article 8. doi: 10.1186/s41073-018-0051-5

Huisman, J., & Smits, J. (2017). Duration and quality of the peer review process: The author’s perspective. *Scientometrics*, 113, 633–650. doi: 10.1007/s11192-017-2310-5

Larivière, V., Haustein, S., & Mongeon, P. (2015). The oligopoly of academic publishers in the digital era. *PLOS ONE*, 10(6), e0127502. doi: 10.1371/journal.pone.0127502

Lee, C. J., Sugimoto, C. R., Zhang, G., & Cronin, B. (2013). Bias in peer review. *Journal of the American Society for Information Science and Technology*, 64(1), 2–17. doi: 10.1002/asi.22784

Piwowar, H., Priem, J., Larivière, V., Alperin, J. P., Matthias, L., Norlander, B., Farley, A., West, J., & Haustein, S. (2018). The state of OA: A large-scale analysis of the prevalence and impact of open access articles. *PeerJ*, 6, e4375. doi: 10.7717/peerj.4375

Shamseer, L., Moher, D., Maduekw, O., Turner, L., Barbour, V., Burch, R., Clark, J., Galipeau, J., Roberts, J., & Shea, B. J. (2017). Potential predatory and legitimate biomedical journals: Can you tell the difference? A cross-sectional comparison. *BMC Medicine*, 15, Article 28. doi: 10.1186/s12916-017-0785-9

Shu, F., Mongeon, P., Haustein, S., Siler, K., Alperin, J. P., & Larivière, V. (2018). Is it such a big deal? On the cost of journal use in the digital era. *College & Research Libraries*, 79(6), 785–798. doi: 10.5860/crl.79.6.785

Siler, K., Lee, K., & Bero, L. (2015). Measuring the effectiveness of scientific gatekeeping. *Proceedings of the National Academy of Sciences*, 112(2), 360–365. doi: 10.1073/pnas.1418218112

Siler, K., & Strang, D. (2017). Peer review and scholarly originality: Let 1,000 flowers bloom, but don’t step on any. *Science, Technology, & Human Values*, 42(1), 29–61. doi: 10.1177/0162243916656919

Smith, R. (2006). Peer review: A flawed process at the heart of science and journals. *Journal of the Royal Society of Medicine*, 99(4), 178–182. doi: 10.1177/014107680609900414

Strinzel, M., Severin, A., Milzow, K., & Egger, M. (2019). Blacklists and whitelists to tackle predatory publishing: A cross-sectional comparison and thematic analysis. *mBio*, 10(3), e00411-19. doi: 10.1128/mBio.00411-19

Tennant, J. P., Waldner, F., Jacques, D. C., Masuzzo, P., Collister, L. B., & Hartgerink, C. H. J. (2016). The academic, economic and societal impacts of open access: An evidence-based review. *F1000Research*, 5, 632. doi: 10.12688/f1000research.8460.3

Tennant, J. P., Dugan, J. M., Graziotin, D., Jacques, D. C., Waldner, F., Mietchen, D., Elkhatib, Y., Collister, L. B., Pikas, C. K., Crick, T., Masuzzo, P., Caravaggi, A., Berg, D. R., Niemeyer, K. E., Ross-Hellauer, T., Mannheimer, S., Rigling, L., Katz, D. S., Greshake Tzovaras, B., ... Colomb, J. (2017). A multi-disciplinary perspective on emergent and future innovations in peer review. *F1000Research*, 6, 1151. doi: 10.12688/f1000research.12037.3

Tomkins, A., Zhang, M., & Heavlin, W. D. (2017). Reviewer bias in single- versus double-blind peer review. *Proceedings of the National Academy of Sciences*, 114(48), 12708–12713. doi: 10.1073/pnas.1707323114

Tumin, D., & Tobias, J. D. (2019). The peer review process. *Saudi Journal of Anaesthesia*, 13(Suppl. 1), S52–S58. doi: 10.4103/sja.SJA_544_18Peer review is an academic quality-control system designed to screen, critique, and revise research before it is published (Burnham, 1990). For people just learning about research, it is crucial to understand that peer review is not a magical stamp of absolute truth, but an evolving, imperfect human process.

Burnham, J. C. (1990). The evolution of editorial peer review. *JAMA*, 263(10), 1323–1329. https://doi.org/10.1001/jama.1990.03440100023003

Lee, C. J., Sugimoto, C. R., Zhang, G., & Cronin, B. (2013). Bias in peer review. *Journal of the Association for Information Science and Technology*, 64(1), 2–17. https://doi.org/10.1002/asi.22784

Merriman, B. (2021). Peer review as an evolving response to organizational constraint: Evidence from sociology journals, 1952–2018. *The American Sociologist*, 52(2), 341–366. https://doi.org/10.1007/s12108-020-09473-x

Smith, R. (2006). Peer review: A flawed process at the heart of science and journals. *Journal of the Royal Society of Medicine*, 99(4), 178–182. https://pmc.ncbi.nlm.nih.gov/articles/PMC1420798/

H1: Overview of [Topic] / Overview Main Idea 3

Main idea box

Drafting Note:” For this chapter, I would present research types as **different ways of making sense**, not as a ladder from “better” to “worse.” That fits the chapter’s larger argument: students should not treat one study as a magic answer machine; they should learn to ask what a study can help them understand, what it cannot tell them, and how its method shapes the kind of truth it can offer. Your notes already frame the big shift this way: the common-sense approach treats research as truth transferred from researchers to readers, while the (un)common-sense approach treats research as sense-making with partial, contextual, and local evidence.

Types of Peer-Reviewed Communication Research

Where peer-reviewed communication research is published

Peer-reviewed communication research is most often published in **academic journals**. Major communication organizations publish or sponsor many of these journals. The National Communication Association publishes 12 scholarly journals, and the International Communication Association publishes six major peer-reviewed journals. Examples include *Communication Monographs*, *Human Communication Research*, *Journal of Communication*, *Communication Theory*, *Quarterly Journal of Speech*, *Journal of Applied Communication Research*, *Communication Education*, *Text and Performance Quarterly*, *Review of Communication*, and *Annals of the International Communication Association*.

Communication research also appears in **edited volumes**, especially scholarly handbooks published by academic presses such as Oxford, Routledge, SAGE, and Wiley/Blackwell. These chapters are often written by experts and reviewed by editors, and some volumes use external peer review, but the review process can vary more than it does for journal articles. Students should not automatically assume every book chapter is peer-reviewed in the same way as a journal article. The safe question is: **What was the review process for this publication?** Examples include Oxford and Routledge handbooks that gather state-of-the-field chapters by communication scholars.

Conference papers may also be reviewed by conference divisions or interest groups, but they are usually not treated as the same thing as a final peer-reviewed journal article. A conference paper is often an important step in the research process, but students should check whether the work was later published in a peer-reviewed journal or edited volume.

1. Empirical research articles

What they are

An **empirical research article** reports original evidence collected or analyzed by the researchers. The authors do not only argue from existing scholarship; they gather or analyze data. In communication, that data might come from surveys, experiments, interviews, observations, texts, social media posts, videos, organizational documents, speeches, conversations, field notes, or archival materials.

What they accomplish

Empirical articles help answer questions like:

- What is happening?
- How often does this happen?
- What patterns appear?
- How do people experience this?
- What meanings do people make?
- What variables are related?
- What communication practices shape an outcome?
- How does communication work in a particular context?

Example

A study might examine how college roommates talk about conflict, how couples negotiate privacy on social media, how public health messages affect vaccine attitudes, or how managers use feedback in workplace meetings.

Where they are published

Empirical research appears across many communication journals. *Human Communication Research*, for example, describes itself as emphasizing theory-driven empirical work and methods for observing and measuring communication behavior.

2. Quantitative empirical articles

What they are

Quantitative research uses numerical data. Researchers might count, measure, compare, model, or statistically test relationships among variables. Quantitative communication research often uses surveys, experiments, content analysis, computational methods, network analysis, or statistical modeling.

What they accomplish

Quantitative studies are useful when researchers want to examine:

- how much
- how often
- how strongly two things are related
- whether one group differs from another
- whether a communication message changes an outcome
- whether a pattern appears across a large number of people, texts, or cases

Examples

A quantitative article might ask:

- Do people who use more affectionate communication report higher relationship satisfaction?
- Does an apology with responsibility-taking produce more forgiveness than an apology without it?
- How often do mental health TikToks mention professional treatment?
- Are certain kinds of social support messages associated with lower stress?

Where they are published

Quantitative studies are common in journals such as *Human Communication Research*, *Journal of Communication*, *Communication Research*, *Communication Monographs*, *Journal of Applied Communication Research*, *Health Communication*, and *Communication Methods and Measures*. *Communication Methods and Measures* specifically focuses on methodological developments, both qualitative and quantitative, for communication scholars.

Student takeaway

Quantitative research is powerful, but numbers do not automatically mean universal truth. Students should ask what was measured, how it was measured, who was included, what comparison was made, and what the numbers can and cannot support.

3. Qualitative empirical articles

What they are

Qualitative research studies meanings, experiences, practices, relationships, identities, interpretations, contexts, and processes. Instead of primarily asking “How many?” qualitative research often asks “How?” “Why?” “What does this mean?” or “How do people make sense of this?”

Qualitative communication researchers may use interviews, focus groups, participant observation, field notes, textual analysis, discourse analysis, conversation analysis, grounded theory, thematic analysis, narrative analysis, or visual analysis.

What they accomplish

Qualitative studies help students understand:

- how people experience communication
- how people interpret situations
- how meanings are created
- how power operates in everyday interaction
- how relationships, identities, and institutions are made through communication
- how context changes what something means

Examples

A qualitative article might ask:

- How do first-generation college students interpret feedback from professors?
- How do families talk about grief after a death?
- How do coworkers make sense of burnout in a high-stress workplace?
- How do patients describe feeling dismissed in medical appointments?
- How do people narrate friendship breakups?

Where they are published

Qualitative research appears in many communication journals, including *Qualitative Research Reports in Communication*, *Journal of Applied Communication Research*, *Communication Monographs*, *Communication and Critical/Cultural Studies*, *Critical Studies in Media Communication*, and organizational, health, interpersonal, intercultural, and media communication journals. *Qualitative Research Reports in Communication* is a peer-reviewed journal specifically devoted to brief qualitative and critical research essays about human communication.

Student takeaway

Qualitative research is not “just stories.” Strong qualitative research has rigor, evidence, method, analysis, and standards of quality. Tracy’s “big-tent” criteria for excellent qualitative research include worthy topic, rich rigor, sincerity, credibility, resonance, significant contribution, ethics, and meaningful coherence.

4. Ethnographies

What they are

Ethnography is a qualitative research approach where researchers study communication in a cultural or social setting, often through observation, participation, interviews, field notes, and immersion over time.

Ethnography is especially useful when communication cannot be fully understood outside the place, community, organization, or culture where it happens.

What they accomplish

Ethnographies help researchers understand:

- everyday practices
- group norms
- rituals
- insider meanings
- workplace cultures
- community life
- power relations
- how communication feels and functions in context

Examples

An ethnography might study:

- how nurses coordinate care during shift changes
- how a crisis hotline trains volunteers to listen
- how a fandom community creates belonging online
- how a classroom culture shapes who feels comfortable speaking
- how a workplace handles conflict without officially naming it

Where they are published

Ethnographies can appear in qualitative communication journals, applied communication journals, organizational communication journals, critical/cultural journals, and edited volumes. They may also appear in interdisciplinary journals depending on the topic.

Student takeaway

Ethnography helps students see that communication is not just what people say. It is also what people do repeatedly, what a community treats as normal, what is rewarded, what is punished, and what becomes invisible because “that is just how we do things here.”

5. Autoethnographies

What they are

Autoethnography is a qualitative method where researchers use their own lived experience as a site of inquiry and connect that experience to larger cultural, relational, institutional, or political meanings. It is not simply personal storytelling. It is self-reflexive research that links the personal to the cultural.

What they accomplish

Autoethnography can make visible:

- embodied experience
- grief
- trauma
- illness
- disability
- race
- gender
- sexuality
- class
- family history
- academic life
- professional identity
- experiences that are difficult to capture through surveys or experiments

Examples

An autoethnography might examine:

- navigating ADHD and dyslexia in graduate school
- being a caregiver in a medical system
- growing up in a family where conflict was never named
- experiencing racism in classroom communication
- learning to communicate after loss

Where they are published

Autoethnographies may appear in journals such as *Journal of Autoethnography*, *Qualitative Inquiry*, *Text and Performance Quarterly*, *Cultural Studies ↔ Critical Methodologies*, and some communication journals or edited volumes. The *Journal of Autoethnography* describes itself as a peer-reviewed/refereed interdisciplinary journal devoted to autoethnography as a method of inquiry.

Student takeaway

Autoethnography is valuable because some communication knowledge begins in lived experience. The key is that the researcher does not stop at “this happened to me.” They ask what the experience reveals about culture, identity, power, institutions, relationships, or meaning.

6. Mixed-methods articles

What they are

Mixed-methods research combines quantitative and qualitative methods in one study or research project. For example, researchers might survey 500 people and then interview 30 of them, or they might analyze social media posts statistically and then closely interpret a smaller set of posts.

What they accomplish

Mixed-methods research can offer both breadth and depth.

It can help researchers understand:

- how common a pattern is
- what that pattern means to people
- whether numerical trends match lived experience
- where statistical findings need more explanation
- how large-scale patterns and local meanings connect

Examples

A mixed-methods study might ask:

- How common is workplace burnout among communication interns, and how do interns describe the communication practices that make burnout worse?
- How often do influencers cite “research,” and what kind of authority are they constructing when they do?

- Do students report anxiety about public speaking, and how do they narrate that anxiety in interviews?

Where they are published

Mixed-methods work can appear in many communication journals, including empirical journals, applied journals, health communication journals, media journals, and edited volumes.

Student takeaway

Mixed-methods research is useful because communication often needs both pattern and meaning. The number can show that something is happening; the qualitative material can help explain how it is happening and why it matters.

7. Rhetorical, critical, textual, and critical-cultural research

What it is

Not all communication research studies people through surveys, interviews, or experiments. Some communication scholars study **texts, discourse, symbols, performances, images, speeches, media, policies, archives, public controversies, or cultural artifacts**.

Rhetorical and critical-cultural research often asks how communication creates meaning, power, identity, public memory, ideology, norms, resistance, or social reality.

What it accomplishes

This kind of research helps students understand:

- how language shapes public belief
- how media frames groups or problems
- how institutions justify power
- how identities are represented
- how public arguments become persuasive
- how histories are remembered or erased
- how “common sense” gets built through discourse

Examples

A rhetorical or critical article might analyze:

- how political speeches construct immigration as a threat
- how a university talks about diversity after a crisis
- how “civility” discourse can silence marginalized anger
- how true-crime media frames victims and suspects
- how public health campaigns imagine responsibility

Where it is published

This work appears in journals such as *Quarterly Journal of Speech*, *Communication and Critical/Cultural Studies*, *Critical Studies in Media Communication*, *Text and Performance Quarterly*, *Communication and Race*, and edited volumes. *Quarterly Journal of Speech* is a peer-reviewed NCA journal that publishes original scholarship and book reviews, and *Text and Performance Quarterly* is a peer-reviewed NCA journal that publishes work on performance as social, critical, and communicative practice.

Student takeaway

This research reminds students that communication does not only transmit ideas. Communication also builds worlds: norms, categories, identities, institutions, and histories.

8. Theory articles

What they are

A **theory article** develops, revises, critiques, or extends a concept, model, or theory. It may not collect new data. Instead, it offers a new way of understanding communication.

Theory articles are not “less real” because they do not report original data. They help build the frameworks that guide future research.

What they accomplish

Theory articles help scholars:

- define important concepts
- explain relationships among concepts
- challenge old assumptions
- build new frameworks
- clarify what researchers should study next
- make sense of findings across different studies

Examples

A theory article might:

- propose a new model of relational repair
- revise a theory of self-disclosure for social media contexts
- critique the idea that communication is mainly information transfer
- develop a framework for understanding AI-mediated intimacy
- rethink listening as relational labor rather than passive reception

Where they are published

Theory articles often appear in *Communication Theory*, *Communication Monographs*, *Journal of Communication*, *Review of Communication*, *Annals of the International Communication Association*, and edited volumes. *Communication Theory* is an ICA journal focused on theoretical development in communication, while *Annals of the International Communication Association* publishes state-of-the-discipline reviews, theoretical articles, critical essays, and review articles.

Student takeaway

Theory gives students better questions. It helps them notice things they may not have had language for yet.

9. Literature reviews

What they are

A **literature review** synthesizes existing research on a topic. It does not usually collect new data from participants. Instead, it reads across a body of scholarship to explain what researchers have found, where they disagree, what questions remain, and what future research should do.

There are different kinds of literature reviews:

- Narrative reviews
- Integrative reviews
- Systematic reviews
- Scoping reviews
- State-of-the-field essays

What they accomplish

Literature reviews help students and scholars understand:

- what is already known
- what debates exist
- what patterns appear across studies
- what gaps remain
- what theories are commonly used
- what future research should examine

Examples

A literature review might ask:

- What do we know about social media and loneliness?
- How has communication research studied apology?
- What does research say about family communication during illness?
- How have scholars defined misinformation in health communication?
- What theories explain online self-disclosure?

Where they are published

Literature reviews appear in journals such as *Annals of the International Communication Association*, *Review of Communication*, *Journal of Communication*, *Communication Theory*, and edited handbooks. *Annals of the International Communication Association* specifically publishes state-of-the-discipline literature reviews and essays.

Student takeaway

A literature review is useful because one study rarely settles a question. Reviews help students see the conversation around the study.

10. Systematic reviews and scoping reviews

What they are

A **systematic review** uses a transparent, planned process to search for, select, evaluate, and synthesize research on a focused question.

A **scoping review** maps the range of research on a broader topic, often to see what has been studied, what methods have been used, and where gaps exist.

What they accomplish

Systematic and scoping reviews help researchers:

- avoid cherry-picking studies
- make search strategies visible
- identify patterns across research
- locate gaps in knowledge
- clarify how a field has approached a topic
- show where evidence is strong, mixed, or missing

Examples

A systematic review might ask:

- What interventions reduce communication apprehension?
- What does research show about doctor-patient communication and trust?
- What are the effects of social support messages on stress?

A scoping review might ask:

- How has communication research studied TikTok?
- What methods have scholars used to study AI-mediated communication?
- What topics dominate research on remote work communication?

Where they are published

Systematic and scoping reviews may appear in empirical journals, review journals, health communication journals, interdisciplinary media journals, and edited volumes.

Student takeaway

These reviews are helpful because they show students that responsible research reading is not only about finding one study. It is about understanding the body of evidence.

11. Meta-analyses

What they are

A **meta-analysis** is a quantitative review of many studies. Instead of only summarizing what each study says, the researchers statistically combine findings to estimate an overall pattern, relationship, or effect.

What they accomplish

Meta-analyses help answer questions like:

- Across many studies, how strong is this effect?
- Are findings consistent or mixed?
- What conditions make the effect stronger or weaker?
- Does the pattern vary by method, population, context, or measurement?

Examples

A meta-analysis might examine:

- how accurate people are at detecting deception
- whether fear appeals persuade people to change health behavior
- how social support relates to stress
- whether communication accommodation improves intergroup attitudes
- whether media exposure is associated with body image concerns

Where they are published

Meta-analyses may appear in *Annals of the International Communication Association*, *Journal of Communication*, *Human Communication Research*, *Communication Monographs*, *Communication Research*, and other empirical or review journals. *Annals of the International Communication Association* notes that it encourages meta-analytic submissions, while not accepting single-case empirical studies.

Student takeaway

A meta-analysis can be powerful because it looks across many studies, but it still depends on which studies were included, how variables were measured, and whether the research being combined actually belongs together.

12. Methodological articles

What they are

A **methodological article** focuses on how research should be done. It may introduce a new method, improve a measurement tool, critique a common method, compare approaches, or offer guidance for analysis.

What they accomplish

Methodological articles help researchers:

- measure communication more carefully
- design better studies
- code messages more reliably
- improve interviews or observations
- use computational tools responsibly
- think through ethics
- avoid weak claims

Examples

A methodological article might:

- develop a new scale for communication anxiety
- compare human coding and AI-assisted coding of social media posts
- explain how to conduct ethical research with vulnerable communities
- critique overreliance on self-report surveys
- propose better ways to study relational repair

Where they are published

Methodological articles appear in journals such as *Communication Methods and Measures*, *Human Communication Research*, *Journal of Communication*, *Communication Monographs*, and edited methods volumes. *Communication Methods and Measures* is specifically designed to bring methodological developments, qualitative and quantitative, to communication scholars.

Student takeaway

Method articles remind students that research findings do not appear from nowhere. Methods shape what researchers can see.

13. Communication pedagogy research

What it is

Communication pedagogy research studies teaching and learning in communication contexts. It may examine public speaking classes, group projects, online learning, basic course instruction, debate, discussion, classroom climate, or assessment.

What it accomplishes

Communication pedagogy research helps instructors and students understand:

- how people learn communication
- what teaching practices help or harm students
- how classroom communication shapes belonging
- how assignments affect anxiety, confidence, or skill development
- how communication concepts transfer into everyday life

Examples

A pedagogy article might ask:

- Does a low-stakes speaking activity reduce public speaking anxiety?
- How do students experience group projects?
- What feedback practices help students revise speeches?
- How does classroom discussion shape belonging for marginalized students?
- How can AI be used ethically in communication assignments?

Where it is published

Communication pedagogy research appears in *Communication Education*, *Communication Teacher*, *Journal of Communication Pedagogy*, and edited volumes on teaching communication. *Communication Education* is a peer-reviewed NCA journal, and *Journal of Communication Pedagogy* is a peer-reviewed journal sponsored by the Central States Communication Association.

Student takeaway

Pedagogy research matters because the communication classroom is also a communication context. How we teach communication shapes what students think communication is.

14. Book chapters in edited volumes

What they are

A **book chapter in an edited volume** is a scholarly chapter written by one or more authors and included in a book organized by editors. Edited volumes often gather experts around a topic, method, theory, or subfield.

What they accomplish

Edited volume chapters are useful for:

- broad overviews
- state-of-the-field arguments
- theoretical development
- methodological guidance
- historical context
- specialized topics
- emerging areas of research

Examples

A chapter might review:

- the history of interpersonal communication research
- theories of family communication
- methods for studying social media
- research on health misinformation
- critical approaches to organizational communication
- communication and race
- AI-mediated communication

Where they are published

Edited volumes appear through academic presses and series such as Oxford Handbooks, Routledge Handbooks in Communication Studies, SAGE Handbooks, and Wiley/Blackwell Handbooks.

Student takeaway

Edited volume chapters can be excellent scholarly sources, especially for background and synthesis. But students should check the publication process because "edited" and "peer-reviewed" are not always the same thing.

The Bias Toward Quantitative Research

What the bias looks like

There is often a bias, especially in popular and institutional understandings of research, that treats quantitative research as more "real," "objective," "scientific," or "rigorous" because it uses numbers, statistics, large samples, and claims that look generalizable.

This bias shows up when people assume:

- numbers are automatically more objective
- larger samples are always better
- qualitative research is "just anecdotal"
- interviews are less rigorous than surveys
- stories are less valuable than statistics
- generalizability is the only goal of research
- the best research is the research that predicts or proves

That bias fits the **common-sense approach to research** because numbers can look like certainty. They can seem clean, transferable, and universal. Your chapter notes already warn against that move: one of the key problems is treating a study as complete truth rather than partial, contextual evidence.

Why that bias is not helpful

The bias against qualitative research is problematic because it asks every kind of research to answer the same kind of question. But communication research does not only ask "How much?" or "How often?" It also asks "What does this mean?" "How do people experience this?" "How does power move through language?" "How do people make sense of conflict?" "What histories are shaping this interaction?"

A survey can tell us how many students report public speaking anxiety. Interviews can help us understand what that anxiety feels like, where it comes from, how students talk about it, and what classroom practices make it better or worse. Both kinds of research can be rigorous. They are just rigorous in different ways.

Why "sample size" is not the same across methods

One common mistake is judging qualitative research by quantitative standards. For example, a student might say, "They only interviewed 20 people, so the study is bad." But not all research is trying to statistically generalize to a large population. Some research is trying to understand meaning, process, experience, discourse, or context in depth. Your notes flag this exact issue: not all research is designed to be statistically generalizable, and sample size should be evaluated in relation to the method and purpose.

This does not mean sample size never matters. It means sample size matters differently depending on what the study is trying to do.

Why quantitative research is not bias-free

Quantitative research can be excellent, but it is not automatically neutral. Researchers still choose what to measure, how to define variables, what questions to ask, what sample to recruit, what statistical model to use, and how to interpret the results. A number can look clean while hiding many human decisions underneath it.

So the better lesson is not:

Quantitative research is biased, and qualitative research is better.

The better lesson is:

Every method lets us see some things clearly and makes other things harder to see.

Why qualitative research matters for communication

Qualitative research is especially important in interpersonal communication because so much communication depends on meaning, timing, history, identity, emotion, power, and context. Those things do not always show up well when reduced too quickly to variables.

Qualitative research can help us understand:

- what an apology means to someone
- how people narrate betrayal
- how families avoid difficult topics
- how workplace norms silence disagreement
- how students experience belonging
- how marginalized communities create support
- how people interpret care, conflict, or disrespect

Those are not side issues. They are central to communication.

Better chapter language

I would phrase the point this way:

Quantitative research is not better because it uses numbers. Qualitative research is not weaker because it uses words, stories, observations, or interpretation. The better question is not "Which method is more scientific?" The better question is "What kind of question is this study asking, and does the method fit that question?"

That phrasing protects you from sounding anti-quantitative while still clearly challenging the hierarchy.

A student-friendly method-matching guide

Use quantitative research when the question is:

- How many?
- How often?
- How strong is the relationship?
- Does this intervention change an outcome?
- Are two groups different?
- Can this pattern be measured across many cases?

Use qualitative research when the question is:

- What does this mean?
- How do people experience this?
- How do people interpret this situation?
- What practices shape this community?
- How does power show up in interaction?
- What does this look like in context?

Use ethnography when the question is:

- How does communication work inside this community, workplace, group, or culture?
- What norms, rituals, and practices shape everyday interaction?

Use autoethnography when the question is:

- What can lived experience reveal about a larger cultural, relational, or institutional pattern?

Use rhetorical or critical analysis when the question is:

- How does this text, speech, image, policy, or discourse shape public meaning, identity, power, or belief?

Use a literature review when the question is:

- What do scholars already know about this topic?
- What debates or gaps exist?

Use a meta-analysis when the question is:

- Across many studies, what is the overall pattern or effect?

Use a theory article when the question is:

- What framework, concept, or model helps us understand this problem better?

Use a methodological article when the question is:

- How should we study this more carefully?

The main takeaway for the chapter

The chapter should not teach students to rank research types from best to worst. It should teach them to ask:

What kind of question is this research asking?

What kind of evidence does this method produce?

What does this method help us see?

What might this method miss?

Where is this finding useful, and where might it not apply?

That keeps the section aligned with the (un)common-sense approach. Students learn that research does not become stronger by pretending all methods do the same thing. Research becomes stronger when we understand what each method can and cannot help us see.

Possible callout box

DON'T FLATTEN THIS: NUMBERS ARE NOT THE ONLY KIND OF RIGOR

Quantitative research can show patterns across many people, messages, or situations.

Qualitative research can show meaning, context, experience, power, and process.

Theory can give us better concepts.

Reviews can show us the larger conversation.

Meta-analyses can summarize patterns across many studies.

No method sees everything. Good research begins by matching the method to the question.

APA references for drafting

Annotated bibliography box

Association of College and Research Libraries. (2016). *Framework for information literacy for higher education*. Association of College and Research Libraries.

International Communication Association. (n.d.). *Journals*. International Communication Association.

National Communication Association. (n.d.). *NCA journals*. National Communication Association.

NCA/Taylor & Francis. (n.d.). *Communication Monographs: Aims and scope*.

Oxford University Press. (n.d.). *Communication Theory*.

Oxford University Press. (n.d.). *Human Communication Research: Publishing with Human Communication Research*.

Taylor & Francis. (n.d.). *Annals of the International Communication Association: Aims and scope*.

Taylor & Francis. (n.d.). *Communication Methods and Measures: Aims and scope*.

Taylor & Francis. (n.d.). *Qualitative Research Reports in Communication: Aims and scope*.

Taylor & Francis. (n.d.). *Quarterly Journal of Speech: Aims and scope*.

Tracy, S. J. (2010). Qualitative quality: Eight "big-tent" criteria for excellent qualitative research. *Qualitative Inquiry*, 16(10), 837–851. doi:10.1177/1077800410383121.

Callout idea box

a table of the different kinds of research.

H1: Approaching [Topic]

Section goal

Introduce the two approaches explicitly, answer the guiding questions for each approach, and help students see how common sense as knowing and (Un)Common Sense as understanding begin from different assumptions, myths or beliefs, behaviors, and outcomes.

Common Sense definition of perception

From the common-sense approach, research is a method of knowing. It produces findings, packages them into reports, and transfers expert knowledge from researchers to readers.

In this view, the research article is like a container. The researchers discovered something. They wrote it down. The reader opens the article, extracts the finding, remembers it, cites it, and applies it.

Drafting Note: You don't have to say this but it is important to keep in mind. This definition directly extends the Chapter 1 common-sense communication model. Chapter 1 explains that the common-sense approach treats communication as the transfer of information, while the (un)commonsense approach treats communicating as messy, collaborative, material, and situated sense-making. When this transfer model gets applied to research, the research report becomes the message, the researcher becomes the sender, the reader becomes the receiver, and the finding becomes the information being transferred. Your analyzed voice notes name this directly: the common-sense approach treats research reports as containers of expert findings that transfer knowledge to readers.

Thorough description

The common-sense approach treats research as if its main purpose is to deliver a finding. It imagines reading research as a fairly simple process:

A study asks a question.
Researchers collect data.
The article reports the answer.
The reader learns the answer.
The reader can now use the answer.

That model is attractive because it makes research feel useful, clean, and powerful. It gives students something to do with a difficult article: find the point. It also gives them something to do in everyday arguments: cite the point.

This is why the opening story's "Actually, research shows..." fantasy matters. It is not just a joke about graduate-school arrogance. It reveals the pleasure of the common-sense approach. A citation can feel like authority. A study can feel like a weapon. A finding can feel like a rule.

The voice notes repeatedly connect this to online life: Reddit users asking for "Sources?," social media posts saying "research says," and people invoking studies as if the presence of a source settles the question. The problem is not asking for sources. The problem is expecting a source to end interpretation.

What this approach assumes

The common-sense approach previews three assumptions that the chapter will later unpack as discontinuous leaps.

Assumption 1: Research is for reporting

The reader's job is to report what the study found. This creates a reading habit: students look for the finding, conclusion, quote, or statistic that can be repeated. Your raw notes describe this as reading from top to bottom because the truth appears to be communicated in order.

This assumption leads into the first leap:

Reporting to Sense-Making

The problem is not reporting itself. Students do need to report accurately. The problem is when reporting replaces understanding.

Assumption 2: Research gives universal truths

The common-sense approach treats a single article as if it can contain the truth. The reader evaluates the article as true or false, right or wrong, useful or useless. Your voice notes describe this as monologic thinking: the research report becomes either all true or all false, which erases the complexity of good research.

This leads into the second leap:

Universal Truths to Local Truths

Assumption 3: Research should be applied as an algorithm

The common-sense approach wants research to tell us what to do. It treats findings like rules:

If research says X, do X, and Y should happen.

Your notes use the 5-to-1 ratio as a strong example. A common-sense reader might treat "five positive interactions for every negative one" as an exact relational repair formula. But that turns a research-informed pattern into a rigid rule. The rupture may be too large, the "positive" interaction may not feel positive to the other person, or the relationship history may change what the ratio means.

This leads into the third leap:

Summarizing to Synthesizing

A summary gives the rule. Synthesis asks what the rule depends on, where it applies, and how it needs to be adapted.

2. Why the Common-Sense Approach Feels Useful

The section should be careful here. Do not make the common-sense approach sound stupid. It is not.

The common-sense approach helps students:

- find evidence
- identify claims
- summarize articles
- cite sources
- value peer-reviewed research
- avoid relying only on opinion

This matters because the chapter should not sound anti-research. Chapter 0 already makes this move with knowledge: knowing is useful, but not enough. The knowing framework treats communication as information to collect, transfer, correct, or apply; the understanding framework treats communicating as contextual sense-making. Chapter 3 should make the same move with research.

A useful drafting line:

The common-sense approach is not wrong because it asks what a study found. It becomes too small when it thinks that finding is the whole point.

3. Why the Common-Sense Approach Becomes Dangerous

It turns partial findings into certainty

Your analyzed notes argue that misinformation often happens when research is processed through a knowing framework. Research becomes something to extract, memorize, repeat, or use as a conclusive answer; once a finding gets flattened into a slogan, tip, "research says" claim, or universal rule, it becomes brittle knowledge.

That is the key danger: the common-sense approach makes research feel more certain than it is.

This is not only a theoretical problem. The methodological literature supports the same caution. The ASA's p-value guidance, replication literature, causal inference literature, and open science reforms all push against treating one finding, one threshold, or one publication as settled truth. Your verified APA guide summarizes this well: p-values do not prove truth, peer review is screening rather than final certification, one strong study is not definitive, and qualitative rigor should be judged by method-appropriate standards.

It turns real research into misinformation

A major point for the draft: misinformation does not always begin with fake research.

Sometimes it begins with real research that gets stripped of context.

A careful finding becomes a catchy post.

A cautious claim becomes a rule.

A correlation becomes a cause.

A study with limitations becomes "science proves."

A partial truth becomes a myth.

Sumner et al. (2014) support this communication-chain argument. Their BMJ study examined exaggeration in health-related science news and academic press releases and found that exaggeration in news was strongly associated with exaggeration already present in press releases. That gives you a peer-reviewed example of how research can become distorted as it travels from study to public communication.

It shapes interpersonal judgment

This is where the chapter becomes clearly relevant to interpersonal communication.

Your misinformation voice notes emphasize that misinformation about social processes does not stay abstract. It shapes how people judge relationships, credibility, deception, mental health, intelligence, and everyday behavior.

The best example is nonverbal deception.

The voice notes describe communication charlatans who claim that eye movement, facial expression, or body posture can reveal lying. The danger is interpersonal: someone might watch a romantic partner’s eyes during a painful conversation and treat eye movement as evidence of betrayal.

Peer-reviewed deception research supports the caution. Bond and DePaulo’s meta-analysis synthesized 206 documents and 24,483 judges and found average lie-truth judgment accuracy of about 54%, only slightly above chance. DePaulo et al. (2003) also reviewed cues to deception across 1,338 estimates of 158 cues, which supports the broader point that simple cue-based lie detection is far less reliable than popular advice suggests.

It turns social media summaries into “research”

The ADHD TikTok example is especially useful for the chapter because it is concrete, current, and recognizable. Yeung et al. (2022) analyzed the 100 most popular TikTok videos about ADHD and classified 52% as misleading, 27% as personal experience, and 21% as useful.

This does not mean every ADHD TikTok is bad. It means students need research literacy because social media can make partial, personal, clinical, and research-based claims look equally authoritative.

A useful drafting line:

The problem is not that social media talks about research. The problem is that social media often rewards the version of research that sounds the most confident, portable, and emotionally useful.

It makes AI summaries feel like understanding

AI intensifies the common-sense approach because it makes information transfer feel effortless. The student asks a question, the system produces a clean answer, and the answer feels like understanding.

But automation bias research cautions against over-reliance on automated systems. Goddard et al. define automation bias as the tendency to over-rely on automation and review how people may fail to monitor automated outputs carefully. Logg et al. (2019) also show “algorithm appreciation”: across six experiments, people sometimes adhered more to advice when they believed it came from an algorithm rather than a person.

The chapter should phrase this carefully:

AI does not automatically make students less critical. But when students treat AI output as a substitute for reading, evaluating, and connecting evidence, AI can strengthen the common-sense fantasy that understanding can simply be delivered.

Chu-Ke and Dong (2024) support the AI-literacy connection, arguing that generative AI changes misinformation environments and requires users to critically evaluate AI-generated content. Long and Magerko (2020) define AI literacy around the competencies people need to understand, use, and critically evaluate AI technologies.

(Un)Common Sense definition of perception

From the (un)common-sense approach, research is a method of understanding. It helps us make sense of questions, patterns, problems, contexts, relationships, and claims by working with partial, contextual evidence.

This definition shifts research from a truth-transfer model to a sense-making model.

The article is not a magic answer machine.

The reader is not a passive receiver.

The finding is not a universal rule.

The study is one resource for understanding.

Your analyzed notes define this strongly: the (un)common-sense approach treats research reading as interaction. The reader interacts with the report in relation to existing understanding, other research, lived experience, and the context where the finding might be applied.

Thorough description

The (un)common-sense approach does not ask students to distrust research. It asks them to use research more responsibly.

Research becomes a way to ask:

- What does this study help me understand?
- What question were the researchers asking?
- Who or what did they study?
- What methods did they use?
- What does this finding explain well?
- What does it not explain?
- Where might it apply?
- Where might it not apply?
- What other research complicates it?
- How might this guide action without becoming a rigid rule?

This is where the chapter can connect research literacy to the larger textbook framework. Chapter 0 defines understanding as contextual sense-making: figuring out how facts, people, relationships, histories, and meanings connect while interaction is still unfolding. Chapter 3 can say research literacy is the information-world version of that same practice.

A useful drafting line:

Literacies are sense-making practices. They are not just what we know about information; they are what we can do when information shows up in a messy world.

This is well supported by your voice notes, which frame information literacy, scientific literacy, research literacy, data literacy, digital literacy, and AI literacy as forms of understanding rather than memorization. It is also supported by peer-reviewed literacy research. Mackey and Jacobson (2011) define metaliteracy as an expanded information-literacy framework for participatory digital environments where people produce, share, and evaluate information. Norris and Phillips (2003) argue that scientific literacy cannot be separated from reading, writing, interpreting, and evaluating scientific texts. Klucevsek (2017) argues that scientific information literacy should include understanding scholarship as conversation and publication as part of the scientific process.

5. What the (Un)Common-Sense Approach Assumes

Assumption 1: Research reading is sense-making, not reporting

The (un)common-sense approach assumes that students do not read research merely to report what the article says. They read to make sense of what the article helps them see.

That changes reading behavior. Your raw notes say that once students understand they are “making sense with the research article,” they are freed from reading it as passive listeners. They can move from abstract to discussion, to rationale, to conclusion, back to the literature review, and so on.

This previews the first discontinuous leap:

Reporting to Sense-Making

Drafting emphasis:

Reporting asks:

What did the study say?

Sense-making asks:

What does this study help me understand?

Assumption 2: Research produces local truths, not universal truths

The (un)common-sense approach assumes that a study can be true and limited at the same time.

A finding may be true:

- for this population
- in this setting
- using this method
- under these conditions
- for this question
- with these limits

That is what this chapter can call **local truth**.

Student-facing definition:

A local truth is a supported finding with its conditions still attached.

This is not relativism. It does not mean “anything can be true.” It means a finding becomes more usable when readers understand where it came from and how far it can responsibly travel.

Peer-reviewed sources support this move. Haraway’s concept of situated knowledges argues against the fantasy of a view from nowhere and emphasizes partial, located knowledge. Simons et al. (2017) propose “constraints on generality” statements so empirical articles identify the populations, settings, materials, and conditions to which findings are expected to apply. Rothwell (2005) asks the external-validity question directly: to whom do trial results apply? Deaton and Cartwright (2018) similarly caution that randomized controlled trials answer defined causal questions under assumptions, but results do not automatically travel to new populations or settings.

This previews the second discontinuous leap:

Universal Truths to Local Truths

Drafting emphasis:

Universal-truth reading asks:

Is this true?

Local-truth reading asks:

When, where, how, and for whom is this useful?

Assumption 3: Research should be synthesized and applied heuristically

The (un)common-sense approach assumes that one study is rarely enough to understand a complex communication problem. Students need to synthesize.

Summary says:

Here is what this article found.

Synthesis asks:

How does this finding connect to the research question, methods, participants, limitations, other studies, lived experience, and the communication problem I am trying to understand?

This previews the third discontinuous leap:

Summarizing to Synthesizing

This is also where the algorithm/heuristic thread should become explicit.

The common-sense approach wants research to work algorithmically:

Do X and Y will happen.

The (un)common-sense approach uses research heuristically:

This finding can guide what I notice, but I still have to interpret the situation.

Your notes name this as one of the strongest conceptual distinctions in the chapter. Algorithms are rule-based and promise predictable outputs. Heuristics are flexible guides that help readers notice possibilities, connect new information, anticipate outcomes without pretending to predict them, and contextualize information for the situation.

6. Real-World Examples to Use in This Section

Example 1: “Research says”

Use this early, right after defining the common-sense approach.

Common-sense reading:

Research says this, so the matter is settled.

(Un)common-sense reading:

Which research? What did it study? What claim can it support? What are its limits?

This example comes directly from the opening story and voice notes. It is accessible and funny, but it also teaches the central problem: research gets personified as a final authority instead of interpreted as evidence.

Example 2: Nonverbal deception myths

Use this to show interpersonal stakes.

Common-sense reading:

Research says body language reveals lying, so I can watch someone’s eyes and know whether they are deceiving me.

(Un)common-sense reading:

Research on deception suggests people are not very accurate lie detectors, so I should be cautious about turning one nonverbal behavior into evidence of dishonesty.

This example is strong because it moves the chapter from abstract research literacy to relational harm. It also has strong peer-reviewed support from deception meta-analyses.

Example 3: ADHD TikTok and self-diagnosis content

Use this to show how social media can make research feel intimate, useful, and misleading at the same time.

Common-sense reading:

This video explains my behavior using clinical language, so now I know what is happening.

(Un)common-sense reading:

This video may help me notice a possible pattern, but I need to know what evidence it is drawing from, whether it is overgeneralizing, and what professional or scholarly context is missing.

Yeung et al. (2022) provide the peer-reviewed grounding here: in their sample of the 100 most popular ADHD TikTok videos, 52% were classified as misleading.

Example 4: The 5-to-1 ratio

Use this to show the difference between algorithms and heuristics.

Common-sense reading:

For every negative interaction, do five positive things, and the relationship should improve.

(Un)common-sense reading:

Relationships often need more positive, affirming, repairing interaction than damaging interaction, but what counts as positive depends on the people, relationship history, rupture, timing, and context.

This is a clean bridge into the later assumptions because it shows how a summarized finding can become a rigid rule if students do not synthesize.

Example 5: AI summaries

Use this to show contemporary urgency.

Common-sense reading:

AI summarized the study, so now I understand it.

(Un)common-sense reading:

AI may help me enter the study, but I still need to check the original source, evaluate what was left out, and decide what the study can support.

This connects the chapter to automation bias, algorithm appreciation, and AI literacy.

7. How This Section Should Lead into the Discontinuous Leaps

The section should end by telling students that the chapter is not just going to give them tips for reading research. It is going to ask them to make three discontinuous leaps in how they relate to research.

A strong structure:

Leap 1: Reporting to Sense-Making

The common-sense approach asks students to report what research says.

The (un)common-sense approach asks students to make sense with research.

Leap 2: Universal Truths to Local Truths

The common-sense approach asks whether the finding is true.

The (un)common-sense approach asks where, when, how, and for whom the finding is useful.

Leap 3: Summarizing to Synthesizing

The common-sense approach asks students to summarize the answer.

The (un)common-sense approach asks students to connect findings, methods, limits, contexts, other studies, and lived experience.

The closing bridge can say:

These leaps matter because research is not only something we use in school. Research claims shape how we understand ourselves, diagnose problems, judge other people, use AI, consume social media, and communicate in everyday life.

Specific help request box

drafting note: the following are the three discontinuous leaps:

Section title	Common-sense assumption	Core myth	(Un)common-sense assumption	Core belief
1. Reporting to Sense-Making	Research gives findings to report.	Having the finding means understanding the research.	Research helps readers make sense.	Literacies are sense-making practices.
2. Universal Truths to Local Truths	A good study gives a general truth.	If research is true, it should be true everywhere.	Research offers partial, contextual, local truth.	A finding is useful when we understand its conditions.
3. Summarizing to Synthesizing	Reading research means summarizing what the study says.	A clean summary is the same thing as understanding.	Reading research means synthesizing connections.	Understanding grows by connecting claims, methods, contexts, limits, and other evidence.

H1: Discontinuous Leaps

Section goal

Reporting asks, “What did the study say?” Sense-making asks, “What does this study help me understand, what does it not help me understand, and what do I need to connect it to before I use it?” This leap matters because many students have been trained to prove that they read something by summarizing it. That habit is useful, but it is incomplete. A student can accurately report a finding and still misunderstand what the finding can support. The leap helps students realize that research literacy is not just having information. It is knowing how to move with information: how to evaluate it, connect it, contextualize it, and use it without turning it into a rule. Your notes make this connection directly by framing literacies as forms of understanding rather than memorization; literacy is what people do when they encounter information, interpret it, connect it to experience, and figure out what to do with it.

H1: Discontinuous Leaps / Leap 1

Framing box

The leap from **reporting to sense-making** moves students from treating research as something they have to **repeat correctly** toward treating research as something they have to **interact with**.

1. Reporting to Sense-Making

What the leap is and does

The leap from **reporting to sense-making** moves students from treating research as something they have to **repeat correctly** toward treating research as something they have to **work with carefully**. Reporting asks, “What did the study say?” Sense-making asks, “What does this study help me understand, what does it not help me understand, and what do I need to connect it to before I use it?” This leap matters because many students have been trained to prove that they read something by summarizing it. That habit is useful, but it is incomplete. A student can accurately report a finding and still misunderstand what the finding can support. The leap helps students realize that research literacy is not just having information. It is knowing how to move with information: how to evaluate it, connect it, contextualize it, and use it without turning it into a rule. Your notes make this connection directly by framing literacies as forms of understanding rather than memorization; literacy is what people do when they encounter information, interpret it, connect it to experience, and figure out what to do with it.

Common-sense assumption: Research gives findings to report

Description of the assumption

The common-sense approach assumes that research works like a transfer system. Researchers discover something, write it down, and the reader receives it. The article becomes a container of expert knowledge. The student’s task is to extract the finding, report it correctly, and maybe cite it later. This mirrors the larger common-sense approach to communication in Chapter 1, where communication is treated as the transfer of information from sender to receiver.

For students, this assumption sounds like:

“The article has the answer. My job is to find it and say it back.”

That assumption is understandable. Students do need to know what a study found. They need to summarize accurately. They need to cite carefully. The problem is that reporting can become a substitute for understanding. Students may think they understand the research because they can quote the conclusion, name the finding, or say “research says.” But they may not know what the study measured, who was included, what the method can support, what the limitations were, or how the finding connects to other evidence.

Example

The opening story captures this assumption perfectly. The narrator reads the article as if the article “knows things” and the reader does not. Research feels like the place where facts, truths, and certainty live before the student collects them and carries them into the world.

A real-world version is the online “Sources?” exchange. Someone makes a claim. Someone else asks for sources. That can be a good practice. But from the common-sense approach, the source can become a truth token. If there is a source, the claim feels settled. If there is no source, the claim feels dismissed. The point becomes whether someone can produce a citation, not whether anyone understands what the citation actually supports. Your notes identify Reddit “Sources?” culture as a strong example of treating research as a weapon, proof object, or truth token rather than something to interpret.

Core myth

Core myth: If I can report the finding, I understand the research.

This myth is powerful because reporting looks responsible. It sounds academic. A student can say, “The researchers found X,” and that sounds like research literacy. But the myth hides the difference between having a sentence and understanding a study. It hides questions like: What was the research question? What counted as evidence? What was measured? Who was studied? What limits did the authors name? What would be an overclaim?

How the myth shapes behavior

This myth shapes students into **answer extractors**. They read to locate the result. They treat the abstract as the article. They underline sentences without necessarily understanding how those sentences connect. They cite studies to win arguments. They may use AI to summarize a study and then treat the summary as understanding.

It also encourages **algorithmic reading behavior**. The student follows an implicit recipe:

- 1. Find a source.
- 2. Pull the main finding.
- 3. Put it in the paper.
- 4. Use it to prove the claim.

That behavior is not malicious. It is often exactly what school has rewarded. But it makes students vulnerable to shallow source use, especially in AI-rich environments where summaries are easy to generate. AI can help students access and organize information, but your notes are clear that AI cannot do understanding for the reader. Understanding still requires interpretation, credibility evaluation, and connection across contexts.

(Un)common-sense assumption: Research is sense-making

Definition and description

The (un)common-sense approach assumes that research is not simply something readers receive. Research is something readers **interact with**. A study gives evidence, but that evidence only becomes useful when the reader makes sense of it in relation to the question, method, participants, context, limits, other research, lived experience, and the situation where the claim might be applied.

A student-facing definition:

The (un)common-sense approach treats research as a sense-making practice. A study does not hand us the final answer. It helps us understand a question, problem, pattern, relationship, or context more carefully.

This approach does not reject reporting. It places reporting inside a larger practice. Students still ask, “What did the study find?” But they do not stop there. They also ask, “What does this finding help me notice?” “What does it leave out?” “What could I responsibly do with this?” “What would be too much to claim?”

The belief this assumption informs

The belief underneath this assumption is:

Literacies are sense-making practices.

This is one of the strongest ideas in your materials. Information literacy, scientific literacy, digital literacy, data literacy, research literacy, and AI literacy should not be treated as piles of facts. They are ways of making sense of information. Your notes explicitly say literacies are not about memorizing science or research facts; they are about evaluating, interpreting, using previous information to understand current information, and deciding what information can be used for.

That matters because the chapter should not make students feel like they need to become experts in every field before they can read research responsibly. The goal is not to know everything. The goal is to become more literate: more capable of asking better questions, noticing missing context, checking credibility, connecting evidence, and revising judgment.

This connects well to the annotated bibliography’s literacy sources. Lloyd (2006) frames information literacy as situated practice rather than a neutral checklist. Mackey and Jacobson (2011) reframe information literacy as metaliteracy, emphasizing that digital environments require people to evaluate, create, share, and reflect on information. Norris and Phillips (2003) argue that scientific literacy depends on reading, writing, interpreting, and evaluating texts, not simply knowing scientific facts.

How this shapes behavior

The (un)common-sense assumption changes what students actually do.

Instead of reading an article as a passive receiver, they read as an active interpreter. They can move around the article strategically: abstract, discussion, conclusion, method, literature review, findings, and back again. Your notes say this directly: when readers understand that they are making sense with the article, they are freed from reading top-to-bottom as if truth must be transferred in order.

Instead of treating AI as an answer machine, students use AI as a support tool. They might ask AI to define a term, generate questions about the methods, or summarize a difficult paragraph. But then they return to the article and ask, "Does this match the source?" "What did the AI leave out?" "Did it overstate the finding?" "What context disappeared?"

Instead of saying, "This study proves my point," they say, "This study helps me ask a better question."

Developed example: The pug eye/David Bowie example

The pug eye/David Bowie example is especially useful for this section because it makes literacy concrete. The narrator connects a pug’s nonreactive pupil, a childhood head injury, remembered information about David Bowie’s blown pupil, concern about whether the eye needed to be removed, and further investigation. The point is not that the narrator becomes a veterinarian. The point is that literacy is not passive fact storage. It is active connection-making. The narrator does not simply accept or reject expert information. They connect prior knowledge, lived experience, curiosity, and new evidence to make sense of a situation.

This is the kind of example students can understand immediately. It shows that literacy is not "I know all the facts." Literacy is "I can connect enough pieces to ask better questions."

Developed example: AI summary of a research article

A common-sense student might upload an article to AI, receive a summary, and think, "Great, now I understand it." The (un)common-sense student treats the summary as a starting point. They ask the AI for a list of claims, then check those claims against the article. They ask what method was used. They look for limitations. They compare the AI’s confident language with the authors’ more cautious language. They ask whether the summary turned "suggests" into "proves." The behavior changes because the student no longer treats information access as understanding. They treat information access as the beginning of sense-making.

Drafting emphasis

The section should make this promise to students:

You do not have to become an expert before research can help you. But you do have to stop treating research as something that understands for you.

H1: Discontinuous Leaps / Common Sense Structure

Assumption

Description of the assumption

The common-sense approach assumes that a research report works like a delivery system. Researchers discover something, place the finding into an article, and readers receive the finding. The study becomes the "message," the researchers become the "senders," and the student becomes the "receiver." This maps directly onto the common-sense communication model from Chapter 1, where communication is treated as the transfer of information.

Student-facing definition:

The common-sense approach to research treats a study like a container of expert knowledge. The researchers found something, wrote it down, and the reader’s job is to report it correctly.

This assumption is not useless. Students do need to know what a study found. They need to summarize accurately. They need to cite responsibly. They need to avoid making things up. But this becomes a problem when reporting becomes the whole goal.

Example from the voice notes

The "Reddit sources?" example is a strong everyday entry point. Someone makes a claim, and another person demands "Sources?" That can be a good thing. But the notes point out that sources often get treated as proof objects: if a source exists, the claim feels settled.

The same habit shows up in "research says" claims. A person online says "research says" or "studies show" as if Research were a person in a lab coat carrying the final answer. The problem is not that people want evidence. The problem is that a vague appeal to research can become a shortcut around interpretation.

Research example from the bibliography

The ACRL Framework helps support this section because it defines information literacy around concepts such as **Authority Is Constructed and Contextual**, **Information Creation as a Process**, **Research as Inquiry**, and **Scholarship as Conversation**. That framework works against the idea that research is just something students retrieve and report; it treats research as an ongoing, contextual process of inquiry (Association of College and Research Libraries, 2016).

Norris and Phillips (2003) are also useful because they argue that scientific literacy depends on reading, interpreting, and evaluating scientific texts, not just knowing science facts. That supports the chapter’s claim that reading research is an interpretive practice, not a fact-retrieval task.

Myth

Core myth

Core myth: If I can report the finding, I understand the research.

This myth feels reasonable because reporting is visible. A student can quote the conclusion. They can name the finding. They can cite the article. They can say, "According to this study..." That looks like research literacy. But it may only be research reporting.

The myth hides the difference between:

- knowing what the article says
- understanding what the article means
- knowing what the study found
- understanding what the study can support
- having a citation
- knowing whether the citation fits the claim

How it impacts communication

How the myth informs behavior

This myth shapes student behavior in several predictable ways.

Students may read articles linearly because they assume truth must be received in order. They start at the first sentence, move through each section, and try to survive until the conclusion. Your voice notes name this directly: the common-sense approach shapes how students read by encouraging linear reading, as if the truth must be transferred in sequence.

Students may also overvalue the abstract or conclusion because those sections seem to contain "the answer." They may quote the finding without checking the method, sample, measures, limitations, or research question.

They may use research as debate ammunition. The opening story's "Actually, research shows..." fantasy captures this well. Research becomes a way to win rather than understand.

They may treat AI summaries as understanding. AI can produce a clean explanation, but a clean explanation is not the same thing as interpretive judgment. Automation bias research is useful here because Goddard et al. (2012) define automation bias as the tendency to over-rely on automation, and Logg et al. (2019) show that people may sometimes adhere more to advice when they believe it came from an algorithm.

Drafting move

The section should not shame students for reporting. Instead, say:

Reporting is a starting point. Sense-making is the goal.

A draftable transition:

Reporting asks, "What did the study say?" Sense-making asks, "What does this study help me understand, and what would I need to know before using it?"

Assumption

The common-sense approach assumes that a research finding should be complete, stable, and broadly applicable. A good study should tell us what is true, not just what was found in one context.

Myth

Core myth: If research is true, it should be true everywhere.

How it impacts communication

This myth shapes behavior by encouraging **overgeneralization** and **rigid application**.

Assumption

The common-sense approach assumes that if I can summarize the study, I know the study.

Myth

A clean summary is the same thing as understanding.

How it impacts communication

This myth shapes behavior by encouraging students to stop too early.

H1: Discontinuous Leaps / (Un)Common Sense Structure

Assumption

(Un)common-sense assumption: Research is sense-making

Definition

The (un)common-sense approach treats research as a **sense-making practice**. A research article does not simply transfer truth to a passive reader. It gives the reader a structured opportunity to interpret evidence, examine limits, connect ideas, and ask better questions.

Student-facing definition:

The (un)common-sense approach to research treats a study as something we interact with to make better sense of a question, problem, relationship, or communication situation.

This is consistent with Chapter 0's distinction between knowing and understanding. Knowing means having accurate information one can state, recall, identify, or correct. Understanding means grasping how pieces of information connect so a situation, relationship, or interaction makes more sense.

Drafting Note: Use voice notes and resources to help develop this assumption more.

Belief

The belief this assumption informs

This assumption informs the belief that **literacies are forms of sense-making**.

This is the connection you specifically asked to emphasize. Research literacy, information literacy, scientific literacy, data literacy, digital literacy, and AI literacy are not just piles of facts. They are practiced capacities for interpreting information, evaluating credibility, noticing limits, connecting evidence, and deciding how a claim should or should not be used.

Your brainstorming notes say this strongly: literacies are not about memorizing facts; they are about evaluating, interpreting, using previous information to understand current information, and judging credibility and use. The notes also offer the "street smart" metaphor: literacies are like street smarts for information environments.

This connects well to Mackey and Jacobson's (2011) argument that information literacy should be understood as metaliteracy in participatory digital environments, where people do not simply consume information but also create, share, remix, and evaluate it.

How it changes our behavior

How this shapes behavior

If students read research as sense-making, they behave differently.

They read recursively instead of obediently. They can move from abstract to discussion, to conclusion, to methods, to literature review, and back again. Your notes explicitly recommend this: from a sense-making perspective, readers can move around the article because they are interacting with it rather than passively receiving it.

They ask different questions:

- What question were the researchers asking?
- What does this study help me see?
- Who or what was studied?
- What was measured?
- What was left out?
- What does the article explain well?
- What does it complicate?
- How does it connect to other evidence?
- What should I avoid claiming from this?

They use AI differently. Instead of asking AI to replace understanding, they use it to support sense-making: clarify vocabulary, generate questions, explain a method, or help organize notes. Then they return to the article and check whether the output fits the source. Long and Magerko (2020) define AI literacy as competencies for understanding, using, and critically evaluating AI, which aligns with this chapter’s anti-outsourcing-understanding stance.

Real-world examples to use

Use the **pug eye/David Bowie** example as the “literacies are street smarts” case. Your notes describe it as an example where the narrator connects a pug’s eye issue, prior experience with a head injury, remembered knowledge about David Bowie’s pupil, curiosity, and further investigation. The point is not that the narrator becomes a medical expert. The point is that literacy means connecting information carefully instead of accepting or rejecting a claim too quickly.

Use the **AI summary** example as the contemporary version. A student asks AI to summarize a study. The common-sense behavior is to treat the summary as the answer. The (un)common-sense behavior is to ask: What did the AI leave out? Did it invent or overstate anything? Does the summary fit the method and limitations? What do I still need to check?

Assumption

The (un)common-sense approach assumes that research findings are **partial, contextual, and local**.

Belief

A study becomes more useful, not less useful, when we understand its limits.

How it changes our behavior

The (un)common-sense approach changes evaluation behavior. Instead of asking only, “Is this true?”

Assumption

The (un)common-sense approach assumes that reading research well means **synthesizing**.

Student-facing definition:

Synthesis means connecting a study to other evidence, context, methods, limitations, lived experience, and the question you are trying to understand.

Summary says:

Belief

Understanding grows through connection.

How it changes our behavior

Students also connect research to lived experience without reducing research to personal opinion. Lived experience becomes part of sense-making, not a replacement for evidence.

Idea development box

2. Universal Truths to Local Truths

What the leap is and does

The leap from **universal truths to local truths** moves students away from treating research findings as complete, everywhere-applicable truths and toward treating them as evidence that works under particular conditions. This leap is crucial because it helps students avoid two equally bad habits: over-trusting research and dismissing research. The common-sense approach says, "If a study is true, it should be true everywhere." When students discover limitations, they may think the study is weak or useless. The (un)common-sense approach gives them a better middle position: a finding can be true, useful, and rigorous without being universal. Local truth keeps the conditions attached. It asks students to notice when, where, for whom, and under what circumstances a finding helps. This leap does not make research weaker. It makes research more usable because it teaches students not to rip findings out of the contexts that made them meaningful.

Common-sense assumption: Good research gives universal truth

Description of the assumption

The common-sense approach assumes that a good study should give a universal answer. If the study is true, it should be true for everyone. If it is not true for everyone, then it must not be a good study. This assumption is appealing because universal truth feels stable. It gives students something to hold onto. It also makes research easier to use in arguments, social media posts, and everyday advice.

Student-facing definition:

The common-sense approach assumes that a research finding should be complete, stable, and broadly applicable. A good study should tell us what is true, not just what was found in one context.

This assumption fits the common-sense tendency toward monologic thinking: true or false, right or wrong, useful or useless. Chapter 0's framework helps here because the knowing framework asks for the right fact, rule, answer, or script, while the understanding framework asks how pieces fit together in context.

Example

The small-sample-size critique is a strong student example. A student reads a qualitative study with 20 participants and says, "That sample is too small. You can't generalize from that." Sometimes sample size is a real issue. But the common-sense mistake is assuming that every study is trying to generalize statistically. Some qualitative studies are trying to understand meaning, experience, process, context, or theory. Judging them only by sample size applies the wrong standard. Your notes specifically identify this problem: people often reject studies that challenge them by saying the sample is too small, even though not all research seeks generalizability in the same way.

Another example is the 5-to-1 ratio. A common-sense reader may take the idea that healthy relationships need more positive than negative interactions and convert it into a universal rule: do five positive things for every negative thing and the relationship should recover. Your notes caution that the problem is not that the ratio is simply false. The problem is applying it algorithmically. What counts as positive? How large was the rupture? What does this partner need? What is the history of the conflict? A relationship does not become repaired because someone completed five emotional chores and went to bed.

Core myth

Core myth: If research is true, it should be true everywhere.

This myth causes students to detach research from its conditions. They want the finding without the method, the conclusion without the sample, the rule without the context. But social research usually does not work that way. A finding depends on who was studied, what was measured, how the study was designed, what assumptions were made, what the researchers were asking, and what body of evidence surrounds it.

How the myth shapes behavior

This myth shapes behavior by encouraging **overgeneralization** and **rigid application**.

Students may take one finding and turn it into a rule for all relationships, all workplaces, all families, all students, or all communicators. They may accept a study too quickly when it agrees with them and pick it apart when it does not. Your notes connect this directly to confirmation bias: when a study agrees with us, we may treat it as universal fact; when it disagrees, we may suddenly become very interested in limitations.

It also shapes social media misinformation. A careful finding becomes a general claim. The general claim becomes advice. The advice becomes a rule. The rule becomes a myth. The chapter notes explain that misinformation is not always invented from nothing. Sometimes it begins with something partly true that becomes oversimplified, overgeneralized, or made too certain.

Research examples from the bibliography

The master bibliography gives several examples that fit this section. Hill (1965) is useful for the myth that correlation implies causation: association can be important evidence, but causal reasoning requires more than noticing that two things move together. The ASA statements on p-values are useful for the myth that statistical significance equals truth: p-values are one input to inference, not a truth machine. Deaton and Cartwright (2018) are especially important for local truth because they show that even randomized controlled trials answer defined causal questions under assumptions and do not automatically "travel" to new populations or settings. The bibliography summarizes this as a distinction between internal validity and transportability.

The verified APA guide also supports careful phrasing: say "correlation does not by itself establish causation," not "correlation never supports causation"; say "a p-value is one input to inference," not "p-values are useless"; say "peer review is screening, critique, and revision," not "peer review is worthless."

(Un)common-sense assumption: Research offers local truth

Definition and description

The (un)common-sense approach assumes that research findings are **partial, contextual, and local**.

Student-facing definition:

A local truth is a finding that is supported under particular conditions: with these participants, in this context, using these methods, for this question. Local truth is not fake truth. It is truth with its conditions still attached.

This is the most important point to protect from misunderstanding. Local truth does **not** mean "anything can be true." It does not mean research is just opinion. It means research becomes more useful when we understand the boundaries of what it can claim.

Your notes say this clearly: no single piece of research produces universal truth; findings may have evidentiary support, but that truth is local because it is always in the context of the study.

Peer-reviewed grounding for local truth

"Local truth" can be grounded in several peer-reviewed traditions even if those authors use different language.

Haraway's (1988) concept of **situated knowledges** argues that knowledge is partial, located, and accountable rather than a "view from nowhere." This supports the idea that all knowledge is produced from somewhere.

Flyvbjerg (2006) defends context-dependent knowledge in case-study research and challenges the assumption that abstract, context-independent knowledge is always superior.

Rothwell (2005) asks the external-validity question directly: to whom do the results apply?

Deaton and Cartwright (2018) argue that causal evidence, including RCT evidence, does not automatically travel from one setting to another. That is very close to the student-facing question: "Will this finding still work here?"

Simons, Shoda, and Lindsay (2017) propose "constraints on generality" statements so researchers specify the populations, settings, materials, and contexts to which their findings are expected to apply.

Together, these sources support the chapter’s claim that responsible research reading means keeping conditions attached to claims.

The belief this assumption informs

The belief underneath local truth is:

A study becomes more useful, not less useful, when we understand its limits.

This belief changes how students experience limitations. Instead of reading the limitations section as a confession that the study failed, students learn to see limitations as instructions for responsible use. A limitation tells us where to be careful. It helps us avoid overclaiming. It shows us what the study can help us understand and where we need more evidence.

That belief also helps students avoid cynicism. The goal is not to say, “Every study has limits, so research is pointless.” The goal is to say, “Every study has limits, so I need to understand what kind of help this study can actually give me.”

How this shapes behavior

The (un)common-sense approach changes evaluation behavior. Instead of asking only, “Is this true?” students ask:

- When is this true?
- Where is this true?
- For whom is this true?
- Under what conditions is this true?
- What would make this finding travel poorly?
- What would I need to know before applying it?

This is the local-truth heuristic. It helps students read research with curiosity rather than certainty.

Developed example: Nonverbal deception myths

The nonverbal deception example is one of the strongest local-truth cases because it shows interpersonal harm. Your voice notes describe the common myth that people can detect lying by watching eye movement, facial expressions, or body language. The notes then connect that myth to everyday relational danger: a person might watch their partner’s eyes during a conversation about cheating and treat an eye movement as evidence of deception.

The (un)common-sense approach does not say, “Nonverbal behavior never matters.” That would just create another universal rule. Instead, it asks: What does deception research actually support? What kinds of cues have been studied? Under what conditions? How accurate are people, on average? What should we not infer from one behavior in one tense relational moment?

This is local truth at work. It prevents students from turning weak or context-bound patterns into accusations.

Developed example: Mirroring in sales

Mirroring is a useful example because it may appear helpful in one context and manipulative in another. The common-sense approach wants one universal answer: mirroring works or mirroring does not work. The (un)common-sense approach asks when, how, for whom, and under what relational conditions mirroring might be useful or harmful. Your notes identify this exact caution: mirroring may help in some contexts and backfire in others when perceived as manipulative or inauthentic.

This teaches students that contradictory findings do not always mean one study is “right” and the other is “wrong.” Sometimes contradiction reveals boundary conditions. It tells us that context matters.

Developed example: ADHD TikTok and self-diagnosis

Social media self-diagnosis and self-treatment examples can also fit here if handled carefully. The chapter should avoid shaming self-advocacy or communities that use social media for support. The issue is not that people learn about ADHD, autism, depression, or anxiety online. The issue is that complex clinical and scientific research can become overly broad identity scripts or self-treatment rules. Your notes flag this caution directly: social media can spread research-based misinformation that shapes self-diagnosis or self-treatment, but the chapter should avoid shaming self-advocacy.

A local-truth reading might ask: Is this creator describing a lived experience, a clinical criterion, a research finding, or a personal interpretation? Does the post cite evidence? Does it turn “can be associated with” into “means you have”? Does it offer community support, or does it sell certainty?

Drafting emphasis

The section should make this sentence feel obvious by the end:

A finding without context is easier to share, but harder to use responsibly.

H1: Discontinuous Leaps / Leap 3

Idea development box

3. Summarizing to Synthesizing

What the leap is and does

The leap from **summarizing** to **synthesizing** moves students from treating research as an isolated article to treating research as part of a larger conversation. Summarizing asks students to condense: “What did this article say?” Synthesizing asks students to connect: “How does this article relate to other evidence, methods, contexts, lived experience, and the question I am trying to understand?” This leap matters because summaries are seductive. They are clean, quick, and portable. But summary alone can erase the complexity that makes research useful. The leap to synthesis helps students stop treating one source as a final answer. It teaches them to build understanding across sources and situations. In the larger chapter arc, this is where students begin to practice research literacy as communication judgment.

Common-sense assumption: Reading research means summarizing what the article says

Description of the assumption

The common-sense approach assumes that the goal of reading research is to produce a clear summary. The student wants to know what the article says, what the authors found, and how to state it in a paper or conversation.

Student-facing definition:

The common-sense approach assumes that if I can summarize the study, I know the study.

Again, summary is not bad. It is necessary. Students should be able to identify the topic, research question, method, and finding. But summary becomes a problem when it replaces synthesis. A student may summarize a study accurately while still not knowing how to connect it to other research, evaluate its method, notice its limits, or apply it responsibly.

Example

The opening story’s “ritual of reading” is the emotional version of this assumption. The narrator moves sentence by sentence, underlines, writes question marks, and performs reading. The goal seems to be getting through the article and extracting what it says.

The AI version is even more current. A student asks AI to summarize a research article. The output is clear, polished, and confident. The student feels relief. But if they stop there, they may only have a cleaned-up version of the article’s surface. They may not know whether the AI preserved the limitations, represented the method correctly, exaggerated the findings, or fabricated connections.

Core myth

Core myth: A clean summary is the same thing as understanding.

This myth is powerful because summaries feel like comprehension. They reduce friction. They make hard research feel manageable. They are especially tempting for students who feel overwhelmed by academic prose, jargon, statistics, or article structure.

But clean summaries often remove the very things students need most:

- uncertainty
- limitations
- method
- sample
- context
- competing evidence
- caveats
- contradictions
- practical boundaries

How the myth shapes behavior

This myth shapes behavior by encouraging students to stop too early.

They may read the abstract and conclusion but skip the methods. They may use one article because it supports their argument. They may cite a finding without explaining what kind of evidence produced it. They may use AI to produce a summary and then never return to the original source. They may treat a literature review, meta-analysis, qualitative interview study, and experiment as if all research types do the same thing.

It also encourages algorithmic application:

“The study says X, so I should do X.”

That is how research becomes advice content. A finding becomes a tip. The tip becomes a rule. The rule becomes a script.

Your notes frame this as one of the strongest distinctions in the chapter: algorithms are associated with rule-based application, while heuristics are flexible guides that help readers notice, connect, anticipate, and contextualize.

Research examples from the bibliography

Sumner et al. (2014) are useful here because they show how research can become exaggerated as it moves from article to press release to news coverage. That is a synthesis problem: students need to understand that public-facing summaries are not neutral windows into research. They are part of a communication chain.

The AI literacy sources also fit here. Long and Magerko (2020) define AI literacy as the ability to understand, use, and critically evaluate AI. Chu-Ke and Dong (2024) connect generative AI to misinformation and literacy needs. These sources support the chapter’s claim that AI makes synthesis more important, not less important.

(Un)common-sense assumption: Reading research means synthesizing connections

Definition and description

The (un)common-sense approach assumes that reading research well means **synthesizing**.

Student-facing definition:

Synthesis means connecting a study to other evidence, context, methods, limitations, lived experience, and the question you are trying to understand.

Summary says:

“Here is what this article said.”

Synthesis asks:

“How does this article connect to what else I know, what else has been studied, and the situation where this claim might matter?”

This is where the chapter can make the strongest connection to understanding. Chapter 0 defines understanding as figuring out how facts, people, relationships, histories, and meanings connect while interaction is still unfolding. In Chapter 3, synthesis is the research-reading version of that same practice.

The belief this assumption informs

The belief underneath synthesis is:

Understanding grows through connection.

One article rarely gives the whole answer. One summary rarely gives enough context. One source rarely captures a full conversation. Synthesis teaches students that research is cumulative, relational, and dialogic. A study matters not only because of what it found, but because of how it fits with other studies, theories, methods, histories, and contexts.

This belief aligns with the information literacy idea that scholarship is conversation. It also aligns with scientific literacy because students must learn to interpret scientific texts, methods, and evidence rather than relying only on someone else’s simplified explanation.

How this shapes behavior

Synthesis changes student behavior in several concrete ways.

Students compare sources instead of collecting them. They ask whether studies agree, disagree, or apply to different contexts. They notice method differences: a survey, experiment, interview study, ethnography, rhetorical analysis, and meta-analysis all produce different kinds of evidence. They stop asking one method to answer every question.

Students connect claims to methods. If a study interviewed 25 people about workplace burnout, it may help explain how people experience burnout, what language they use, and what patterns appear in their stories. It may not estimate the exact national prevalence of burnout. If a survey of 2,000 people finds a relationship between communication frequency and satisfaction, it may show a pattern, but it may not explain what those conversations meant to the people having them.

Students also connect research to lived experience without reducing research to personal opinion. Lived experience becomes part of sense-making, not a replacement for evidence.

Developed example: ADHD craft cart

The ADHD craft-cart example is one of the clearest ways to explain synthesis as heuristic thinking. Algorithmically, the advice becomes:

“Buy this craft cart.”

Heuristically, the research-informed insight becomes:

“Create a visible, accessible container that helps organize scattered materials related to crafts, hobbies, tasks, or hyperfixations.”

That shift matters. The common-sense reader copies the solution. The (un)common-sense reader synthesizes the function. Once the student understands the function, they can adapt the advice. The answer might be a craft cart, but it might also be a laundry basket, shelf, bin, phone reminder, checklist, visual cue, or shared household system. Your notes identify this as the difference between applying advice algorithmically and applying it heuristically.

This example is especially strong because it does not mock the advice. The advice may be helpful. The problem is not the cart. The problem is mistaking the cart for the principle.

Developed example: 5-to-1 ratio

The 5-to-1 ratio can also show synthesis. Summary gives the student the ratio. Synthesis asks what the ratio is trying to help us notice: relationships often need more positive, affirming, connecting interaction than negative or damaging interaction. But synthesis also asks what the ratio cannot decide for us. What counts as positive for this person? Was the negative interaction minor irritation or major betrayal? Is the relationship already strained? Does one partner experience the “positive” gesture as repair or avoidance? Is the person doing five nice things to restore connection, or to complete the algorithm?

This is where the heuristic is much richer than the rule:

“Repair usually requires more than balancing the scoreboard . Pay attention to the size of the rupture, the history of the relationship, and what the other person actually experiences as care.”

Developed example: Nonverbal deception research

A common-sense summary of deception research might become:

“Body language reveals lying.”

A slightly more research-sounding but still flattened summary might become:

“Some nonverbal cues are associated with deception.”

A synthesis asks better questions:

- What does the broader evidence say about deception detection accuracy?
- Are nonverbal cues reliable enough to use in real interpersonal judgment?
- What happens when someone applies a weak cue inside a high-stakes relationship?
- How might anxiety, disability, neurodivergence, culture, trauma, or power affect someone's nonverbal behavior?
- What harm can happen when we treat a cue as evidence?

This example keeps the chapter anchored in interpersonal communication because the consequence is relational. Research misuse changes what people notice, accuse, trust, and misread.

Developed example: AI-generated synthesis vs. AI-generated summary

This is a useful AI-era classroom example. Ask students to compare two AI tasks:

Summary prompt:

“Summarize this article.”

Synthesis prompt:

“Identify the study’s research question, method, participants, main finding, limitations, and what kinds of claims the study can and cannot support. Then give me three questions I should ask before applying this finding to interpersonal communication.”

The second prompt still does not outsource understanding, but it invites sense-making. The student has to check the output against the article, revise it, and decide what is justified. This turns AI from answer machine into thinking support.

Drafting emphasis

This section should end by helping students feel the difference:

A summary gives you the finding. A synthesis helps you understand what the finding can do.

Annotated bibliography box

Below is a **source-backed drafting guide** for the section that comes right after the opening story and before the three discontinuous leaps. I would use this section to define both approaches, show why the common-sense approach feels natural but becomes dangerous, and preview the three shifts: **Reporting to Sense-Making, Universal Truths to Local Truths, and Summarizing to Synthesizing**.

I verified the major peer-reviewed sources used below through publisher pages, PubMed, ACM, SAGE, Wiley, journal pages, or university-hosted journal records. A few sources in your broader bibliography, such as ACRL, are official frameworks rather than peer-reviewed articles; they are useful for background, but I rely mainly on peer-reviewed sources for research claims.

Section Goal

This section should answer a simple question for students:

What is research for?

The opening story shows one answer: research can feel like something experts know and students are supposed to absorb. The 30-page article seems to “know things,” while the reader does not. The narrator treats research as a place where “answers,” “facts,” “truths,” and “certainty” live before being collected and used. That is the common-sense approach before it is formally named.

The section should then make the conceptual move:

The problem is not that research gives us knowledge. The problem is what happens when we treat knowledge as the whole purpose of research.

That leads directly into the two approaches.

1. Defining Research from the Common-Sense Approach

Student-facing definition

From the common-sense approach, research is a method of knowing. It produces findings, packages them into reports, and transfers expert knowledge from researchers to readers.

In this view, the research article is like a container. The researchers discovered something. They wrote it down. The reader opens the article, extracts the finding, remembers it, cites it, and applies it.

This definition directly extends the Chapter 1 common-sense communication model. Chapter 1 explains that the common-sense approach treats communication as the transfer of information, while the (un)commonsense approach treats communicating as messy, collaborative, material, and situated sense-making. When this transfer model gets applied to research, the research report becomes the message, the researcher becomes the sender, the reader becomes the receiver, and the finding becomes the information being transferred. Your analyzed voice notes name this directly: the common-sense approach treats research reports as containers of expert findings that transfer knowledge to readers.

Thorough description

The common-sense approach treats research as if its main purpose is to deliver a finding. It imagines reading research as a fairly simple process:

- A study asks a question.
- Researchers collect data.
- The article reports the answer.
- The reader learns the answer.
- The reader can now use the answer.

That model is attractive because it makes research feel useful, clean, and powerful. It gives students something to do with a difficult article: find the point. It also gives them something to do in everyday arguments: cite the point.

This is why the opening story’s “Actually, research shows...” fantasy matters. It is not just a joke about graduate-school arrogance. It reveals the pleasure of the common-sense approach. A citation can feel like authority. A study can feel like a weapon. A finding can feel like a rule.

The voice notes repeatedly connect this to online life: Reddit users asking for “Sources?,” social media posts saying “research says,” and people invoking studies as if the presence of a source settles the question. The problem is not asking for sources. The problem is expecting a source to end interpretation.

What this approach assumes

The common-sense approach previews three assumptions that the chapter will later unpack as discontinuous leaps.

Assumption 1: Research is for reporting

The reader’s job is to report what the study found. This creates a reading habit: students look for the finding, conclusion, quote, or statistic that can be repeated. Your raw notes describe this as reading from top to bottom because the truth appears to be communicated in order.

This assumption leads into the first leap:

Reporting to Sense-Making

The problem is not reporting itself. Students do need to report accurately. The problem is when reporting replaces understanding.

Assumption 2: Research gives universal truths

The common-sense approach treats a single article as if it can contain the truth. The reader evaluates the article as true or false, right or wrong, useful or useless. Your voice notes describe this as monologic thinking: the research report becomes either all true or all false, which erases the complexity of good research.

This leads into the second leap:

Universal Truths to Local Truths

Assumption 3: Research should be applied as an algorithm

The common-sense approach wants research to tell us what to do. It treats findings like rules:

If research says X, do X, and Y should happen.

Your notes use the 5-to-1 ratio as a strong example. A common-sense reader might treat “five positive interactions for every negative one” as an exact relational repair formula. But that turns a research-informed pattern into a rigid rule. The rupture may be too large, the “positive” interaction may not feel positive to the other person, or the relationship history may change what the ratio means.

This leads into the third leap:

Summarizing to Synthesizing

A summary gives the rule. Synthesis asks what the rule depends on, where it applies, and how it needs to be adapted.

2. Why the Common-Sense Approach Feels Useful

The section should be careful here. Do not make the common-sense approach sound stupid. It is not.

The common-sense approach helps students:

- find evidence
- identify claims
- summarize articles
- cite sources
- value peer-reviewed research
- avoid relying only on opinion

This matters because the chapter should not sound anti-research. Chapter 0 already makes this move with knowledge: knowing is useful, but not enough. The knowing framework treats communication as information to collect, transfer, correct, or apply; the understanding framework treats communicating as contextual sense-making. Chapter 3 should make the same move with research.

A useful drafting line:

The common-sense approach is not wrong because it asks what a study found. It becomes too small when it thinks that finding is the whole point.

3. Why the Common-Sense Approach Becomes Dangerous

It turns partial findings into certainty

Your analyzed notes argue that misinformation often happens when research is processed through a knowing framework. Research becomes something to extract, memorize, repeat, or use as a conclusive answer; once a finding gets flattened into a slogan, tip, "research says" claim, or universal rule, it becomes brittle knowledge.

That is the key danger: **the common-sense approach makes research feel more certain than it is.**

This is not only a theoretical problem. The methodological literature supports the same caution. The ASA's p-value guidance, replication literature, causal inference literature, and open science reforms all push against treating one finding, one threshold, or one publication as settled truth. Your verified APA guide summarizes this well: p-values do not prove truth, peer review is screening rather than final certification, one strong study is not definitive, and qualitative rigor should be judged by method-appropriate standards.

It turns real research into misinformation

A major point for the draft: misinformation does not always begin with fake research.

Sometimes it begins with real research that gets stripped of context.

A careful finding becomes a catchy post.

A cautious claim becomes a rule.

A correlation becomes a cause.

A study with limitations becomes "science proves."

A partial truth becomes a myth.

Sumner et al. (2014) support this communication-chain argument. Their BMJ study examined exaggeration in health-related science news and academic press releases and found that exaggeration in news was strongly associated with exaggeration already present in press releases. That gives you a peer-reviewed example of how research can become distorted as it travels from study to public communication. ([PubMed](#))

It shapes interpersonal judgment

This is where the chapter becomes clearly relevant to interpersonal communication.

Your misinformation voice notes emphasize that misinformation about social processes does not stay abstract. It shapes how people judge relationships, credibility, deception, mental health, intelligence, and everyday behavior.

The best example is nonverbal deception.

The voice notes describe communication charlatans who claim that eye movement, facial expression, or body posture can reveal lying. The danger is interpersonal: someone might watch a romantic partner's eyes during a painful conversation and treat eye movement as evidence of betrayal.

Peer-reviewed deception research supports the caution. Bond and DePaulo's meta-analysis synthesized 206 documents and 24,483 judges and found average lie-truth judgment accuracy of about 54%, only slightly above chance. ([Sage Journals](#)) DePaulo et al. (2003) also reviewed cues to deception across 1,338 estimates of 158 cues, which supports the broader point that simple cue-based lie detection is far less reliable than popular advice suggests. ([PubMed](#))

It turns social media summaries into "research"

The ADHD TikTok example is especially useful for the chapter because it is concrete, current, and recognizable. Yeung et al. (2022) analyzed the 100 most popular TikTok videos about ADHD and classified 52% as misleading, 27% as personal experience, and 21% as useful. ([PubMed](#))

This does not mean every ADHD TikTok is bad. It means students need research literacy because social media can make partial, personal, clinical, and research-based claims look equally authoritative.

A useful drafting line:

The problem is not that social media talks about research. The problem is that social media often rewards the version of research that sounds the most confident, portable, and emotionally useful.

It makes AI summaries feel like understanding

AI intensifies the common-sense approach because it makes information transfer feel effortless. The student asks a question, the system produces a clean answer, and the answer feels like understanding.

But automation bias research cautions against over-reliance on automated systems. Goddard et al. define automation bias as the tendency to over-rely on automation and review how people may fail to monitor automated outputs carefully. ([PubMed](#)) Logg et al. (2019) also show "algorithm appreciation": across six experiments, people sometimes adhered more to advice when they believed it came from an algorithm rather than a person. ([IDEAS/RePEc](#))

The chapter should phrase this carefully:

AI does not automatically make students less critical. But when students treat AI output as a substitute for reading, evaluating, and connecting evidence, AI can strengthen the common-sense fantasy that understanding can simply be delivered.

Chu-Ke and Dong (2024) support the AI-literacy connection, arguing that generative AI changes misinformation environments and requires users to critically evaluate AI-generated content. ([Sage Journals](#)) Long and Magerko (2020) define AI literacy around the competencies people need to understand, use, and critically evaluate AI technologies. ([ACM Digital Library](#))

4. Defining Research from the (Un)Common-Sense Approach

Student-facing definition

From the (un)common-sense approach, research is a method of understanding. It helps us make sense of questions, patterns, problems, contexts, relationships, and claims by working with partial, contextual evidence.

This definition shifts research from a truth-transfer model to a sense-making model.

The article is not a magic answer machine.

The reader is not a passive receiver.

The finding is not a universal rule.

The study is one resource for understanding.

Your analyzed notes define this strongly: the (un)common-sense approach treats research reading as interaction. The reader interacts with the report in relation to existing understanding, other research, lived experience, and the context where the finding might be applied.

Thorough description

The (un)common-sense approach does not ask students to distrust research. It asks them to use research more responsibly.

Research becomes a way to ask:

- What does this study help me understand?
- What question were the researchers asking?
- Who or what did they study?
- What methods did they use?
- What does this finding explain well?
- What does it not explain?
- Where might it apply?
- Where might it not apply?
- What other research complicates it?
- How might this guide action without becoming a rigid rule?

This is where the chapter can connect research literacy to the larger textbook framework. Chapter 0 defines understanding as contextual sense-making: figuring out how facts, people, relationships, histories, and meanings connect while interaction is still unfolding. Chapter 3 can say research literacy is the information-world version of that same practice.

A useful drafting line:

Literacies are sense-making practices. They are not just what we know about information; they are what we can do when information shows up in a messy world.

This is well supported by your voice notes, which frame information literacy, scientific literacy, research literacy, data literacy, digital literacy, and AI literacy as forms of understanding rather than memorization. It is also supported by peer-reviewed literacy research. Mackey and Jacobson (2011) define metaliteracy as an expanded information-literacy framework for participatory digital environments where people produce, share, and evaluate information. ([College & Research Libraries](#)) Norris and Phillips (2003) argue that scientific literacy cannot be separated from reading, writing, interpreting, and evaluating scientific texts. ([Wiley Online Library](#)) Kluecvsek (2017) argues that scientific information literacy should include understanding scholarship as conversation and publication as part of the scientific process. ([PDXScholar](#))

5. What the (Un)Common-Sense Approach Assumes

Assumption 1: Research reading is sense-making, not reporting

The (un)common-sense approach assumes that students do not read research merely to report what the article says. They read to make sense of what the article helps them see.

That changes reading behavior. Your raw notes say that once students understand they are “making sense with the research article,” they are freed from reading it as passive listeners. They can move from abstract to discussion, to rationale, to conclusion, back to the literature review, and so on.

This previews the first discontinuous leap:

Reporting to Sense-Making

Drafting emphasis:

Reporting asks:

What did the study say?

Sense-making asks:

What does this study help me understand?

Assumption 2: Research produces local truths, not universal truths

The (un)common-sense approach assumes that a study can be true and limited at the same time.

A finding may be true:

- for this population
- in this setting
- using this method
- under these conditions
- for this question
- with these limits

That is what this chapter can call **local truth**.

Student-facing definition:

A local truth is a supported finding with its conditions still attached.

This is not relativism. It does not mean “anything can be true.” It means a finding becomes more usable when readers understand where it came from and how far it can responsibly travel.

Peer-reviewed sources support this move. Haraway’s concept of situated knowledges argues against the fantasy of a view from nowhere and emphasizes partial, located knowledge. ([PhilArchive](#)) Simons et al. (2017) propose “constraints on generality” statements so empirical articles identify the populations, settings, materials, and conditions to which findings are expected to apply. ([Sage Journals](#)) Rothwell (2005) asks the external-validity question directly: to whom do trial results apply? ([PubMed](#)) Deaton and Cartwright (2018) similarly caution that randomized controlled trials answer defined causal questions under assumptions, but results do not automatically travel to new populations or settings. ([PubMed](#))

This previews the second discontinuous leap:

Universal Truths to Local Truths

Drafting emphasis:

Universal-truth reading asks:
Is this true?

Local-truth reading asks:
When, where, how, and for whom is this useful?

Assumption 3: Research should be synthesized and applied heuristically

The (un)common-sense approach assumes that one study is rarely enough to understand a complex communication problem. Students need to synthesize.

Summary says:

Here is what this article found.

Synthesis asks:

How does this finding connect to the research question, methods, participants, limitations, other studies, lived experience, and the communication problem I am trying to understand?

This previews the third discontinuous leap:

Summarizing to Synthesizing

This is also where the algorithm/heuristic thread should become explicit.

The common-sense approach wants research to work algorithmically:

Do X and Y will happen.

The (un)common-sense approach uses research heuristically:

This finding can guide what I notice, but I still have to interpret the situation.

Your notes name this as one of the strongest conceptual distinctions in the chapter. Algorithms are rule-based and promise predictable outputs. Heuristics are flexible guides that help readers notice possibilities, connect new information, anticipate outcomes without pretending to predict them, and contextualize information for the situation.

6. Real-World Examples to Use in This Section

Example 1: “Research says”

Use this early, right after defining the common-sense approach.

Common-sense reading:

Research says this, so the matter is settled.

(Un)common-sense reading:

Which research? What did it study? What claim can it support? What are its limits?

This example comes directly from the opening story and voice notes. It is accessible and funny, but it also teaches the central problem: research gets personified as a final authority instead of interpreted as evidence.

Example 2: Nonverbal deception myths

Use this to show interpersonal stakes.

Common-sense reading:

Research says body language reveals lying, so I can watch someone’s eyes and know whether they are deceiving me.

(Un)common-sense reading:

Research on deception suggests people are not very accurate lie detectors, so I should be cautious about turning one nonverbal behavior into evidence of dishonesty.

This example is strong because it moves the chapter from abstract research literacy to relational harm. It also has strong peer-reviewed support from deception meta-analyses. ([Sage Journals](#))

Example 3: ADHD TikTok and self-diagnosis content

Use this to show how social media can make research feel intimate, useful, and misleading at the same time.

Common-sense reading:

This video explains my behavior using clinical language, so now I know what is happening.

(Un)common-sense reading:

This video may help me notice a possible pattern, but I need to know what evidence it is drawing from, whether it is overgeneralizing, and what professional or scholarly context is missing.

Yeung et al. (2022) provide the peer-reviewed grounding here: in their sample of the 100 most popular ADHD TikTok videos, 52% were classified as misleading. ([PubMed](#))

Example 4: The 5-to-1 ratio

Use this to show the difference between algorithms and heuristics.

Common-sense reading:

For every negative interaction, do five positive things, and the relationship should improve.

(Un)common-sense reading:

Relationships often need more positive, affirming, repairing interaction than damaging interaction, but what counts as positive depends on the people, relationship history, rupture, timing, and context.

This is a clean bridge into the later assumptions because it shows how a summarized finding can become a rigid rule if students do not synthesize.

Example 5: AI summaries

Use this to show contemporary urgency.

Common-sense reading:

AI summarized the study, so now I understand it.

(Un)common-sense reading:

AI may help me enter the study, but I still need to check the original source, evaluate what was left out, and decide what the study can support.

This connects the chapter to automation bias, algorithm appreciation, and AI literacy. ([PubMed](#))

7. How This Section Should Lead into the Discontinuous Leaps

The section should end by telling students that the chapter is not just going to give them tips for reading research. It is going to ask them to make three discontinuous leaps in how they relate to research.

A strong structure:

Leap 1: Reporting to Sense-Making

The common-sense approach asks students to report what research says.

The (un)common-sense approach asks students to make sense with research.

Leap 2: Universal Truths to Local Truths

The common-sense approach asks whether the finding is true.

The (un)common-sense approach asks where, when, how, and for whom the finding is useful.

Leap 3: Summarizing to Synthesizing

The common-sense approach asks students to summarize the answer.

The (un)common-sense approach asks students to connect findings, methods, limits, contexts, other studies, and lived experience.

The closing bridge can say:

These leaps matter because research is not only something we use in school. Research claims shape how we understand ourselves, diagnose problems, judge other people, use AI, consume social media, and communicate in everyday life.

Draftable Version of the Section

Research can feel like something experts have and students are supposed to receive. That is part of what made the first 30-page article so intimidating. The article seemed to know things. I did not. So I assumed the point of reading research was to get the knowledge out of the article and into my head.

That assumption is not random. It comes from the common-sense approach to research.

From the common-sense approach, research is a method of knowing. A study produces findings. A research article reports those findings. The reader receives them, remembers them, cites them, and applies them. In this view, the article works like a container of expert knowledge. The researchers found something, wrote it down, and now the reader is supposed to know it too.

This approach is not useless. We do need evidence. We do need sources. We do need to know what a study found. The problem is what happens when that becomes the whole purpose of research. When research is treated only as knowing, a study can become a truth machine. A citation can become a weapon. A finding can become a rule.

That is how research gets flattened.

The common-sense approach encourages us to report what a study says, treat findings as universal truths, and apply research like an algorithm. If the study says X, then we should do X, and Y should happen. That kind of thinking is attractive because it promises clarity and control. It also shows up everywhere: in “research says” posts, social media health advice, communication hacks, AI summaries, workplace trainings, and everyday arguments where a citation is expected to end the conversation.

But research is usually more complex than that. Good research rarely gives us a universal rule for human behavior. It supports, complicates, qualifies, and raises new questions. A study may be useful and limited at the same time. A finding may be true in one context but not automatically true in another. A summary may be accurate but still incomplete.

The (un)common-sense approach begins there.

From the (un)common-sense approach, research is a method of understanding. A research article is not a magic answer machine. It is a resource for sense-making. It helps us ask better questions, notice patterns, evaluate evidence, understand limits, connect findings, and decide how a claim might or might not apply to real life.

This means reading research is not just about becoming more informed. It is about practicing literacies. Research literacy, information literacy, scientific literacy, data literacy, digital literacy, and AI literacy are all forms of sense-making. They help us ask where a claim came from, how it was produced, what evidence supports it, what it leaves out, and how it changes when the context changes.

So this chapter will ask us to make three discontinuous leaps. First, we will move from reporting to sense-making. Second, we will move from universal truths to local truths. Third, we will move from summarizing to synthesizing.

Together, these leaps change what research is for. Research is not something we use to become more certain as quickly as possible. Research is something we use to become more careful, more flexible, and more responsible with what we think we know.

APA References for the Drafting Packet

Bond, C. F., Jr., & DePaulo, B. M. (2006). Accuracy of deception judgments. *Personality and Social Psychology Review*, 10(3), 214–234. https://doi.org/10.1207/s15327957pspr1003_2

Chu-Ke, C., & Dong, Y. (2024). Misinformation and literacies in the era of generative artificial intelligence: A brief overview and a call for future research. *Emerging Media*, 2(1). <https://doi.org/10.1177/27523543241240285>

Deaton, A., & Cartwright, N. (2018). Understanding and misunderstanding randomized controlled trials. *Social Science & Medicine*, 210, 2–21. <https://doi.org/10.1016/j.socscimed.2017.12.005>

DePaulo, B. M., Lindsay, J. J., Malone, B. E., Muhlenbruck, L., Charlton, K., & Cooper, H. (2003). Cues to deception. *Psychological Bulletin*, 129(1), 74–118. <https://doi.org/10.1037/0033-2909.129.1.74>

Goddard, K., Roudsari, A., & Wyatt, J. C. (2012). Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association*, 19(1), 121–127. <https://doi.org/10.1136/amiajnl-2011-000089>

Haraway, D. (1988). Situated knowledges: The science question in feminism and the privilege of partial perspective. *Feminist Studies*, 14(3), 575–599. <https://doi.org/10.2307/3178066>

Klucsevsek, K. M. (2017). The intersection of information and science literacy. *Communications in Information Literacy*, 11(2), 354–365. <https://doi.org/10.15760/comminfoit.2017.11.2.7>

Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151, 90–103. <https://doi.org/10.1016/j.obhdp.2018.12.005>

Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–16. <https://doi.org/10.1145/3313831.3376727>

Mackey, T. P., & Jacobson, T. E. (2011). Reframing information literacy as a metaliteracy. *College & Research Libraries*, 72(1), 62–78. <https://doi.org/10.5860/crl-76r1>

Norris, S. P., & Phillips, L. M. (2003). How literacy in its fundamental sense is central to scientific literacy. *Science Education*, 87(2), 224–240. <https://doi.org/10.1002/sce.10066>

Rothwell, P. M. (2005). External validity of randomised controlled trials: “To whom do the results of this trial apply?” *The Lancet*, 365(9453), 82–93. [https://doi.org/10.1016/S0140-6736\(04\)17670-8](https://doi.org/10.1016/S0140-6736(04)17670-8)

Simons, D. J., Shoda, Y., & Lindsay, D. S. (2017). Constraints on generality: A proposed addition to all empirical papers. *Perspectives on Psychological Science*, 12(6), 1123–1128. <https://doi.org/10.1177/1745691617708630>

Sumner, P., Vivian-Griffiths, S., Boivin, J., Williams, A., Venetis, C. A., Davies, A., Ogden, J., Whelan, L., Hughes, B., Dalton, B., Boy, F., & Chambers, C. D. (2014). The association between exaggeration in health related science news and academic press releases: Retrospective observational study. *BMJ*, 349, g7015. <https://doi.org/10.1136/bmj.g7015>

Yeung, A., Ng, E., & Abi-Jaoude, E. (2022). TikTok and attention-deficit/hyperactivity disorder: A cross-sectional study of social media content quality. *The Canadian Journal of Psychiatry*, 67(12), 899–906. <https://doi.org/10.1177/07067437221082854>

Specific help request box

How to make the (un)common-sense side feel richer throughout

1. Make it a practice, not just a definition

Do not only define the (un)common-sense approach. Show what students actually do differently.

They ask different questions.
They read in a different order.
They treat limitations differently.
They use AI differently.
They respond to “research says” differently.
They apply findings differently.

2. Keep the student payoff visible

The (un)common-sense approach helps students:

- read research with less panic
- evaluate social media claims
- use AI without over-trusting it
- avoid turning studies into myths
- communicate with more humility
- make better personal and professional judgments
- ask stronger questions in relationships and workplaces

3. Make heuristics the thread across all three leaps

The heuristic thread should appear everywhere.

In **Reporting to Sense-Making**, the heuristic is:

“Ask what this study helps me understand.”

In **Universal Truths to Local Truths**, the heuristic is:

“Ask when, where, for whom, and under what conditions this finding applies.”

In **Summarizing to Synthesizing**, the heuristic is:

“Connect this study to methods, limits, other research, lived experience, and the situation where I might use it.”

4. Do not make the common-sense side stupid

The common-sense approach should feel tempting because it is useful in limited ways. Reporting matters. General truths are appealing. Summaries help. Students should not feel attacked for doing these things.

The problem is not reporting, truth, or summary.

The problem is stopping there.

H1: Practice Bridge

Section goal

The goal of this section is to act as the **conceptual hinge** between the opening story and the rest of the chapter.

In one sentence:

This section should help students realize that the problem is not just that research is hard to read; the problem is that many of us have been taught to relate to research as if it gives us certainty, when reading research well actually requires sense-making, literacy, context, and judgment.

Expanded goal

This section should take the feeling from the opening story — staring at a dense article, hoping it will give answers, treating research as a source of facts, truths, and certainty — and name it as the **common-sense approach to research**. In the opening draft, the narrator treats research as something that “knows things,” promises answers, and can be cited to win arguments; the story then connects that habit to “research says,” “studies show,” and the idea that a study becomes a truth machine, a citation becomes a weapon, and a finding becomes a rule.

From there, the section should explain that this common-sense approach is not foolish or useless. It makes sense because it mirrors the larger common-sense model of communication: information moves from sender to receiver. Chapter 1 defines the common-sense approach as treating communication like the transfer of information, while the (un)common-sense approach treats communicating as messy, collaborative, material, situated sense-making. Applied to research, that becomes: researchers found something, wrote it down, students read it, and now students know it.

But the section’s real job is to show why that approach becomes dangerous when it is the only approach students have.

What the section should accomplish

1. It should name the common-sense problem

The section should make clear that the common-sense approach treats research as a **method of knowing**. A research report appears to transfer expert findings from researchers to readers. The reader’s job becomes extracting the finding, deciding whether it is true or false, and applying it directly to life. Your analysis notes describe this as a deceptively simple model: read the article, receive the truth, decide whether the report is right or wrong, and apply the finding directly.

The problem is that this creates brittle certainty. Students may accept research that agrees with them, reject research that challenges them, or turn one finding into a rule. The chapter notes describe this as the shift from research as a linear transfer of expert truth to research as interactional sense-making.

2. It should explain why this matters for misinformation

The section should show that misinformation is not only fake information. Sometimes misinformation begins with **real research that gets flattened**.

- A careful finding becomes a slogan.
- A limited study becomes universal advice.
- A contextual claim becomes a TikTok rule.
- A citation becomes a weapon.
- A summary becomes “what research says.”

Your notes are especially clear that research misinformation often happens when research is treated through a knowing framework: the finding gets extracted, repeated, overgeneralized, and made more certain than the evidence allows.

This is why the section belongs in an interpersonal communication textbook. Misinformation about communication affects how people interpret partners, friends, coworkers, students, bodies, behavior, credibility, deception, mental health, intelligence, and everyday decision-making.

3. It should connect social media and AI to the same underlying problem

The section should not treat social media misinformation and AI overreliance as separate side topics. They are both intensified versions of the same common-sense fantasy:

Someone or something can hand me the answer.

Social media does this with “research says” claims, communication hacks, deception myths, dopamine explanations, ADHD advice, self-diagnosis content, and wellness rules. AI does it by producing fluent, organized summaries that can feel like understanding before the student has actually interpreted or evaluated the evidence.

The goal is not to make the chapter anti-social-media or anti-AI. The goal is to show students that in an information environment full of claims, summaries, screenshots, statistics, and AI-generated answers, they need stronger literacy practices, not just faster access to information. Your notes explicitly frame AI as useful for access and support, while also insisting that AI cannot do understanding for the reader.

4. It should introduce the (un)common-sense alternative

After students see the problem, the section should offer the alternative: the **(un)common-sense approach to research**.

This approach treats research as a **method of understanding**, not just knowing. A study is not a magic answer machine. It is a resource for sense-making. Students interact with the article, ask what it helps them understand, notice its limits, connect it to other evidence, and apply it flexibly rather than algorithmically.

Your notes define the (un)common-sense approach as reading research through partial, contextual, local evidence. Instead of treating the article as a complete truth, students ask what the study can and cannot support.

This also connects directly to Chapter 0. The knowing framework treats communication as information to collect, transfer, correct, or apply. The understanding framework treats communicating as contextual sense-making: figuring out how facts, people, relationships, histories, and meanings connect while interaction is still unfolding. This section simply applies that same distinction to research.

5. It should set up the three major leaps

This section should prepare students for the three organizing moves that follow:

Reporting to Sense-Making

Students move from reporting what research says to asking what research helps them understand.

Universal Truths to Local Truths

Students move from treating findings as universal rules to asking when, where, for whom, and under what conditions a finding applies.

Summarizing to Synthesizing

Students move from producing clean summaries to connecting studies with methods, limits, other research, lived experience, and actual communication problems.

Those three leaps give the chapter structure. They also keep the chapter from becoming a generic “how to read a research article” lesson. The notes say the chapter’s deeper purpose is to change students’ relationship to research: students should stop treating studies as fixed answers and start treating them as tools for understanding.

What students should leave this section believing

By the end of this section, students should believe:

- Research matters because it shapes everyday life, not just academic papers.**
- Reading research well does not require becoming an expert, but it does require better questions.**
- A source is not a magic truth stamp.**
- A study can be useful and incomplete at the same time.**
- AI can help students work with information, but it cannot understand for them.**

Literacies are sense-making practices.

Good research does not remove judgment; it helps students practice judgment.

What this section should not do

- It should not become a full methods chapter yet.
- It should not teach every type of research yet.
- It should not overwhelm students with metascience before they know why it matters.
- It should not make research sound impossible.
- It should not make AI sound evil.
- It should not imply that common sense is always wrong.

Instead, it should say:

Common sense is useful, but too small. Research as knowing helps us collect findings. Research as understanding helps us use those findings responsibly.

That phrasing fits the larger textbook pattern from Chapter 1, where the problem is not that common sense is always wrong; the problem is that common sense becomes too small when it is the only approach students have.

Best goal statement for your drafting packet

The goal of this section is to help students shift their relationship to research. After the opening story, students should see that many of us approach research through the common-sense logic of knowing: we expect studies to transfer truth, settle arguments, and give us rules. This section names that habit, shows how it contributes to misinformation, social media overgeneralization, and AI overreliance, and then introduces the (un)common-sense alternative: reading research as a practice of understanding. The section prepares students to treat research as partial, contextual, connected evidence that can guide judgment without becoming a weapon, a rulea rule, or a substitute for sense-making.

Practice narrative box

DRAFTING NOTE: Below I AM PROVIDING THE OUTLINE AND CONTENT FOR THIS SECTION. **Common sense asks us to read to know and collect.**
(Un)common sense asks us to read to understand and connect.

Collecting to Connecting

Most of us were taught to read research as if the article were a locked room full of facts.

Our job was to get in, gather as much as we could, and leave with the important pieces: the definition, the finding, the citation, the theory, the sample size, the conclusion, the sentence that sounded smart enough to underline. If we collected enough pieces, we could say we “knew” the article.

That is the common-sense approach to reading research.

It treats reading like transfer. The article has information. The reader needs information. Reading is the process of moving information from the article into the reader’s brain.

No wonder research articles feel so rude.

If that is what reading means, then every unfamiliar word becomes a wall. Every citation you do not recognize becomes evidence that everyone else arrived earlier. Every methods section feels like a locked basement. You start underlining sentences not because they make sense, but because they look like the kind of sentences a responsible person would underline.

You are not really reading anymore.

You are collecting.

And collecting can become exhausting because research articles contain far more than any reader can carry at once. They include arguments, concepts, theories, literature, methods, findings, limitations, implications, and references to other conversations that began long before you arrived. Trying to memorize all of that is not reading. It is academic hoarding.

The problem is not that facts are bad. You do need facts. You need to know what the article studied, what the authors found, what concepts they used, and how they supported their claims. But if you treat reading only as fact collection, you can miss the thing that makes research useful in the first place.

Research is not useful because it gives you more things to store.

Research is useful because it helps you make connections.

That is the leap this section asks you to make:

From collecting facts to connecting ideas.

The article is not as linear as it looks

One reason students often read research articles from the first word to the last word is that articles look like they are supposed to be read that way. They usually begin with an introduction, move into previous research, describe a method, present findings, and end with a discussion or conclusion. That structure makes research look tidy, almost like the authors calmly began at the beginning, followed a perfect plan, and arrived at the end with the findings they expected.

But research rarely happens that neatly.

Sarah Tracy argues that many qualitative research articles are forced into a conventional deductive writing style that makes research appear more linear, predictable, and rule-bound than it actually was. She describes the familiar journal article structure as a kind of “four-act play”: introduction and literature review, methods, findings or discussion, and conclusion. The problem is that this form can make inductive, interpretive, and iterative research look as if it unfolded in a straight line, even when the actual process involved circling back, revising questions, rethinking theory, following unexpected findings, and making meaning over time.

That matters for you as a reader.

If articles are often written into a cleaner order than the research actually followed, then reading them in that same clean order may not be the best way to understand them. You are allowed to move around. You are allowed to start with the discussion. You are allowed to look at the conclusion before you understand the literature review. You are allowed to skim the abstract, jump to the findings, go back to the method, and then reread the introduction once you finally know what the article is doing.

Research articles are not mystery novels.

You are allowed to find out where they are going before you decide how to travel with them.

The *Friend Not Foe* reading guide makes this point directly: articles are not mystery novels, so it is okay to start in the middle or even near the end. Skimming first helps you see what is familiar, what is unfamiliar, and where you need to spend more time.

That is not cheating.

That is reading strategically.

Stop trying to conquer the article

When research feels intimidating, it is easy to treat the article like an enemy. You sit down ready to defeat it. You sharpen your pen. You open the PDF. You prepare for battle.

The article, unfortunately, does not care.

It sits there with its tiny font and its theory section, completely unbothered.

The problem with treating an article like an enemy is that enemies invite conquest. You try to beat the article. You try to force it to surrender its facts. You try to extract the correct answer and move on.

But the (un)common-sense approach asks for a different relationship.

Do not conquer the article.

Get to know it.

That is the central metaphor of the *Friend Not Foe* guide. Reading a research article is more like getting to know a person than defeating a machine. When you make a friend, you do not learn everything about them in one minute. You ask questions. You spend time. You introduce them to other parts of your life. You notice what you share, where you disagree, what you want to understand better, and whether this is someone you will return to often or only occasionally.

Research articles work the same way.

Some articles will become close friends. You will return to them again and again because they shape your thinking, your project, your writing, or your sense of the world.

Some articles will be more like classmates you recognize but do not need to hang out with every weekend.

Some articles will be people you meet once, learn one useful thing from, and then politely move on.

That is okay.

Not every article needs the same level of intimacy.

Decide what kind of relationship you need

Before you read deeply, ask one question:

Why am I spending time with this article?

That question changes everything.

If you are reading for class discussion, you may need the main argument, one useful example, and one question you can bring to class.

If you are reading for a methods assignment, you may need to spend most of your time figuring out how the authors collected and analyzed data.

If you are reading for a literature review, you may need to summarize the article carefully and synthesize it with other studies.

If the article is central to your project, you may need to read it slowly, return to it, trace its citations, argue with it in the margins, and let it change how you understand the topic.

The *Friend Not Foe* guide makes this point in a practical way: the reason you are reading an article should determine how many of the 6 S's you need to complete. If you are just beginning a literature review, skimming may be enough. If the article is central to your work, you will probably need to summarize and synthesize.

This should be a relief.

You do not need to understand every article at the deepest possible level. Part of becoming research literate is learning how much understanding the situation requires.

Sometimes you need a map.

Sometimes you need a close reading.

Sometimes you need one useful connection.

The goal is not perfect understanding. Perfect understanding is not available anyway. You are not the author. You are not standing in the same place, asking the same questions, or bringing the same history to the article. Your understanding will always be situated.

That does not make it wrong.

It makes it yours.

The point is connection

When you read to collect, you ask:

What did the article say?
What are the key terms?
What did the authors find?
What citation can I use?
What should I remember?

When you read to connect, you ask:

What does this article help me understand?
What does this remind me of?
What does this complicate?
Where have I seen this pattern before?
How does this connect to another article?
What does this article help me ask next?

That shift matters because understanding is not the same as possession. You do not understand an article just because you possess its facts. You understand it when you can connect its pieces: the problem, the method, the findings, the limits, the context, and your own purpose for reading.

That is why summary alone is not enough.

A summary tells us what the article says.

Synthesis tells us what happens when the article enters a larger conversation.

The *Friend Not Foe* framework puts this clearly: summarizing means putting important ideas into your own words, while synthesizing takes summary further by making connections between existing knowledge and creating a new or original insight.

That is the move from collecting to connecting.

The 6 S's: Six moves, not six commandments

One practical way to read research through the (un)common-sense approach is to use the **6 S's**:

Skim
Scour
Scrawl
Seek
Summarize
Synthesize

These are not steps you must follow in order.

They are moves.

Think of them like moves in a conversation. Sometimes you listen first. Sometimes you ask a question. Sometimes you interrupt yourself because you realize you misunderstood something. Sometimes you circle back to something said earlier because now it makes more sense.

Reading research works the same way.

You might skim the article, then jump to the discussion, then seek a definition, then scour the findings, then scrawl a question in the margin, then summarize the main argument, then synthesize it with another reading from class.

Or you might start by seeking background information because the theory is unfamiliar.

Or you might begin with the conclusion because you need to know where the article is headed before the introduction makes sense.

Or you might read the methods first because the assignment asks you to evaluate research design.

The point is not to obey the 6 S's.

The point is to use them.

Skim: Get the shape of the article

Start by getting a feel for the article.

Read the title. Read the abstract. Look at the headings. Find the research questions or hypotheses. Glance at the method. Look at the findings. Read the discussion or conclusion. Notice any tables, figures, or models.

At this stage, you are not trying to understand everything. You are trying to answer:

What kind of article is this, and where will I need to slow down?

Skimming gives you a map. It helps you see what looks familiar, what looks confusing, and where the article is likely to matter for your purpose.

A good skim might produce a note like:

This article studies how romantic partners talk about jealousy. I understand the topic, but I need to spend more time on the theory and findings.

That is already understanding beginning to form.

Scour: Slow down where it matters

After you skim, choose where to slow down.

You do not need to scour every sentence equally. That is how articles become swamps.

Scour the parts that matter most for why you are reading. If the article is assigned because of its findings, spend time with the findings. If it is assigned because of its method, slow down in the method. If it uses a theory you need for your paper, sit with the theory section. If the discussion explains why the article matters, spend time there.

The *Friend Not Foe* guide notes that scouring often includes results or findings because those sections usually present new information, but students may also need to scour theory or method depending on what is unfamiliar.

A useful question:

Where is this article asking for my attention?

Scouring is not panic-rereading.

It is targeted attention.

Scrawl: Leave evidence of the interaction

Write on the article. Write beside it. Write in a notebook. Use digital comments. Make a messy document. Talk into a voice memo. Draw arrows. Circle confusion. Ask rude but academically productive questions.

Your notes do not need to be beautiful.

They need to show interaction.

The *Friend Not Foe* guide calls this “scrawl” because the point is not polished note-taking. The point is engagement. A blank copy of a research article can be a sign that you have not interacted with it yet.

Scrawling might look like:

- This reminds me of the chapter on communication recipes.
- I do not understand how they define “support.”
- This finding seems useful, but does it apply outside college students?
- This complicates “just be direct.”
- Why did they choose interviews instead of surveys?

That is what connecting looks like in real time.

Seek: Find what you need

Research articles are full of unfamiliar terms, theories, methods, and references. That does not mean you are failing. It means you are entering a conversation already in progress.

Seek what you need.

Look up a concept. Ask your professor. Ask a classmate. Check a textbook. Read a source the article cites. Use AI carefully to explain an unfamiliar term, then verify that explanation with credible sources. Search for a review article if you need a broader map.

Seeking turns confusion into action.

Instead of thinking:

- I do not know this, so I cannot read this.

Try:

- I do not know this yet, so I need to find a handle.

A useful question:

What do I need to understand before this section can make sense?

Summarize: Put the article in your own words

A summary is not a quote collection. It is not the abstract with a different hat.

A useful summary puts the article into your own words for your own purpose.

You might use this template:

- This article studies _____ by using _____. The authors argue or find _____. This matters because _____. For my purpose, the most useful part is _____.

For example:

- This article studies how friends avoid conflict by interviewing college students. The authors argue that avoidance is not always a failure to communicate; sometimes it is a way people try to protect the friendship. This matters because it complicates the advice to “just say what you mean.” For my purpose, the most useful part is the idea that avoidance can function as care, fear, or self-protection depending on context.

Notice what that summary does.

It does not collect everything.

It connects the article to a communication problem.

Synthesize: Connect the article to something beyond itself

Synthesis is where the article becomes part of your understanding.

To synthesize, connect the article to:

- another article
- a class concept
- a theory
- a personal experience
- a workplace example
- a social media claim
- a contradiction
- a limitation
- a new question

Synthesis might sound like:

- This article helps explain why “communicate more” is not always good advice.
- This study agrees with the previous article that support matters, but it defines support differently.
- This finding might apply to roommate conflict, but I am less sure it applies to parent-child conflict because the power dynamic is different.
- This article makes me wonder whether avoidance is sometimes a relationship maintenance strategy rather than simply a conflict problem.

That is the difference between carrying facts and building understanding.

Make the framework fit you

The 6 S's are useful, but they are not sacred.

You can change them.

You can reorder them. You can rename them. You can add to them. You can turn them into a worksheet, a spreadsheet, a color-coding system, a mind map, a voice memo routine, a reading group, or a one-page article template.

The *Friend Not Foe* guide explicitly encourages this kind of adaptation: the framework is one approach, not an exhaustive system, and students can make minor or major changes that help them engage research more productively.

That matters because your brain does not work exactly like anyone else's brain.

Some people understand by writing.

Some understand by talking.

Some need to draw.

Some need structure.

Some need permission to move around.

Some need to start with the ending.

Some need to read with a timer.

Some need to print the article because PDFs make their soul leave the room.

Some need to listen to text-to-speech while walking around because sitting still with a 30-page article is a little too much like being punished by furniture.

Use what works.

If a framework helps you connect, keep it.

If it makes you feel smaller, change it.

The goal is not to become good at a framework. The goal is to become more flexible in your reading.

A better way to know you are done

When are you "done" reading an article?

The common-sense answer is:

When I know it.

That sounds simple, but it is not very helpful. What does "know it" mean? Memorize it? Repeat it? Agree with it? Cite it? Win an argument with it?

The (un)common-sense answer is more flexible:

I am done for now when I understand what I need from this article for this purpose.

That might mean you can explain the main argument. It might mean you can identify the method. It might mean you can use one finding responsibly. It might mean you can connect the article to another study. It might mean you know enough to decide this article is not actually useful for your project.

That counts.

Reading research is not a test of whether you can absorb everything. It is a practice of figuring out what kind of understanding you need and how to build it.

So the next time you open an article and feel the familiar dread rising, try not to begin with:

I have to know all of this.

Begin with:

What kind of connection am I trying to make?

Then skim. Scour. Scrawl. Seek. Summarize. Synthesize.

Not necessarily in that order.

Not necessarily all at once.

Not because the framework is magic.

Because reading research is not about collecting enough facts to survive the article.

It is about connecting well enough that the article can help you understand something beyond itself.

Why collecting makes reading harder

Collecting sounds responsible. After all, research articles do contain information. They include definitions, arguments, methods, findings, citations, theories, tables, examples, limitations, and implications. Students do need to learn how to identify those pieces.

The problem is what happens when collecting becomes the whole goal.

When you read only to collect, you can start treating the article like a storage unit full of facts. Your job becomes finding all the important pieces and carrying them away. This can make reading feel like memorization instead of sense-making.

You might ask:

What does this word mean?

What did the authors find?

What is the definition?

What is the hypothesis?

What is the citation?

What sentence should I highlight?

What fact will be on the quiz?

Those questions can help. But they are incomplete. They keep your attention on pieces rather than relationships among pieces.

Understanding asks different questions:

What problem are the authors trying to understand?

Why did they study it this way?

What does this finding help me notice?

How does this connect to something I already know?

Where might this finding apply?

Where might it not apply?

What does this article help me ask next?

The difference matters because reading is not just information transfer. Learning is not just copying content from one container into another container. Your brain is not a tiny academic moving truck.

Reading research is an interaction.

That is one of the reasons Sarah Tracy’s critique of conventional journal writing matters here. Tracy argues that many research articles, especially in mainstream journals, are written as if research unfolded in a neat, linear, deductive order: introduction, literature review, methods, findings, conclusion. She calls this conventional structure a “four-act play” and argues that forcing inductive qualitative research into that deductive style can create ethical problems, pedagogical confusion, constrained theory building, and inappropriate standards for judging quality (Tracy, 2012).

That matters for student readers because the article you read may look much cleaner, straighter, and more linear than the research process actually was.

Researchers often do not write articles in the same order students read them. They may move from findings to introduction, back to methods, into theory, back to analysis, back to literature, and then into revision. In qualitative research especially, researchers may not know the specific contribution of the study until they are deep into data collection, analysis, and writing. Tracy argues that when this messy, iterative process is forced into a neat deductive order, readers can get a misleading picture of how research happens.

So if you read an article from beginning to end and feel like you are not understanding it in the “right” order, the problem may not be you.

The problem may be the assumption that there is one right order.

Research articles are not mystery novels

One of the most useful things to remember is this:

Research articles are not mystery novels.

You are allowed to skip to the ending.

Actually, you probably should.

A mystery novel depends on suspense. If you read the last page first, you ruin the experience. A research article is different. The goal is not surprise. The goal is understanding.

This means you do not have to start with the first sentence and march obediently to the final sentence. You can move around. You can begin with the abstract. You can read the discussion before the methods. You can skim the conclusion and then return to the introduction. You can look at the findings and then go backward to figure out why those findings matter.

The “Friend Not Foe” research reading framework makes this point clearly: research articles are often difficult, even for people who write them, and students should approach them by spending time with them, asking questions, making connections, and learning how to interact with them rather than trying to conquer them.

That is the core idea of this section.

You are not trying to defeat the article.

You are trying to develop a workable relationship with it.

Choose the level of understanding you need

Here is another piece of good news: you do not need the same depth of understanding from every article.

Some articles will become close friends. You will return to them again and again. They may shape a paper, a project, a literature review, a research question, or the way you understand a communication problem.

Other articles will be more like acquaintances. You need to know who they are, what they do, and why they showed up in the conversation. You may not need to spend three hours getting emotionally attached to their methods section.

That is not laziness. That is strategy.

The level of understanding you need depends on why you are reading.

If you are reading an article because your professor assigned it for class discussion, you may need to understand the main argument, the key finding, and one or two questions you can bring to class.

If you are reading an article for a methods assignment, you may need to spend most of your time on the research design, sample, measures, interview protocol, coding process, or analytic choices.

If you are reading an article for a literature review, you may need to summarize it carefully and synthesize it with other research.

If you are reading an article because it is central to your own project, you may need to spend serious time with it. You may read it more than once. You may annotate it heavily. You may trace its citations. You may argue with it in the margins.

The “Friend Not Foe” handout makes this point directly: the reason you are reading an article should shape how many parts of the reading framework you need to complete. Sometimes skimming is enough. Other times, especially when an article is central to your project, you will need to summarize and synthesize.

So before you read, ask:

What kind of relationship do I need with this article?

A quick introduction?

A working familiarity?

A deep understanding?

A long-term scholarly friendship?

That question can make research reading much less intimidating.

Frameworks help because they give you handles

A reading framework is not a law. It is not a personality test. It is not a sacred ritual.

A framework is a set of handles.

It gives you something to grab when the article feels too big, too dense, or too unfamiliar. It helps you decide where to start, where to slow down, what to ignore for now, what to return to later, and how to make sense of what you are reading.

Frameworks are helpful because they reduce the pressure to understand everything at once.

Instead of thinking, “I have to understand this whole article,” you can think:

I am skimming right now.

I am scouring this section.

I am scrawling questions.

I am seeking background.

I am summarizing the main point.

I am synthesizing this with something else I know.

That shift matters. It turns reading from a vague demand into a flexible practice.

But here is the most important part:

You can change the framework.

Actually, you should.

Your brain does not work exactly like everyone else’s brain. A reading strategy that helps one student may frustrate another. A framework that works for a short quantitative article may not work as well for a long ethnography. A strategy that helps during finals week may not help when you are drafting a literature review.

If a framework makes reading harder, that does not mean you are failing.

It may mean the framework does not fit the task, the article, or your brain.

So use frameworks. Borrow them. Try them. Bend them. Combine them. Make your own. The goal is not to become loyal to the framework. The goal is to use the framework to understand.

The 6 S's: A nonlinear framework for reading research

One useful framework for reading communication research articles is the **6 S's**:

Skim
Scour
Scrawl
Seek
Summarize
Synthesize

These do not have to happen in order. They are not a staircase. They are not a recipe. They are more like six moves you can use as needed.

You might skim, then seek, then skim again.

You might scour the methods, scrawl questions, then jump to the discussion.

You might summarize before you fully synthesize.

You might synthesize with one other article and then return to scour the findings more carefully.

That is the point. Reading research is nonlinear because understanding is nonlinear.

1. Skim

Skimming means getting a first map of the article.

You are not trying to understand every sentence yet. You are trying to figure out what kind of article you are dealing with.

Start by looking at:

the title
the abstract
the section headings
the first few pages
the research questions or hypotheses
the method
the findings or results
the discussion or conclusion
any tables, figures, or models

Skimming helps you identify what is familiar and what is not. It also helps you decide where you need to spend more time. The "Friend Not Foe" framework says skimming lets readers see what they already recognize and where they need to slow down and scour.

A practical skimming question:

What is this article probably about, and where will I need to slow down?

You might write a quick note like:

This article studies friendship maintenance on social media. I understand the topic, but I need to spend more time on the theory and results.

That is already progress.

You have started connecting.

2. Scour

Scouring means slowing down in the parts of the article that matter most or feel least familiar.

You do not scour everything equally. That would be exhausting and possibly illegal under several laws of emotional sustainability.

Instead, you choose where to slow down based on why you are reading.

If you are reading for the main findings, scour the results and discussion.

If you are reading for methods, scour the methods section.

If you are reading for theory, scour the literature review and theoretical framework.

If you are reading for a class conversation, scour the parts your professor emphasized.

The "Friend Not Foe" handout notes that scouring often includes the results or findings because those sections present new information and may take more time to engage. It may also include theory or method if those are unfamiliar.

A practical scouring question:

Where does this article ask me to slow down?

You might write:

I need to reread the findings because this is where the authors explain their contribution.

Or:

The method is unfamiliar, so I need to spend more time figuring out how they collected and analyzed the data.

Scouring is not rereading everything because you are panicking. Scouring is targeted attention.

3. Scrawl

Scrawling means writing on, beside, around, or about the article as you read.

This might mean writing in the margins. It might mean using digital comments. It might mean keeping a reading notebook. It might mean making a messy document full of fragments, questions, arrows, reactions, and half-formed thoughts.

The point is not to produce beautiful notes. The point is to interact.

Scrawling can include:

questions
confusion
definitions
examples
connections
agreements
disagreements
"this reminds me of..." notes
"come back to this" notes
"what???" notes
small summaries
arrows between ideas

The "Friend Not Foe" framework emphasizes that notes do not need to be perfect, complete, or polished. Their value is that they help cement the article in memory and show engagement.

But remember: scrawling is not the same thing as understanding.

Underlining a sentence is not comprehension. Highlighting half a page in yellow is not, by itself, intellectual transformation. It may be the beginning of engagement, but the real question is:

What did this mark help me understand?

A practical scrawling move is to write short connection notes:

- This reminds me of Chapter 1’s critique of communication recipes.
- This finding might connect to roommate conflict.
- I disagree because the sample seems too narrow for the claim.
- I need to look up “relational turbulence.”

These notes help move you from collecting to connecting.

4. Seek

Seeking means looking for what you do not know yet.

Research articles are full of unfamiliar terms, theories, methods, acronyms, and references. That does not mean you are not smart enough to read them. It means you are entering a conversation that has already been going on for a while.

When you encounter something unfamiliar, seek help.

You can:

- look up a term
- ask your professor
- ask a classmate
- check a textbook
- look at a cited source
- search for background information
- look for a review article
- use AI carefully to clarify vocabulary, then verify with credible sources

The “Friend Not Foe” framework encourages students to seek out what they do not know rather than letting jargon stop them. It specifically recommends asking professors and using outside searching to contextualize unfamiliar theories or terms.

Seeking is not cheating. Seeking is part of reading.

Experts do this too. Researchers look things up constantly. The difference is not that experts never get confused. The difference is that experts have more practice figuring out what to do with confusion.

A practical seeking question:

What do I need to know before this part can make sense?

You might write:

- I need a basic definition of social penetration theory before this section will make sense.

Or:

- I need to understand what “thematic analysis” means before I can evaluate the method.

Seeking turns confusion into a task.

5. Summarize

Summarizing means putting the article’s important ideas into your own words.

A good summary is not a pile of quotations. It is not copying the abstract. It is not replacing every third word with a synonym and hoping the citation gods look away.

A good summary explains what matters for your purpose.

Two people can read the same article and summarize different things because they are reading for different reasons. If you are writing a paper on relationship maintenance, you may summarize the article’s findings about daily affection. If someone else is writing about methods, they may summarize the same article’s survey design.

The “Friend Not Foe” framework explains that summaries should paraphrase important ideas relevant to your project and should be in your own words because putting an idea into your own words helps you connect what you are reading to what you already know.

A practical summary template:

- In this article, [authors] study [topic/problem] by using [method] with [participants/texts/context]. They argue/find that [main point]. This matters because [contribution]. For my purposes, the most useful part is [specific idea].

For example:

- In this article, the authors study how college roommates manage conflict by interviewing first-year students. They argue that roommates often avoid direct conflict to preserve daily comfort, even when avoidance creates longer-term tension. This matters because it complicates the advice to “just be direct.” For my purposes, the most useful part is the connection between avoidance, politeness, and shared living spaces.

Notice what happened there. You did not collect the whole article. You connected the article to a communication problem.

6. Synthesize

Synthesizing means connecting the article to something beyond itself.

This is the heart of the (un)common-sense approach.

Summary asks:

What does this article say?

Synthesis asks:

What happens when I connect this article to another article, another theory, another experience, another example, or another question?

The “Friend Not Foe” framework describes synthesis as taking summary one step further by making connections between existing knowledge to create a new or original insight.

Synthesis can connect an article to:

- another research article
- a theory from class
- your own lived experience
- a social media example
- a workplace problem
- a relationship pattern
- a contradiction
- a limitation
- a new research question

Synthesis sounds like:

This article helps explain why “just communicate more” may fail when people keep repeating the same conflict pattern.

This study agrees with the previous article that support matters, but it defines support differently.

This finding might apply to workplace feedback, but I am not sure it applies to family conflict because the power dynamics are different.

This article complicates my assumption that directness is always more honest.

This makes me wonder whether the same pattern would appear in online friendships.

That is connecting.

You are no longer just holding information. You are using information to understand.

A nonlinear example

Here is what using the 6 S's might actually look like.

You open a 28-page research article on conflict avoidance in friendships.

First, you **skim** the title, abstract, headings, and conclusion. You figure out that the article is about why friends avoid conflict even when avoidance causes tension.

Then you **seek** a term you do not know: “facework.” You look it up and realize it has to do with protecting identity, dignity, or social image in interaction.

Then you **scrawl** in the margin:

Avoidance may not just be fear. It may be a way of protecting the friendship.

Then you **scour** the findings section because that is where the authors explain the patterns they found in interviews.

Then you **summarize** the main point:

The authors argue that conflict avoidance can function as care, fear, politeness, or self-protection depending on the friendship context.

Then you **synthesize** it with something from class:

This complicates the common-sense advice to “be direct.” Directness might help in some friendships, but in others, people may avoid conflict because they are trying to preserve the relationship.

Notice the order:

Skim → Seek → Scrawl → Scour → Summarize → Synthesize.

But another article might require a different order.

You might start with the discussion. You might scour the methods first. You might summarize before seeking more background. You might synthesize after reading three related articles.

That is fine.

The framework exists to help you read. You do not exist to obey the framework.

Make the framework work for your brain

The 6 S's are one possible framework. They are not the only way to read.

You can modify them.

You might add:

Sort — organize articles by theme, method, or usefulness.

Sketch — draw a concept map or diagram.

Speak — explain the article out loud to someone else.

Save — keep a reusable note with the citation, summary, and key connections.

Share — bring one question or connection to class.

Sleep — step away and return later because sometimes your brain needs to stop being aggressively academic.

You might rename the steps in language that makes more sense to you.

You might turn the 6 S's into a checklist, a mind map, a spreadsheet, a sticky-note system, a digital template, a voice memo, or a conversation with a classmate.

One professor in the “Friend Not Foe” handout recommends using a spreadsheet to track the why, how, where, and main findings of each article. Another recommends starting with the discussion, then skimming the results, then reading from beginning to end because the discussion helps frame why the article matters.

The point is not that one strategy is correct. The point is that experienced readers develop strategies.

You can too.

Ask yourself:

Where do I usually get stuck?

Do I need structure, flexibility, or both?

Do I understand better by writing, talking, drawing, or mapping?

Do I need to start with the conclusion so I know where the article is going?

Do I need to read with a timer so I do not spend 45 minutes on one paragraph?

Do I need a glossary of key terms?

Do I need to read with someone else?

Do I need to make a one-paragraph summary before I move on?

Your framework should help you interact with research. If it makes you feel smaller, more panicked, or more confused, change it.

Reading to connect changes what counts as success

From the common-sense approach, success means:

I know the article.

I collected the facts.

I found the answer.

I can repeat the definition.

I can cite the finding.

From the (un)common-sense approach, success sounds different:

I understand what the article helps me see.

I know where I am still confused.

I can explain the main point in my own words.

I can connect it to another idea.

I can identify what it does and does not show.
I can decide how deeply I need to know it.
I can use it without pretending it says everything.
That last part matters.

You will never understand an article perfectly. You are not the author. You are not standing in the same place, with the same background, the same history, the same purpose, or the same relationship to the project. Even the authors’ own understanding of the article likely changed as they wrote and revised it.

That does not make your understanding wrong.

It makes understanding situated.

Your job is not to become the article. Please do not become the article. That sounds uncomfortable and probably full of footnotes.

Your job is to build a useful relationship with the article.

Sometimes that relationship will be brief. Sometimes it will be deep. Sometimes it will be frustrating. Sometimes it will change the way you see a communication problem. Sometimes it will mostly teach you that you need a different article.

All of that counts.

Because reading research is not about collecting everything.

It is about connecting enough to understand what the article can help you do next.

Possible callout box: Try This Reading Plan

TRY THIS: THE 20-MINUTE FIRST DATE WITH A RESEARCH ARTICLE

Set a timer for 20 minutes. You are not trying to understand everything. You are just getting introduced.

1. **Skim** the title, abstract, headings, and conclusion.
2. **Scrawl** three margin notes: one thing you understand, one thing you do not understand, and one thing the article reminds you of.
3. **Seek** one unfamiliar term.
4. **Summarize** the article in two sentences.
5. **Decide** whether this article needs a second date.

Not every article is your best friend. Some are acquaintances. Some are classmates you respect but do not need to hang out with every weekend.

Possible callout box: Collecting vs. Connecting

COLLECTING ASKS:
What did the article say?
What facts can I pull out?
What quote can I use?
What did the authors find?
Can I remember this?

CONNECTING ASKS:
What does this article help me understand?
How does it connect to what I already know?
What does it complicate?
Where might it apply?
Where might it not apply?
What should I ask next?

References for this section

Allen, J. (n.d.). *Friend not foe: An approach to reading communication research articles* [Course handout].

Tracy, S. J. (2012). The toxic and mythical combination of a deductive writing logic for inductive qualitative research. *Qualitative Communication Research*, 1(1), 109–141.

Closing: What to Carry Forward

Section goal

Return to the opening story, name the core myth again, summarize the big shift in one or two vivid sentences, and leave students with a question or challenge that motivates continued practice.